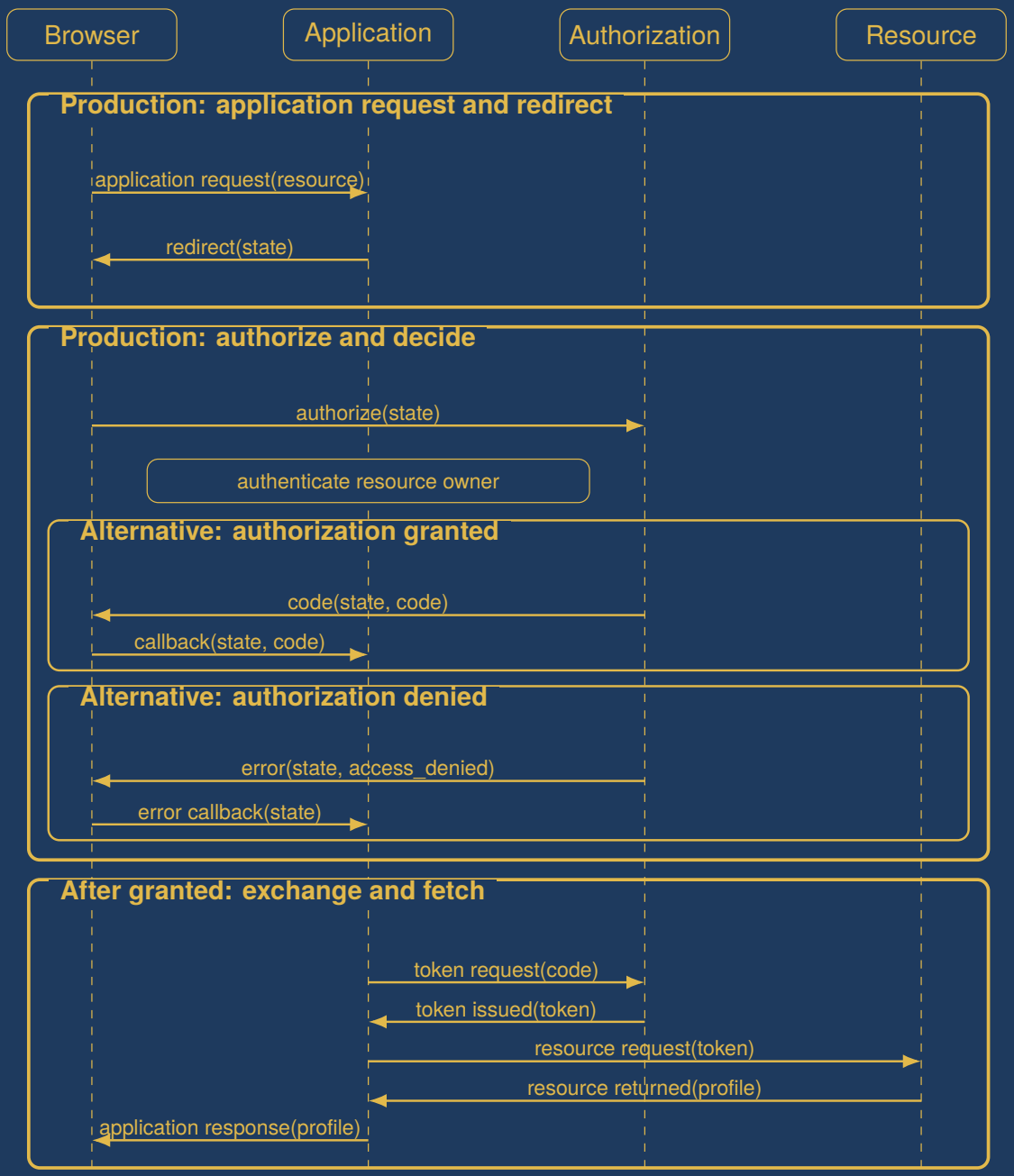
LeanFM Software Engineering



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Preface

LeanFM is a software-engineering methodology for arguing requirements into existence with an LLM. Stakeholders challenge assumptions, add scenarios and constraints, and inspect generated diagrams, traces and measurements. The argument produces durable project artifacts rather than relying on chat memory.

The book is the project's durable design argument and LLM context; `book/main-dark.pdf` is its accessible companion. Figures are generated from the shared specification. The Bootstrap chapter defines the project-root handoff.

A conversation is provisional. The proposed compaction transaction checkpoints decisions into durable artifacts and clears only the captured message prefix after artifact and view validation and recoverable commit; failure preserves prior artifacts and messages. It accepts neither gate. Answers attach to identified questions; the same semantic question is not repeated unless reopened, though a new ambiguity may be asked. Ordinary prompts may proceed while clarification is deferred. The Chat chapter specifies these policies.

Each project root contains its complete resumable state; this repository is itself one such project. The optional examples under `projects/` may be deleted without breaking LeanFM startup, project creation or required checks. Conversation files are disposable too: each turn loads the selected project's book, requirements, protobuf schema, implementation mapping, manifest and source inventory. Missing inputs remain explicit gaps, never invented acceptance or fallbacks to another project. Bootstrap defines stable paths and the two gates: observable requirements, then their implementation mapping. Discovery remains iterative; one-pass implementation is ready only when generated source passes increasingly realistic in-project tests. This requirements-to-PR workflow remains under development.

LeanFM centers a shared, executable account of actor communication down to protobuf bytes. Parameterized multiparty grammars describe legal sequences; generated views and Lean theorems check the model. Not every requirement is an observable message: referential integrity and cross-object constraints are predicates over decoded values and reachable states. Protobuf can encode an ID, but does not enforce its relationship to another object. Requirements.lean states and proves such predicates; Implementation.lean maps them to storage, transactions, or validation. For example, if A must reference B and B requires a C that policy forbids, the requirements are inconsistent and must change before implementation.

Chapter 1 develops this argument into a practical method; the remaining chapters present LeanFM's evolving language, tooling, examples, and implementation [2].

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Part I

The Grammar-Based Method

Chapter 1

Formal Methods as the Default Path

Formal methods are usually presented as an extra stage: first build a system, then translate a simplified version into a model, and finally ask a specialist tool a few questions. That workflow taxes correctness. The model drifts, the difficult translation is postponed, and proofs concern yesterday's architecture.

Our design target is the reverse: the shortest path to a running service passes through an executable behavioural specification. The specification is intended to supply message types, scenario tests, model-checking input, documentation, and the service skeleton. This is the target workflow, not a claim that every generator is already implemented.

Evidence labels used throughout: proposed notation (uncompiled); executable code (builds or runs, not proof); finite-model check (exhaustive only within declared bounds); test (named finite cases); generated analysis (named inputs and reducer); Lean proof (kernel-checked under stated assumptions); future interface (planned, unavailable). Labels scope evidence, not rank it; several may apply to one artifact.

1.1 Behaviour before machinery

The primary object is a set of permitted, visible histories: who communicated with whom, which event occurred, what it carried, and which predecessor order it extends. Protocol bytes, ordering and persistent disk formats become durable compatibility boundaries once another process, customer or stored data set depends on them.

Threads, actors, queues, databases, caches, continuations and retry machinery are realization choices unless they change that observable contract. Modeling every internal choice couples requirements to harmless refactoring and creates a product state space that grows exponentially with independently varying components. Begin with the durable boundary; introduce internal state only when it changes an observable promise.

1.2 Comparison with TLA+

LeanFM is not an extension of TLA+ and does not try to replace it. TLA+ commonly describes a system with variables, an initial-state predicate, a next-state relation, and temporal formulas over the resulting state behaviors [6]. A message queue can be represented as a sequence- valued variable,

and an action can append to or remove from that sequence. This is expressive and makes internal state transitions available to invariants and model checking.

1.2.1 An executable queue model

This TLA+ module sends an order to Bob's inbox, then lets Bob consume it and change his control state:

```
---------------------------- MODULE OrderQueue ----------------------------
EXTENDS Integers, Sequences, FiniteSets, TLC
CONSTANTS Clients, Servers, Orders, QueueBound, CacheBits, ClockBound
Actors == Clients \cup Servers
Messages == [kind : {"placeOrder", "placedOrder"},
             order : Orders, replyTo : Clients]
NoMessage == [kind |-> "none", order |-> "none", replyTo |-> "none"]
VARIABLES inbox, pc, current, sent, placed, received, scratch, tick
vars == <<inbox, pc, current, sent, placed, received, scratch, tick>>

Init == /\ inbox = [a \in Actors |-> <<>>]
        /\ pc = [b \in Servers |-> "idle"]
        /\ current = [b \in Servers |-> NoMessage]
        /\ sent = {} /\ placed = {} /\ received = {}
        /\ scratch = [i \in 1..CacheBits |-> FALSE]
        /\ tick = 0

Send(a, b, o) ==
  /\ o \notin sent /\ Len(inbox[b]) < QueueBound
  /\ inbox' = [inbox EXCEPT ![b] = Append(@,
       [kind |-> "placeOrder", order |-> o, replyTo |-> a])]
  /\ sent' = sent \cup {o}
  /\ UNCHANGED <<pc, current, placed, received, scratch, tick>>

Receive(b) ==
  /\ pc[b] = "idle" /\ Len(inbox[b]) > 0
  /\ current' = [current EXCEPT ![b] = Head(inbox[b])]
  /\ inbox' = [inbox EXCEPT ![b] = Tail(@)]
  /\ pc' = [pc EXCEPT ![b] = "busy"]
  /\ UNCHANGED <<sent, placed, received, scratch, tick>>
```

The excerpt above defines actions but does not yet connect them to a behavior. That connection appears later in the same module:

```
Next == (\E a \in Clients, b \in Servers, o \in Orders: Send(a,b,o))
     \/ (\E b \in Servers: Receive(b) \/ Finish(b))
     \/ (\E a \in Clients: ReadReply(a)) \/ CacheStep \/ ClockStep
Spec == Init /\ [][Next]_vars
```

TLA+ does not call `Send` or `Receive` like procedures. `Next` is a relation between the current variables and their primed next values. Its disjunction says that one step may be a `Send` for some client, server and order; a `Receive` or `Finish` for some server; a `ReadReply` for some client; a cache step; or a clock step. A candidate action whose guards are false is disabled. TLC explores every enabled choice.

Read `Spec == Init /\ [][Next]_vars` as: "the first state satisfies `Init`; thereafter, every state transition satisfies `Next` or leaves every variable in `vars` unchanged." The configuration closes the final loop with `SPECIFICATION Spec`; it tells TLC which temporal formula to explore.

Read `inbox'` as "the next value of inbox," `@` as "the old value at this selected entry," and `UNCHANGED` as "these variables retain their values." `Append` adds a message; `Head` selects the first message; `Tail` removes it. Alice's send updates Bob's inbox, not Bob's private control state. Bob's receive performs the latter transition. The configuration instantiates `Clients = {"alice"}` and `Servers = {"bob"}`; functions indexed by those sets hold instance state. The full module commits the order, sends a reply, and checks that no reply precedes placement. No eventual-reply property is checked.

1.2.2 The cost of irrelevant state

The module includes optional cache-bit toggles and a clock increment that cannot affect an order message. TLC nevertheless distinguishes their values:

Model	Cache bits	Clock bound	Distinct states
Baseline	0	0	26
Cache4	4	0	416
Cache8	8	0	6,656
Clock7	0	7	208

These exhaustive finite-model safety-check counts come from TLC 1.7.4 via `make tla-check`; evidence and provenance are in `book/generated/tla-state-space.json`. With S protocol states, b independent Boolean cache bits and a counter ranging from zero through k, the example has

$$S\,2^b(k+1)$$

states. Read: "S times two to the b times k plus one." S is the protocol state count, b the cache-bit count and k the counter bound. Each irrelevant bit doubles this model's state count. `ClockBound = -1` allows ticks 0, 1, 2 and onward: infinitely many states with the same order behavior. Exhaustive explicit-state search cannot finish that variant. Unbounded queues or retained histories can also make a model infinite.

This is checker state growth, not measured application memory. More states usually require more exploration time and storage; the table gives no fixed bytes-per-state or timing law. TLA+ supports abstraction, hiding and refinement. The lesson is to omit irrelevant state, not to blame the language for its inclusion.

1.2.3 Proposed message-first notation

This is invented TLA+-inspired syntax, not standard TLA+ or a compiled LeanFM feature:

```
// Proposed TLA+-inspired notation. Not standard TLA+, not compiled by LeanFM.
message placeOrder(orderId: OrderId, replyTo: ClientId)
message placedOrder(orderId: OrderId)

actor class Client {
  input placedOrder
  output placeOrder
}
actor class Shop {
  input placeOrder
  output placedOrder
```

```
  inbox FIFO(capacity = 2)
  on placeOrder(orderId, replyTo) {
    require commitRecorded(orderId)
    self:replyTo ! placedOrder(orderId)
  }
}

alice = new Client()
bob = new Shop()
alice:bob ! placeOrder(o1, replyTo = alice)
bob ? placeOrder(o1, replyTo = alice)
// Shop handler records the commit and sends placedOrder.
alice ? placedOrder(o1)

web = instances(WebServer, count = 3, route = roundRobin)
documents = shards(DocumentStore, count = 8,
                   key = documentPartition, replicas = 3,
                   writeQuorum = 2)
```

Read `alice:bob ! placeOrder` as "Alice sends placeOrder to Bob," and `bob ? placeOrder` as "Bob receives placeOrder." Bob consumes the request `placeOrder`; Alice consumes the reply `placedOrder`. The proposal treats inboxes as first-class actor-owned queue state, not arbitrary private variables. Blocking and capacity semantics are specified in Chapter 5; this proposed syntax does not implement them. The handler's commit precondition is an observable obligation, not a database algorithm.

Actor classes describe behavior; instances have distinct identities, queues and local state. WebServer instances share a class, not an inbox. DocumentStore shards partition ownership by key; replicas protect each shard, and a quorum controls write acknowledgment. These deployment roles are placeholders, not classes implemented by the order module. First-class syntax still needs capacity, routing, failure and scheduling rules; it does not eliminate their state space.

1.2.4 Requirements at the message boundary

LeanFM starts from decoded production events: correlate messages and classify their order as conforming or a counterexample without pretending to observe private variables. Its grammar is intended to describe legal traces and generate scenarios; state machines project visible histories. The executable grammar limits are documented in Chapter 5. A property dependent on hidden state needs an observable event or justified abstraction; otherwise its status is indeterminate.

This is a workflow choice, not a claim that TLA+ cannot express queues or traces. Our initial TLA+-style attempt required modeling internal state before an ordinary observed scenario could be recorded, so LeanFM makes production-visible messages the starting point.

The trade is deliberate:

Question	TLA+-style state specification	LeanFM event-language specification
Primary behavior	sequence of states	sequence or partial-order graph of events
Queue representation	usually an explicit variable	observable enqueue, send, receive, blocking, and capacity facts
Conformance input	modeled states or action correspondence	decoded production event stream
Hidden variables	available when declared	omitted unless they affect an observable contract
Generation	behaviors satisfying the state relation	words/traces generated by the grammar
Probabilities	require an additional probabilistic interpretation or tool	weights and observed distributions are first-class model data
Main risk	state model may drift from instrumentation	event model may omit a hidden cause needed for a property

This boundary separates the durable specifications: `Requirements.lean` defines observable behavior, while `Implementation.lean` maps it to generation and runtime choices. Existing source does not redefine requirements; Chapter 15 defines the project-root handoff.

1.3 Discovery first, consolidation before commitment

During discovery, the book may contain alternatives that are not accepted. Before either acceptance gate and again before implementation, classify every normative idea as a requirement, implementation decision, deferred item, or material to remove. Preserve technical claims and qualifications; replace prose that mainly enumerates actors, messages, states, ordering, dependencies, or quantities with a generated view from typed artifacts and a short interpretation. Each view retains its prompt, question, event source, reducer, unit, and boundary. An idea's survival in the book does not make it binding; only accepted artifacts do.

1.4 Three ingredients

The method combines three traditions.

1. Language theory gives compact generative descriptions of legal histories. Regular and context-free grammars provide the approachable starting point. Dana Richards and Henry Hamburger, the author's college professors, are cited in Richards and Hamburger [11] as the source from which the author learned these methods, not as their originators. For the underlying habits of symbolization, validity, proof, quantification, and modal reasoning, we also draw on the accessible progression of Gensler [3].
2. Protocol analysis contributes explicit participants, adversarial choices, trace semantics, and tool-supported checking. The CSP treatment of Ryan et al. [12], including Steve Schneider, is cited as the source from which the author learned this treatment.

3. Temporal logic turns expectations into queries over the generated transition system. CTL is central because it distinguishes claims about some possible future from claims about every possible future [1].

1.5 A test for every abstraction

Each abstraction in this book should answer a practical question:

> Can a developer state a meaningful behaviour once, compose it with other behaviours, check its consequences, and keep it stable while implementation details change?

If an abstraction does not improve that workflow, it does not belong in the core library.

Chapter 2

Histories Made of Events

Every execution is represented by an event sequence. A deliberately small common envelope makes histories comparable across protocols:

```
structure Event (Actor Session Task Kind Payload : Type) where
  timeAt  : ClockTimestamp
  id      : String
  priorIds : List String
  causeId : Option String
  session : Session
  task    : Task
  from    : Actor
  to      : Actor
  kind    : Kind
  payload : Payload

abbrev Trace (Actor Session Task Kind Payload : Type) :=
  List (Event Actor Session Task Kind Payload)
```

The fields have different jobs. `id` identifies a visible fact; `priorIds` records zero or more immediate predecessor backpointers; `causeId` optionally identifies the one direct causal event; `from` and `to` name the parties; `(session, task)` correlates independently progressing conversations; and `kind` supplies a stable discriminator. The payload is protocol-specific and carries the parameters that flow from one production to the next. `timeAt` is sampled from the model's shared monotonic clock. Its differences measure elapsed time; it is not a civil-time date and its origin is immaterial.

The predecessor links form a directed acyclic graph rather than a singly linked list. An initial event has no prior identifiers. A fork is represented by several later events that each name the same prior event. A join is represented by one event whose `priorIds` names every branch on which it depends. An end event of any kind may also name its corresponding start event. That backpointer makes elapsed time simply `end.timeAt - start.timeAt`, even when several attempts with the same `(session, task)` overlap. A backpointer to an already-ancestral start is a redundant ordering edge, but it is useful correlation data and does not alter the partial order. The edge says "ordered after," not "caused by." A causal interpretation requires a separate grammar or domain assertion.

Causality uses a different backpointer. An event's optional `causeId` names the event that directly caused it and does not add or remove permitted interleavings. It is not restricted to terminal events. A task-start event may cause intermediate messages; those messages may cause nested task starts; and terminal events may point to the event whose work they finish. Following the reverse `causeId`

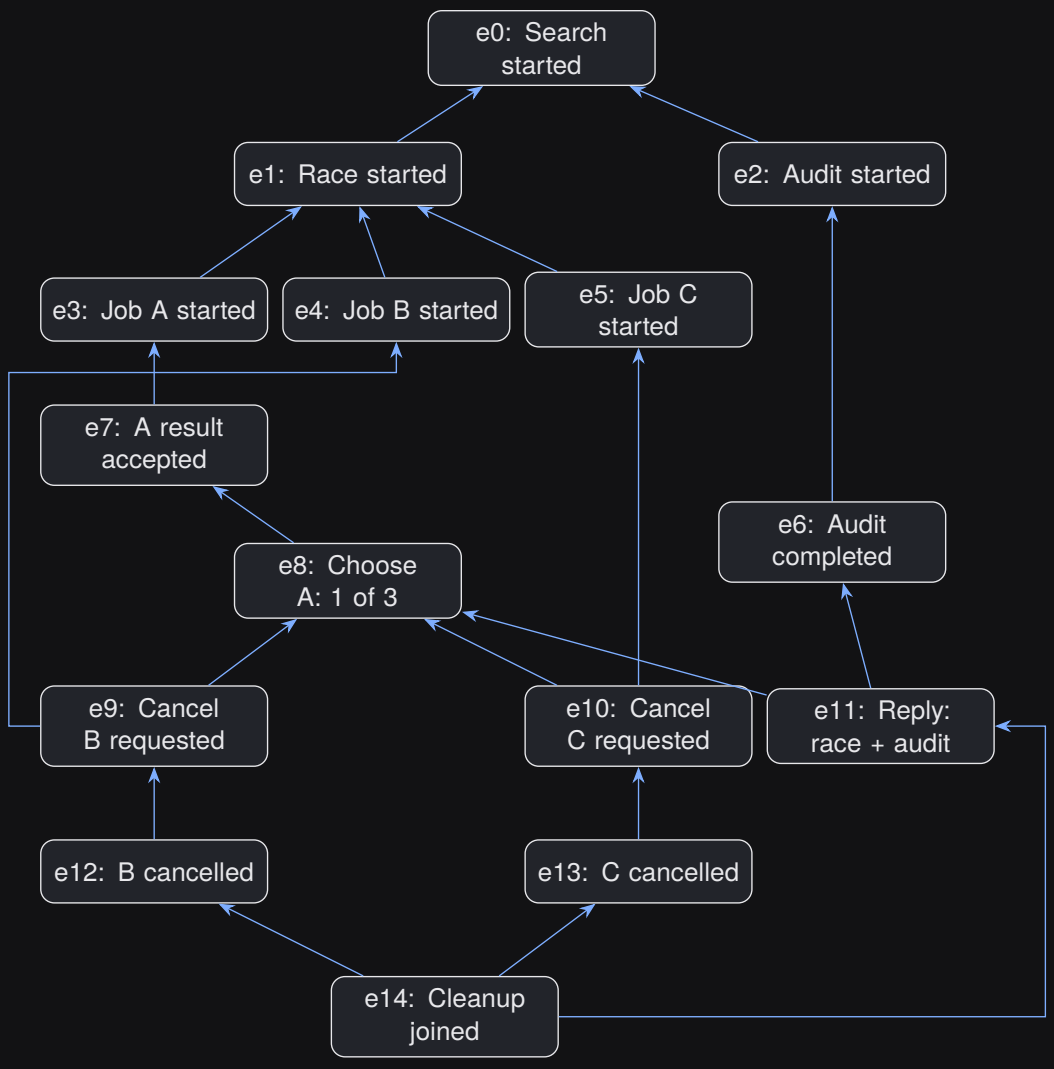
relation produces a causal subtree. A renderer places that subtree inside a labeled box, recursively, so nested work remains visible without pretending that concurrent causal children have a fixed order.

For a request/response production, the response's `causeId` is the request's `id`; read it as "this response answers this request." The rule remains unambiguous when many requests are in flight because correlation follows event identity rather than adjacency or arrival order. `priorIds` remains exclusively the partial-order relation.

For an m-of-n join, the event additionally records m, n, and the exact m branch-terminal IDs that satisfied the threshold. The selected IDs must be distinct members of `priorIds`. Merely seeing two predecessors is not enough to distinguish a two-of-three threshold from an ordinary two-branch all-of join.

2.1 Compute everything from a fork/join event list

2.1.1 Nested joins: choose a winner, then cancel the rest



Arrows show predecessor order with redundant transitive edges removed; full backpointers remain in

the event list. The event list and provenance are generated from `NestedForkJoinExample.lean`; this is not a claim about Google's implementation.

The winner event records threshold 1, population 3 and selected terminal `e7`. Its predecessors are `[e1,e7]`, not the results of all three jobs. The outer reply records `[e0,e8,e6]`: task start, chosen result and completed audit. Reply time is 12 synthetic milliseconds; cleanup ends at 18. These times illustrate ordering, not measured search performance.

For a general n-of-m join, choose n distinct qualifying terminals from m declared branches. Selection is atomic, so racing results cannot select two winners for a 1-of-m task. "First" means the coordinator's observed order, with a declared tie rule, not unsynchronized worker clocks. A cancellation request is not a cancellation acknowledgment; late results must not replace the chosen result. The requirement must say whether a reply waits for losing-task cleanup. Here it does not. Selected-ID and threshold metadata belongs in the join event, not inferred from the picture's number of incoming arrows.

2.1.2 Durations from boundary events

Each event is one observation at one instant. Starting and completing are different messages, so an event does not contain an interval. Durations are derived by selecting the two boundary events and subtracting their timestamps.

```
structure AnalysisEvent where
  id       : EventId
  priorIds : List EventId
  timeAt   : ClockTimestamp
  bytes    : ByteArray
  reveals  : List ValueRef
```

Consider the first retail purchase in `examples/bakery-events.jsonl`. Its bread was baked and shipped before the storefront opened. The purchase itself reserves that inventory, tenders payment, and completes in about a second:

Event	Prior backpointers	`timeAt`	Meaning	Bytes	Reveals
e_0	$[]$	28859991	purchase requested	59	product and quantity
e_1	$[e_0]$	28860343	inventory reserved	62	batch allocation
e_2	$[e_1]$	28861004	payment tendered	55	amount
e_3	$[e_1, e_2]$	28861248	sale completed	72	bread and payment

The list order is one serialization convenient for storage. Event order is the transitive closure of `priorIds`. This sale is sequential; baking and shipping belong to the morning production stream, not to the customer purchase. The following quantities are folds or graph queries over this one list:

$$\begin{aligned}
\mathrm{reservationTime}(e_0, e_1) &= 28860343 - 28859991 = 352 \text{ ms},\\
\mathrm{paymentTime}(e_1, e_2) &= 28861004 - 28860343 = 661 \text{ ms},\\
\mathrm{purchaseLatency}(e_0, e_3) &= 28861248 - 28859991 = 1257 \text{ ms},\\
\mathrm{wireBytes} &= 59 + 62 + 55 + 72 = 248.
\end{aligned}$$

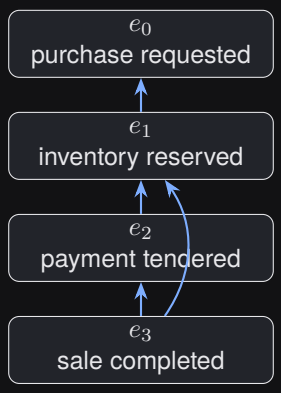


Figure 2.1: The table's immediate predecessor backpointers rendered as an order graph. Each arrow points from an event to an ID in its `priorIds` list: for example, $e_2 \to e_1$. The extra $e_3 \to e_1$ edge preserves direct correlation with the inventory allocation even though e_1 is already an ancestor through e_2.

The subscripted arguments identify each duration's exact start and end events.

Two events are concurrent in this order when neither is an ancestor of the other:

```
def concurrent (events : List AnalysisEvent) (a b : EventId) : Bool :=
  !reachable events a b && !reachable events b a
```

The maximum concurrency permitted by the trace is the width of this partial order: the size of its largest antichain. This purchase is a chain, so its width is one. Other purchases and the morning production and transport streams may still be concurrent. Incomparable events may commute in a linear serialization, provided each event still appears after all of its named predecessors. A timestamped point event has no active interval; runtime concurrency requires separate start and completion messages. This is stronger than counting adjacent list elements and different from measuring average runtime concurrency.

Start and completion boundaries make nested concurrent work especially readable as an order graph. In the following trace, the request forks into two task attempts. Each completion points back to its own start, and the response joins the two completions. The shaded columns are visual grouping only; the arrows are still exactly the stored predecessor pointers.

Timestamps and predecessor links also give elapsed time along a selected path:

```
edgeTime(prior, event) := event.timeAt - prior.timeAt
elapsed(started, completed) := completed.timeAt - started.timeAt
```

The retail critical path is e_0, e_1, e_2, e_3, with elapsed time 1257 ms. The direct $e_3 \to e_1$ correlation edge is redundant for reachability and does not change that duration. Queue occupancy, queued bytes, active `(session,task)` keys, and memory pressure are computed by adding enqueue/dequeue events to the same alphabet and folding them in clock order.

The entire checked-in corpus contains 20,000 orders and 81,039 events. Streaming that file produces the following model inputs; the numbers in this table are generated LaTeX macros, not independently typed constants:

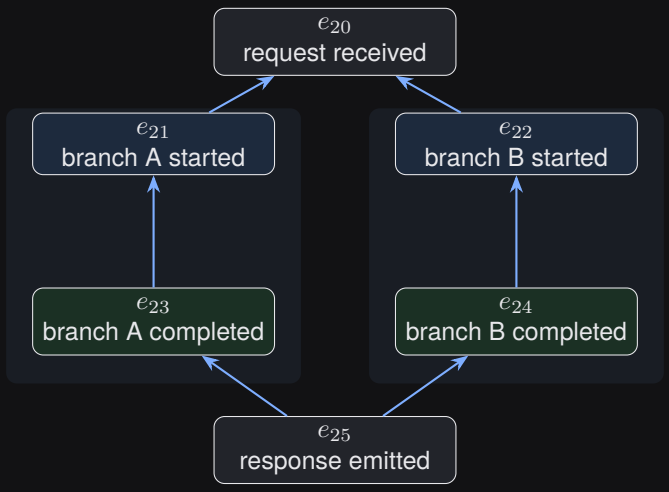


Figure 2.2: A separate synthetic example of task boundaries nested inside a concurrent fork/join. The intervals $e_{21} \ldots e_{23}$ and $e_{22} \ldots e_{24}$ may overlap, while the two branches remain incomparable in predecessor order. Either branch may appear first in a linear serialization without changing the graph.

Reduction over `bakery-events.jsonl`	Observed value
sales completed / purchase requests	20,000/20,000
stockouts / purchase requests	0/20,000
morning batches baked	140
FIFO shipments dequeued / enqueued	140/140
maximum simultaneously active purchases	1

These ratios are suitable probability estimates for this corpus. Regeneration with a different source stream changes the macros and therefore changes every book example that consumes them.

Knowledge is another event reduction. For a value v, calculate the actors that can derive it from initial knowledge and the bytes visible to them:

```
def knowers (actors : List Actor) (v : ValueRef)
    (events : List AnalysisEvent) : List Actor :=
  actors.filter fun actor =>
    Derives (knowledgeFrom actor events) v
```

Encryption, signing, projection, key possession, and compromise contribute rules to `Derives`. The primary result is the set `knowers v events`. "Only these actors know v" is a comparison between that computed set and an allowed set, not a separate primitive called secrecy.

2.2 Derived state

A current session, an authenticated principal, or an outstanding request can often be computed by folding the trace. Such a fact need not also exist as primitive model state. This discipline has two benefits: evidence for a fact remains visible, and the state representation does not multiply independently of the history that justifies it.

Some implementations necessarily maintain private state. The model includes it only when two executions with the same visible trace may make different future behaviour legal. Even then, the first remedy is to expose the relevant distinction as an event or a grammar parameter rather than copying an implementation object graph into the specification.

2.3 Time is more than a timestamp

Timing constraints pair explicit boundaries; changing a timeout bound changes the temporal requirement, not the event-order or causal relations defined above.

For two events from the same `(session, task)`, elapsed time is defined when the first event's `timeAt` value does not exceed the second's:

```
def elapsed (start finish : Event) : Option Duration :=
  if start.session = finish.session &&
     start.task = finish.task &&
     start.timeAt <= finish.timeAt
  then some (finish.timeAt - start.timeAt)
  else none
```

Latency queries name their boundary event kinds and use the terminal's `priorIds` backpointer to select the start. Matching `(session, task)` values alone cannot disambiguate overlapping attempts, so reducers never pair by adjacency. Completion, failure, cancellation, timeout, and other declared terminals use the same explicit pairing for elapsed time, byte deltas, and other measurements.

This pairing is how a view may elide detail without becoming hand-waving. A collapsed production retains its boundary event IDs, complete predecessor links, parameter assignment, aggregate deltas, and validation status. Opening it reveals the intervening events. Closing it again changes only presentation, not the history or the evidence supporting the displayed duration, bytes, or outcome.

2.4 Estimating completion from an event stream

The same boundary-event model can consume observed events. The `LeanFM.StreamStats` implementation reconstructs attempts from stored terminal-to-start predecessor links and computes completion, conditional-success, and completed-latency statistics.

For an observation window containing M starts, C matched terminal events, S successful terminals, and completed latencies $L_1, \ldots, L_C$, the basic estimates are

$$\widehat{P}(\text{complete}) = \frac{C}{M}, \qquad \widehat{P}(\text{success} \mid \text{complete}) = \frac{S}{C}, \qquad \widehat{E}[L \mid \text{complete}] = \frac{\sum_i L_i}{C}.$$

The vertical bar means "given"; hats mark estimates from the observation window. Attempts without a terminal event remain *in flight*. They contribute to the number started and to queue/load measurements, but not to completed-task latency or to the conditional-success denominator. Calling them failures would bias both probability and latency whenever the observation window ends before they do.

The reducer can group these sufficient statistics by task, actor, time bucket, maximum concurrency, or queue occupancy and feed them to charts. Specified probabilities and dwell times are predictions; stream-derived estimates are measurements to compare with them.

2.5 Charts as declarative trace reductions

The proposed chart declaration names an observable event source, a grouping or x expression, and an aggregate or y expression. It describes the reduction, not rendering code; this syntax is illustrative, not an implemented renderer.

```
def outcomeShare : ChartSpec :=
  pie "quote outcomes" do
    source events
    where_ (kind == "quote accepted" || kind == "quote declined")
    groupBy kind
    value count

def responseLatency : ChartSpec :=
  line "quote response latency" do
    source (correlate events by "quoteId")
    x clock
    y (elapsed "quote offered" "customer decision")
    series "p50" median

def latencyByConcurrency : ChartSpec :=
  xyplot "latency by concurrent sessions" do
    source completedRuns
    x maxConcurrentSessions
    y (mean (elapsed firstEvent terminalEvent))

def queueByOfferedLoad : ChartSpec :=
  xyplot "queue occupancy by offered load" do
    source completedRuns
    x requestsPerSecond
    y (mean totalQueueOccupancy)

def propagatedGrantProbability : ChartSpec :=
  xyplot "user probability propagated through deterministic servers" do
    source parameterSweep
    x userGrantProbability
    y terminalGrantProbability
```

An XY plot preserves two numeric axes. It differs from a time series, whose x-axis is specifically an event clock, and from a categorical chart, whose x values are labels. The first two examples reduce each completed model run to one point. The third sweeps a model parameter and checks that deterministic downstream processing transports the user's probability mass.

Apply the provenance and implementation-status rule in Section 3 to each generated chart.

2.6 Round trip through bytes

A generated scenario is not complete until it can cross the byte boundary in both directions. `ScenarioEvent` in `Requirements.proto` carries the event identity, complete `prior` list, correlation key, actors, `timeAt`, and the selected protobuf atom. Encoding a `Scenario` therefore produces bytes for both its ordering envelope and payload. Decoding those bytes must reconstruct the same ordered event list.

The executable round-trip check compares the decoded protobuf value with the generated value and separately checks the reconstructed `prior` graph; loss of a fork or join fails the build.

Chapter 3

Scenarios

A scenario is LeanFM's named, bounded stream of timestamped events for one interaction or execution trace (for example, a quote request, OAuth exchange, or 9×9 game).

3.1 Event envelope and scenario grouping

The shared envelope and semantics of `id`, `priorIds`, `causeId`, actor endpoints, correlation and monotonic `timeAt` are defined in Chapter 2. A serialized `scenario` key or aliases such as `src`/`message` are canonical only if the accepted schema and reducers declare them.

3.2 Derived outputs and provenance

Question-driven outputs include interaction diagrams, residual state machines, latency/throughput charts and outcome distributions. Each `DesiredOutput` must bind its prompt, precise question, event source, reducer, unit and scenario boundary. A listed output is not implemented until its renderer exists; when a generated view carries the explanation, prefer it to repeated prose and add a short interpretation.

3.3 Example: minimal two-player game scenario

Below is an illustrative event fragment for a 9×9 game. Times are milliseconds on the model's monotonic clock:

```
{ "id":"evt-0010", "priorIds":[], "causeId":null,
  "timeAt":30000, "session":"s-g1", "task":"game-g1",
  "from":"alice","to":"server","kind":"MoveRequested",
  "payload":{"move":"E5","color":"B"} }
{ "id":"evt-0011", "priorIds":["evt-0010"], "causeId":"evt-0010",
  "timeAt":30001, "session":"s-g1", "task":"game-g1",
  "from":"server","to":"alice","kind":"MoveAccepted",
  "payload":{"move":"E5"} }
```

The latency pair is `evt-0010` to `evt-0011`: one millisecond.

Prefer runnable scenario traces to prose-only examples, declare their expected derived views and provenance, and keep transport mappings in `Implementation.lean` after requirements acceptance.

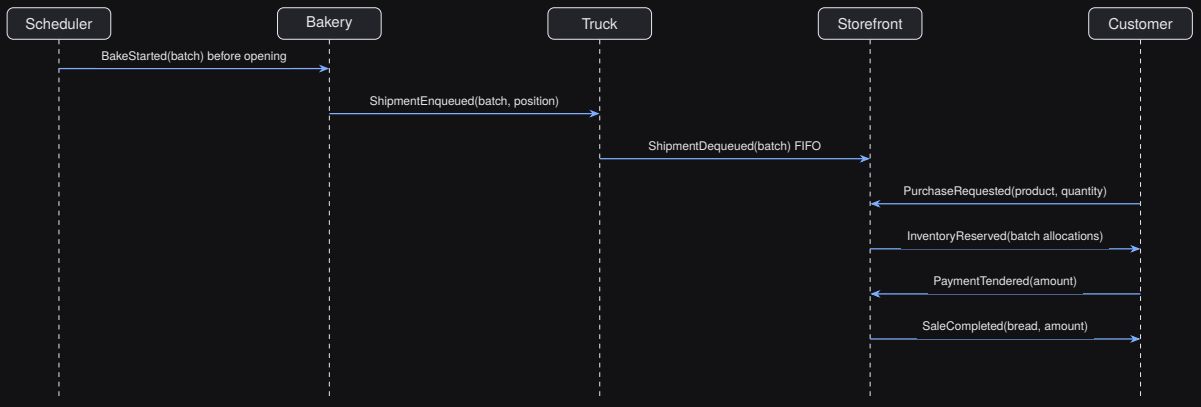
Chapter 4

The Bakery Event Corpus

The bakery is the quantitative worked example. Its source is not a table of percentages copied into this chapter. It is the 81,039-line `examples/bakery-events.jsonl` stream. The reducer reads those events and generates both `examples/bakery-stats.json` and the coordinates plotted below. The checked-in stream is a reproducible synthetic population generated by `scripts/generate_bakery_events.py` with its default seed and 35-day horizon; its statistics describe that corpus, not a real bakery.

4.1 The grammar represented by events

Bread is not baked in response to an individual order. A predictable schedule creates batches by bread kind; each batch is baked, shipped before opening, and stocked at the storefront. Customer orders reserve fresh inventory from those lots. After the declared two- or three-day freshness window, remaining units move to a charity rack and produce a donation receipt. An `InventoryReserved` event names both the paid order and the stocked batch lots that supplied it, making FIFO allocation, age, quality, and stockouts replayable. No event permits a customer to wait for a same-day custom loaf.



The truck is a bounded FIFO queue. `ShipmentEnqueued` records its queue position and capacity; `ShipmentDequeued` backpoints to both the matching enqueue and the preceding dequeue. The retail interaction begins only after `ShipmentDequeued` has added the batch to storefront inventory.

For each calendar-day bucket d, the reducer computes

$$
\begin{aligned}
\text{grossSales}_d &= \sum_{e.kind=\text{SaleCompleted}} e.amount, \\
\text{refunds}_d &= \sum_{e} e.refund, \\
\text{netRevenue}_d &= \text{grossSales}_d - \text{refunds}_d, \\
\text{ingredientCost}_d &= \sum_{e.kind=\text{BatchBaked}} e.ingredientCost, \\
\text{donationValue}_d &= \sum_{e.kind=\text{DonationReceiptRecorded}} e.eligibleValue, \\
\text{employeePay}_d &= \sum_{e.kind=\text{ShiftPaid}} e.pay, \\
\text{profit}_d &= \text{netRevenue}_d - \text{ingredientCost}_d - \text{employeePay}_d .
\end{aligned}
$$

All sums use the same calendar-day bucket d. The equation's event kinds and fields define each amount; donation value is reported separately and is not added to revenue.

Donated bread is already part of batch ingredient cost, so subtracting it again would double count. Donation value is not itself a tax saving; calculate any tax effect only after supplying applicable rules and rates. This is modeled contribution after ingredients and pay, not business profit: costs without events in this corpus are excluded.

4.2 Money by day is the primary XY plot

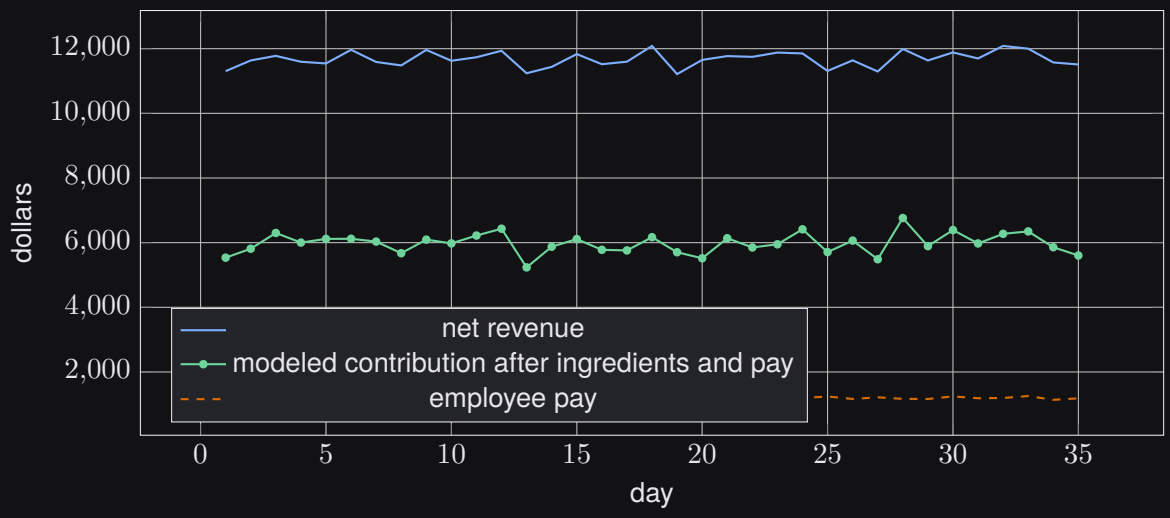


Figure 4.1: Daily bakery economics reduced from morning production, retail sales, payroll and charity-transfer events.

The employee-pay line is not a constant assumption. Each `ShiftPaid` event carries employee, role, minutes, hourly rate, and calculated pay. This lets the same chart answer counterfactual questions after a staffing event stream is changed.

4.3 Waste explains part of the margin

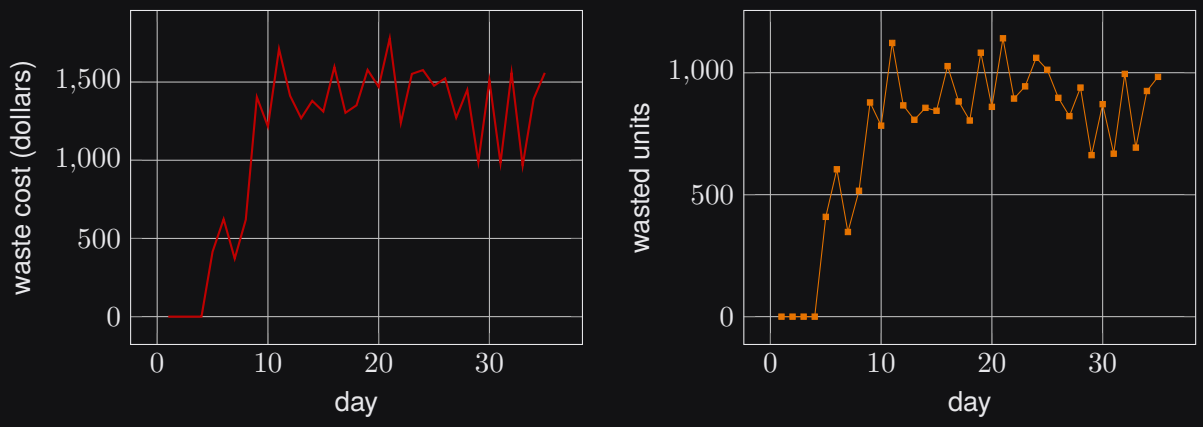


Figure 4.2: Donation-eligible cost and units by day, derived from batch inventory remaining when its freshness window closes.

Cost and units are deliberately separate. A donated sourdough loaf and a donated croissant do not have the same economic effect. Grouping the same events by product, batch age, employee shift, or day produces further XY datasets without changing the scenario.

4.4 XY plots as questions over scenarios

The XY chart is not restricted to time on the horizontal axis. The reducer also generates daily order count against modeled daily contribution:



Figure 4.3: Daily order volume against modeled contribution. Each point is one day reduced from the event corpus.

The same mechanism can measure many scenarios: offered load against latency, staffing hours against modeled contribution, price against cancellation probability, queue capacity against completed orders, or waste against bake batch size. In each case the axes are named reductions over event

streams. A parameter sweep produces several streams; observed operation supplies streams from different days. The chart renderer need not know which kind it received.

The old made-to-order order-lifetime statistic is intentionally absent. Useful service metrics are payment and inventory-reservation time, storefront fulfillment time, stockout rate, inventory age at sale, freshness-window sell-through, and units and eligible value transferred to charity.

4.5 A code-generation boundary

The intended code-generation boundary uses durable files, not the conversation that produced them. Let R be `Requirements.lean`, I be `Implementation.lean`, P the protobuf schema, and C_k an optional existing implementation:

$$(R, I, P) \longrightarrow C_0, \qquad (R, I, P, C_k) \longrightarrow C_{k+1}.$$

Initial generation uses (R, I, P); revision also receives C_k so conforming code can be retained. Missing decisions must be reported rather than recovered from chat. This is the intended boundary, not a claim that the bakery sketch below was generated or that the complete code-generation pipeline is implemented.

The following short Go-like excerpt illustrates the purchase event sequence and its predecessor links. It is schematic: protobuf bindings, event plumbing, and runnable implementations of the helper functions are omitted.

```
// Purchase exchanges pre-made storefront inventory for money. Its messages are
// PurchaseRequested, InventoryReserved, PaymentTendered, and SaleCompleted.
// There is no per-order bake and no courier delivery.
func (s *Storefront) Purchase(ctx context.Context, product string, units uint32,
        priceCents uint64) error {
        requestID := nextID("purchase-requested")
        if err := emit("PurchaseRequested", requestID); err != nil { return err }
        allocation, ok := reserveFIFO(s.Inventory[product], units)
        if !ok { return emitWithPrior("StockoutReported", nextID("stockout"), requestID) }
        reservedID := nextID("inventory-reserved")
        if err := emitWithPrior("InventoryReserved", reservedID, requestID); err != nil { return err }
        paymentID := nextID("payment-tendered")
        if err := tender(ctx, priceCents); err != nil { return err }
        if err := emitWithPrior("PaymentTendered", paymentID, reservedID); err != nil { return err }
        commitAllocation(s.Inventory, product, allocation)
        return emitWithPrior("SaleCompleted", nextID("sale-completed"), reservedID, paymentID)
```

The full 74-line sketch remains at `book/examples/bakery-generated.go`; it is not an executable fixture. The general conformance target is that generated units carry machine-readable requirement references and justifications, with a manifest linking IDs to files, symbols, and tests. Current validation checks requirement references in implementation mappings; symbol-level validation of generated code remains a separate requirement. The grammar should supply legal-trace tests, reject mutations outside its language, and pair byte round trips with the mapped implementation.

Chapter 5

A Multiparty Grammar of Behaviour

A protocol is a language whose terminal symbols are visible events. A production describes a legal continuation. **Proposed notation:** the `Grammar` algebra and BNF below are not an executable API. Durable artifacts use `GrammarExpr` (`event`, `seq`, `choice`, `parallel`, `thresholdJoin`, `guard`, `ref`, `repeat`). Current validation checks grammar/task and message references, atom endpoints, and threshold bounds; it does not compile this notation into trace generation or recognition, or prove actor projection. The queue rules later in this chapter are model semantics, not a proof of equivalence to a Go runtime.

5.1 A small authoring algebra

```
inductive Grammar (Event : Type) where
  | empty
  | event  : Event -> Grammar Event
  | seq    : Grammar Event -> Grammar Event -> Grammar Event
  | choice : List (Grammar Event) -> Grammar Event
  | par    : Grammar Event -> Grammar Event -> Grammar Event
  | where_ : (List Event -> Bool) -> Grammar Event -> Grammar Event
```

`seq` preserves time order. `choice` exposes alternatives rather than burying them in control flow. `par` admits interleavings while preserving the internal order of each side. A `where_` clause checks a fact accumulated from the prior trace and provides the escape from context-free structure when genuine context sensitivity is needed.

A choice does not need a separate annotation naming who makes it. Each branch eventually resolves to terminal events, and each terminal resolves to framed bytes with a source and destination. The first observed terminal that distinguishes two branches identifies both the selected branch and the actor that emitted the distinguishing bytes. If two branches resolve to the same observable bytes, then at this abstraction they are not distinct choices and should be merged. This removes an otherwise ambiguous second account of agency from the grammar.

Simple finite conversations can compile to regular languages. Nested request structure motivates context-free productions. Correlation, freshness, uniqueness, and authorization typically require parameter flow or guarded productions.

5.2 A BNF for composable interactions

The authoring notation can be presented independently of Lean as a small BNF. Named productions are interactions; referring to a production composes that interaction into a larger one.

```
<specification> ::= <production>
                  | <production> <specification>

<production>    ::= <name> "::=" <expression>

<expression>    ::= "empty"
                  | <event>
                  | <name>
                  | "(" <expression> ";" <expression> ")"
                  | "(" <expression> "|" <expression> ")"
                  | "(" <expression> "||" <expression> ")"
                  | "repeat" "(" <expression> ")"
                  | "join" "(" <natural> "of" <expressions> ")"
                  | "when" "(" <predicate> "," <expression> ")"
                  | "after" "(" <name> "," <expression> ")"

<event>         ::= <actor> "->" <actor> ":"
                    <message> "(" <fields> ")"

<fields>        ::= ""
                  | <field>
                  | <field> "," <fields>

<field>         ::= <name> "=" <value>
```

The operators have the same meanings as the grammar AST: semicolon is sequence, vertical bar is alternative, and double vertical bar is parallel composition by order-preserving interleaving. `after(P,E)` includes `E` only on a branch in which production `P` completed. Predicates and repeated field variables supply the context-sensitive correlation that plain context-free syntax does not express by itself.

A threshold join is explicit rather than inferred from the number of observed predecessors:

```
firstTwoReplicas ::= join(2 of replicaA, replicaB, replicaC)
firstTwoThenReply ::= (firstTwoReplicas ; reply)
```

The continuation becomes enabled after any two of the three branches complete. Its first emitted event records `joinRequired=2`, `joinTotal=3`, and exactly two selected branch-terminal IDs. Those IDs also occur in the event's ordinary `prior` list, which may contain other backpointers such as the task start. The threshold declaration says why two predecessors suffice; the selected IDs say which two satisfied this instance. Unselected branches are not implicitly canceled and may still emit events. Cancellation, if required, must be an explicit message or production.

The OAuth example decomposes into independently readable interactions. These generated views show the two complete branches without making a message arrow look like logical implication:

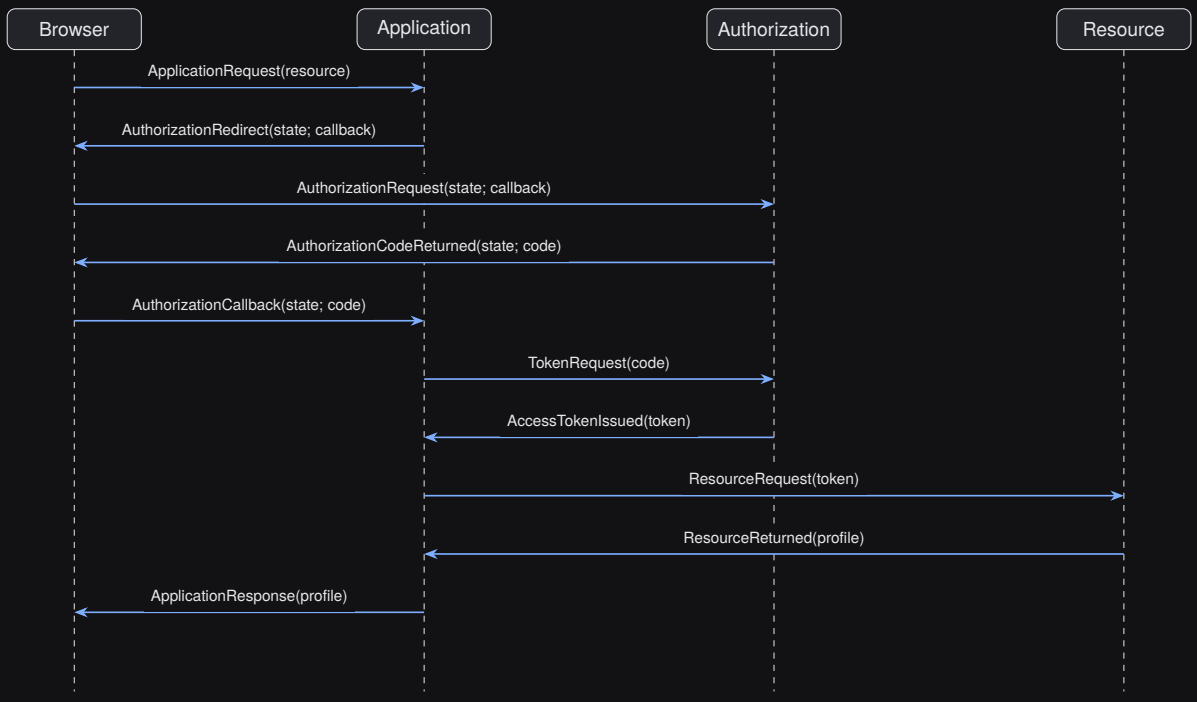


Figure 5.1: Granted OAuth interaction. The resource exchange occurs only after the authorization-code branch succeeds.

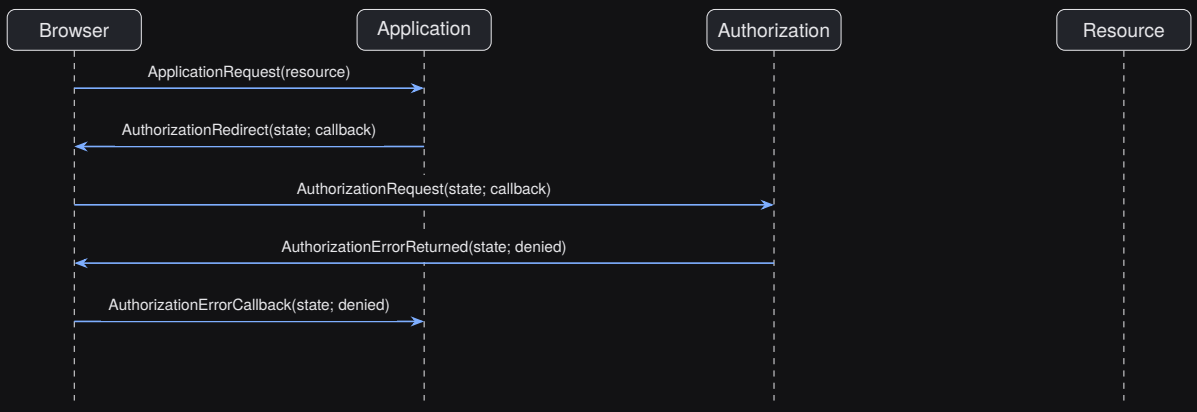


Figure 5.2: Denied OAuth interaction. No token or resource request follows the error.

Chapter 8 gives the complete production definitions, nested alternatives and guarded exchange; these terminal-level diagrams isolate the granted and denied outcomes for comparison.

5.3 Parameters carry information

A production should bind values at one event and reuse them later. Conceptually:

```
def login (browser server provider : Actor) (state : OAuthState) : Grammar Event :=
  event (redirect server browser provider state) >>>
  event (authorize browser provider state) >>>
  callback browser server state
```

The repeated `state` is not decorative. It states the information-flow relationship that distinguishes a matching callback from an unrelated one. Lean's types, constructors, and ordinary function parameters are the surface language; the fixed library supplies grammar composition and semantics.

5.4 Composition without omniscience

A participant's implementation need not know the global execution state. The global grammar states the contract, while projection derives each participant's view. Composition should preserve independently specified modules and make shared events explicit. Proving useful projection and composition theorems is a core research obligation, not an implementation footnote.

5.5 Actors multiplex bounded conversations

An actor has separate positive finite-capacity inbound and outbound message queues. Queue capacity is part of the observable execution model: producing a message may block when the outbound queue is full, and transport may block when the destination's inbound queue is full. Sending is therefore distinct from delivery and from the destination actor consuming the message.

Messages carry a session identifier as well as a task identifier. Their pair is the correlation key for actor-local work:

```
structure WorkKey where
  session : SessionId
  task    : TaskId

structure ActorRuntime where
  inboundCapacity  : Nat
  outboundCapacity : Nat
  inbound          : List Envelope
  outbound         : List Envelope
  inFlight         : List (Prod WorkKey GrammarResidual)
  inboundBounded   : inbound.length <= inboundCapacity
  outboundBounded  : outbound.length <= outboundCapacity
```

An actor may have zero or many distinct `WorkKey`s in flight. Consuming an inbound envelope selects or creates the residual associated with that envelope's `(session, task)` pair; it does not

replace one actor-wide "current task." Messages from different keys may interleave while the grammar preserves the order required within each conversation.

The queues have Go-channel-like operational semantics. Capacity is enforced when an operation is attempted, not checked only after constructing a new state:

```
send(channel, message) =
  if channel.length < channel.capacity
  then sent(channel ++ [message])
  else blocked

receive(channel) =
  if channel = message :: rest
  then received(message, rest)
  else blocked

tryReceive(channel) =
  if channel = message :: rest
  then some(message, rest)
  else none
```

A blocked operation makes no progress and leaves the channel unchanged. A blocking actor may run again when the required space or data becomes available. `tryReceive` is a different operation: `none` reports that the channel is empty now and lets the actor continue. It does not put the actor to sleep, and it must not be confused with receiving a valid optional payload.

The resulting transitions are explicit:

- an actor step appends a produced envelope to its outbound queue;
- a transport step moves the outbound head to the destination's inbound queue when both operations are within capacity;
- a receive step removes an inbound envelope and advances the matching `(session, task)` residual;
- sending to a full outbound queue blocks the sender without enqueueing;
- transport to a full inbound queue blocks without removing the outbound head;
- blocking receive on an empty inbound queue blocks, while nonblocking `tryReceive` returns `none` and permits later work.

Queue order and predecessor order are not identical. FIFO determines which queued envelope is available next, while `priorIds` records the event-order graph, including forks and joins across actors. Incomparable branches may commute; the pointer relation does not itself claim that one event caused another.

Chapter 6

The Scenario Authoring Language

Proposed notation: this chapter develops a concrete surface language using ordinary Lean `def`, function parameters, `let`, typed values, and `do` notation, with library operations for scenarios, endpoints, choices, probability, and rendering. Names may change as semantics are implemented. The worked MDP later in this chapter is a separate **finite-model check**; its figures are **generated analyses** from the named Lean source.

6.1 Proposed scenario notation

A scenario is a named grammar fragment whose terminals are observable message events. This proposed syntax uses `from_` and `to_` (avoiding Lean's `from` syntax) to set the endpoints copied into each event; an `emit` without both endpoints is invalid. Ordinary `let x := e` evaluates a Lean expression, while `let x <- declare T "x"` introduces an immutable typed symbolic value bound by trace generation or matching. Reusing a binding expresses identity, while receiving an event binds its fields for later messages. `emit` and `receive` denote the same grammar-terminal kind; they differ only in participant projection.

```
def requestQuote (customer shop : Actor) : Scenario :=
  scenario "request a quote" do
    let requestId <- declare RequestId "requestId"
    let quoteId <- declare QuoteId "quoteId"
    let amount <- declare Money "amount"
    from_ customer; to_ shop
    emit "quote requested" [field "requestId" requestId]
    from_ shop; to_ customer
    emit "quote offered"
      [field "requestId" requestId, field "quoteId" quoteId,
       field "amount" amount]
```

Early requirements can use stable semantic message names and typed fields without committing to a protobuf schema; these suffice for ordering, correlation and temporal properties. A later refinement maps those fields to a concrete alphabet; its obligation is language inclusion after representation details are erased. Serialization, retries and internal state may be chosen there, but externally visible behavior forbidden by the abstract grammar may not be added.

Named inclusions represent reusable productions rather than textual copying. They can export values for later events; a compiler must resolve references, check exports, reject cycles and build

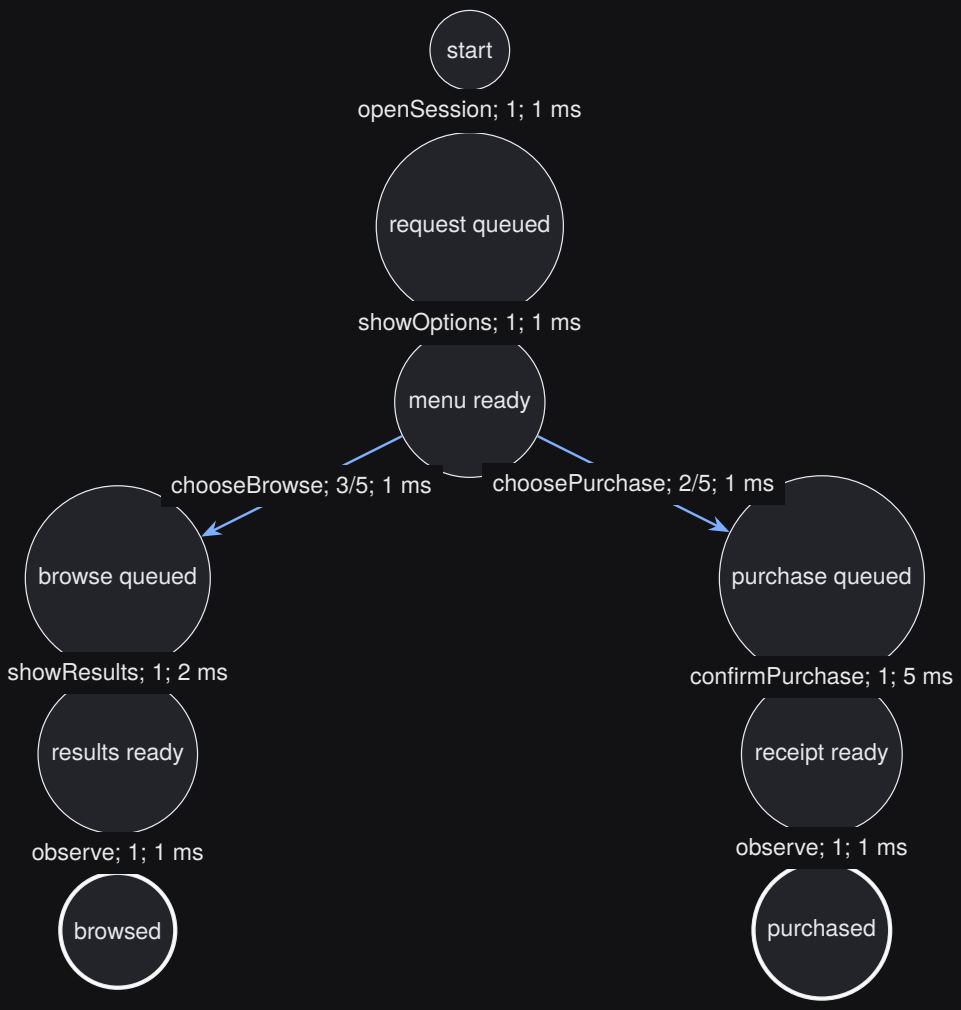
the grammar. For example, a quote outcome can choose between named accepted and declined branches. Keep those branches independently renderable rather than combining unrelated concerns in one story. No quote diagram is generated yet: the event-derived diagram contract is in Chapter 3, with generated interaction examples in Chapter 5.

`oneOf` states nondeterminism without claiming branch frequencies; `weightedOneOf` supplies a distribution when one is supported. Chapter 5 assigns agency to the first distinguishing terminal, and deterministic service transitions carry incoming probability mass forward. Requirements may begin with unresolved choices, symbolic weights or ranges. Refinement must resolve these claims and preserve every abstractly legal branch, including one assigned zero probability by the current implementation.

6.2 Worked MDP: a stochastic user and deterministic server



Weights and probability-one server edges come from the checked `Choice` values.



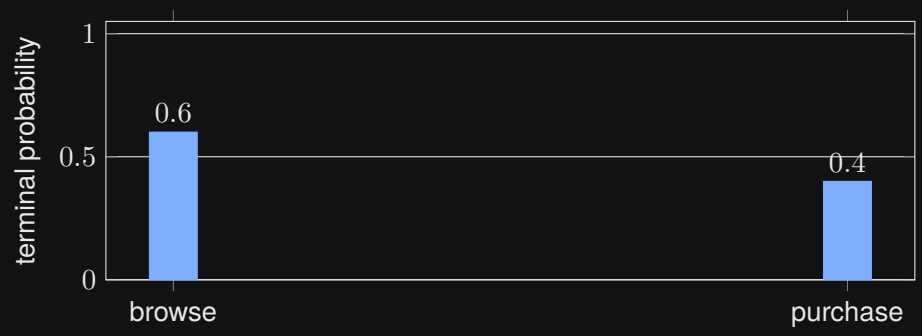
The branch weights are relative integers. Their sum supplies the denominator:

$$\frac{3}{5} + \frac{2}{5} = 1.$$

The checker rejects an empty chance set and every zero-weight branch. Normalization divides each positive branch weight by the total weight; it does not require floating-point arithmetic.

Assign reward 1 to browsing and reward 8 to purchasing. The terminal paths and their durations give

$$\mathbb{E}[R] = \frac{3}{5}(1) + \frac{2}{5}(8) = \frac{19}{5}, \qquad \mathbb{E}[T] = \frac{3}{5}(6) + \frac{2}{5}(9) = \frac{36}{5}\ \mathrm{ms} = 7.2\ \mathrm{ms}.$$



`StochasticUserMDPExample.lean` checks the distribution, deterministic server states, terminal paths, mass conservation and both expectations before generating these views. The present evaluator expands a finite number of steps under a selected policy. It does not yet compute an optimal policy, solve an unbounded fixed point, or take minimum and maximum values over unresolved schedulers. Those are distinct extensions; this example establishes that exact probability propagation already works for a bounded policy. Under this fixed policy, the example is a finite Markov chain: its only branching is the user's specified chance distribution, not a policy choice or unresolved scheduler.

6.3 Actor specifications and instances

An actor name in a grammar denotes a reusable specification, not necessarily one process. A finite population gives that specification concrete identities:

```
actor Client
  |>.instances "client-" 20 (sticky "session")

actor Server
  |>.instances "server-" 2 (shardHash "tenant_id")
```

Generated scenarios resolve grammar roles to concrete instances such as `client-7` and `server-2`. Each instance inherits its actor's behavior and resource bounds and follows the queue and in-flight `(session,task)` semantics of Chapter 5. A population is homogeneous: every member implements the same actor specification, so a cluster cannot mix gateways, databases, and workers merely because they share a deployment boundary.

Each cluster declares message routing: `sticky` keeps the value of an observable affinity field on the same member, `roundRobin` cycles through members in declared order, and `shardHash` hashes an observable shard key. Routing changes legal per-instance histories, queue pressure, hot spots, failure impact, and replay. Kafka and DynamoDB partition keys and MongoDB shard keys motivate shard hashing, but each product has additional replication and routing semantics. Explicit finite populations keep queue, memory, probability, and temporal checks bounded.

6.4 Non-functional requirements: rates and scalability

Using the boundary-event model from Chapter 2, suppose every client repeatedly creates a fresh server task. Each completed unit records a start, a terminal that backpoints to it, and a nonnegative amount of work. For illustration, let one moved byte be one work unit and timestamps be milliseconds; an exhaustive observation list supports two different rates.

The word *load* is too ambiguous to serve as a variable. Let N be the declared upper limit on the number of active clients in the study, and let the parameter n be the active-client population for one observation, with $1 \le n \le N$. Separately, let λ_o be offered work per unit time, λ_a be work admitted per unit time, and X be work completed per unit time. Offering work, accepting it into a queue, and completing it are three different observable boundaries. The client-experienced, properly weighted rate is

$$X_{\text{client}}(n) = \frac{\sum_i \text{work}_i}{\sum_i (e_i.timeAt - s_i.timeAt)}.$$

Long requests contribute in proportion to their duration; this is not an unweighted mean of per-request ratios.

The aggregate server throughput is instead

$$X_{\text{server}}(n) = \frac{\sum_i \text{work}_i}{\max_i(e_i.timeAt) - \min_i(s_i.timeAt)}.$$

Its denominator is the wall-clock interval from the earliest start to latest end, so overlapping client waits are counted once.

For an observation window of duration T, the boundary rates and backlog obey

$$\lambda_o = \frac{W_o}{T}, \qquad \lambda_a = \frac{W_a}{T}, \qquad X = \frac{W_c}{T}, \qquad B_1 = B_0 + W_a - W_c.$$

The backlog identity assumes a conserved work unit; record cancellations, expiry, and failures as terminal dispositions when they remove work. If $\lambda_o > X$, some offered work is rejected, remains queued or in flight, or ends in another recorded disposition; admitted work can also outpace completion while latency and backlog grow. Window throughput differs from the client- and server-task rates above. Fit the Universal Scalability Law against active-client population, not offered work rate λ_o unless the experiment defines an explicit mapping:

$$X(n) = \frac{\gamma n}{1 + \alpha(n-1) + \beta n(n-1)}, \qquad 1 \le n \le N.$$

Gunther's *Guerrilla Capacity Planning* develops the USL as a practical capacity-planning tool for exposing contention, coherency, saturation, and retrograde scaling before deployment supplies those facts [4].

Record per observation its n, selected throughput and event boundary, λ_o, λ_a, window throughput X, latency, backlog, terminal outcomes, server count, and window boundaries. Fit parameters separately and only on exhaustive windows, avoiding bias from omitted or unfinished work.

6.4.1 The three USL regimes and capacity

The scale parameter γ is analogous to clock speed: it is the single-client service-rate scale and multiplies the entire curve vertically. It is not literally a processor clock, because an observed unit of work may include networks, storage, and several actors. Changing γ does not move the population at which the dimensionless USL curve peaks.

The two overhead parameters have distinct effects:

$$\begin{aligned}
\alpha = 0,\ \beta = 0: &\quad X(n) = \gamma n, && \text{ideal linear scaling,}\\
\beta = 0: &\quad X(n) = \frac{\gamma n}{1 + \alpha(n-1)}, && \text{contention-limited saturation,}\\
\beta > 0: &\quad X(n) = \frac{\gamma n}{1 + \alpha(n-1) + \beta n(n-1)}, && \text{coherency-limited retrograde scaling.}
\end{aligned}$$

For $0 \le \alpha < 1$ and $\beta > 0$, differentiation gives the finite capacity population

$$n_* = \sqrt{\frac{1-\alpha}{\beta}}.$$

This fitted peak need not equal the study limit N.

The observed integer capacity is one of the adjacent integer populations, whichever has the larger measured or fitted throughput. When $\beta = 0$ and $\alpha < 1$, the curve increases toward the asymptote γ/α but has no finite USL peak. When both parameters are zero, the mathematical model has no internal capacity limit; a real finite system must eventually reveal an omitted bottleneck or cease to fit that ideal case.

Figure 6.1 uses $\alpha = 0.1$ and, in its third panel, $\beta = 0.005$, marking the fitted peak $n_* \simeq 13.42$.

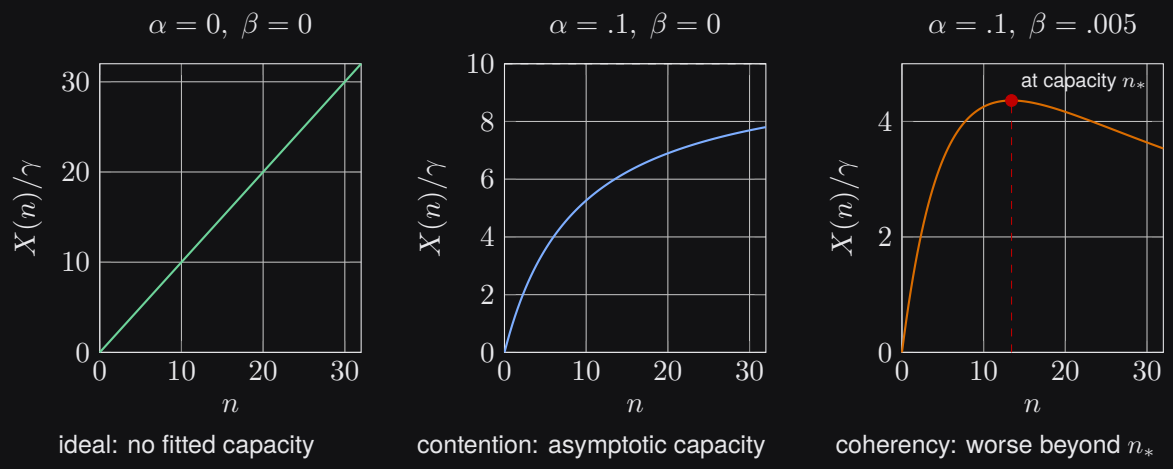


Figure 6.1: Universal Scalability Law regimes. Contention alone saturates; coherency creates a finite peak and declining throughput beyond capacity.

Population capacity is not the only overload boundary. For a fixed deployment with sustainable service rate μ, offered load λ is at capacity when $\lambda = \mu$. Below capacity, a simple queueing illustration has response time proportional to $1/(\mu - \lambda)$ and queued work proportional to $\lambda/(\mu - \lambda)$. At and above capacity these steady-state quantities are unbounded: completed throughput is capped by μ, while unfinished work, latency, queued bytes, and memory pressure accumulate. A finite queue converts that growth into blocking or rejection; exhausting the memory budget is a fatal terminal event rather than another slow sample.

For the queueing foundation behind this illustration—including workload characterization, service demand, utilization, response time, queueing networks, and cost/performance tradeoffs—see Menascé, Almeida, and Dowdy's *Performance by Design* [9]. USL and a single queue answer

different questions and belong together: USL summarizes scaling across populations, while queueing models explain how offered work waits for finite service centers. Neither rescues a requirements model whose demand, service rates, queue bounds, observation window, and overload behavior were never stated.

These examples use the positive finite per-instance inbound/outbound capacities and full/empty blocking semantics defined in Chapter 5.

A queue also has an idempotent `close()` operation. Closing forbids every later push. A sender already sleeping for capacity wakes with a closed rejection and does not mutate the queue or message. Values accepted before close remain available and are received exactly once in order. Receivers report closed only after those buffered values have drained; closing an already empty queue wakes blocked receivers immediately. Consequently, queue iteration has a precise termination condition: the queue is closed and empty. Empty while still open means “wait,” not “end of stream.”

Current costs and resource profile (not optimized). For C queued items, the list-backed channel sends in $O(C)$, receives in $O(1)$, and stores $O(C)$; the close semantics above are not yet implemented. Two-list or ring-buffer channels can make send and receive amortized $O(1)$ at the same capacity.

With H task events and R predecessor references, the current attempt reducer appends starts and linearly searches for terminal backpointers, costing $O(H^2)$ time and $O(H)$ retained attempts in the worst case. An indexed `map[Event][]Event` successor structure plus an open-attempt hash table targets expected $O(H+R)$ construction time and space; hashing is not a worst-case guarantee.

For N declarations and cross-references and grammar size G, repeated list membership and duplicate checks make validation $O(N^2)$ time with $O(N)$ auxiliary data. Grammar traversal is $O(G)$, but repeated concatenation can make skewed trees $O(G^2)$; indexed names and accumulator walks are optimization targets. Enumerating every path is exponential (b^d for branching factor b and depth d) only when all paths are materialized.

The initial ChatUI profile gives browser/session actors capacities 32 inbound and outbound, with at most 4 in flight; the server has capacities 256 and 32 in flight; provider workers have capacities 64 and 16 in flight. Per-instance memory budgets are 64, 512, and 256 MiB, respectively. These replaceable starting values are not measured production claims.

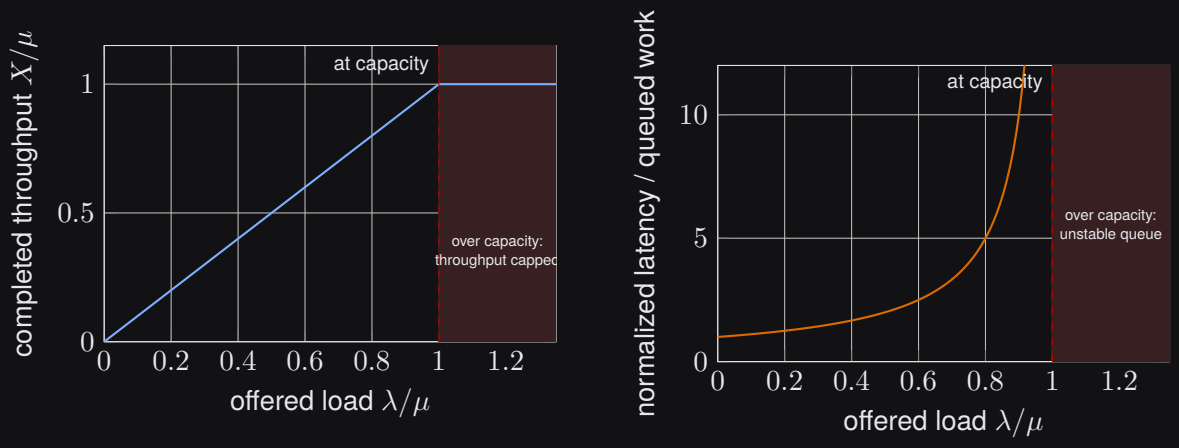


Figure 6.2: Offered-load capacity. Beyond $\lambda/\mu = 1$, throughput cannot increase but queued work and latency continue to worsen until backpressure, rejection, an outage, or fatal memory exhaustion intervenes.

Use the models before implementation to declare demand, sustainable capacity, latency and queue bounds, rejected-work dispositions, replica affordability, and fatal failure modes.

6.5 Outages, recovery, and replica populations

Reliability assumptions belong to actor specifications and are inherited by their instances. A contract may set an outage probability and an expected mean time to repair (MTTR):

```
actor Server
  |>.instances "server-" 2 (roundRobin)
  |>.outageProbability (20 / 10000)
  |>.meanTimeToRepair 45.seconds
```

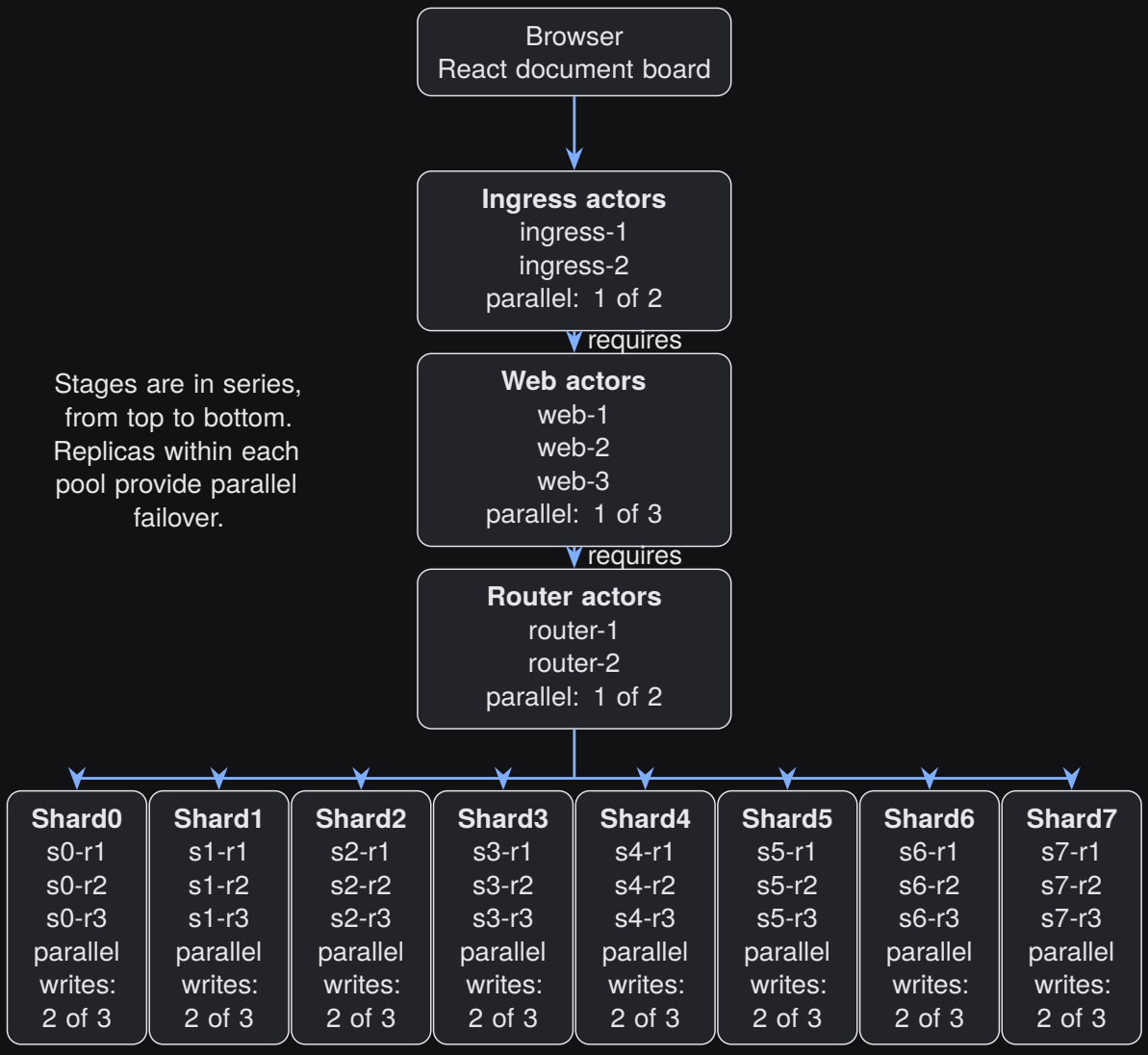
These are specified assumptions. A failed actor does not reliably announce its own failure: there is no ordinary `Unavailable` message to receive. Measurements instead come from evidence visible to an observer. Expected heartbeats stop arriving; an inbound queue ceases to drain; or messages expected from the actor's outbound side cease to appear. Each detector must declare its heartbeat or progress expectation, timeout, clock, observer, and attribution boundary. Silence without such an expectation is not evidence of failure.

The detector may emit derived observations such as `AvailabilityLost` and `AvailabilityRestored`, but these are observer conclusions, not protocol messages sent by the failed actor. A loss observation retains the last known healthy time, the deadline that was missed, the detection time, and the evidence stream. Restoration retains the first new heartbeat or queue/message progress that justifies it and backpoints to the loss observation. The true failure time is generally bounded between the last healthy observation and detection rather than known exactly. If the declared accounting policy uses derived boundary times u_i and r_i, then for R replicas observed from t_0 through t_1, the estimated outage fraction and repair duration are

$$\widehat{O} = \frac{\sum_i (r_i.timeAt - u_i.timeAt)}{R(t_1 - t_0)}, \qquad \widehat{MTTR} = \frac{\sum_i (r_i.timeAt - u_i.timeAt)}{|incidents|}.$$

Availability over the same complete window is $1 - \widehat{O}$. Reports must identify each u_i boundary (last healthy, missed deadline, or detection), since it changes the outage and MTTR estimates. In Kubernetes-style experiments, pods may be actor instances and restarts recovery events; compare configured outage probability and MTTR with observations rather than replacing them.

6.5.1 Redundant, load-balanced runtimes and blast radius



Shards are parallel data partitions with separate availability gates.
Each shard has parallel replicas; writes need a primary and a threshold quorum.
A targeted write selects one shard, not all shards.

Actor topology for a targeted document write. Generated from `RuntimeReliabilityDiagram.lean`, with provenance in `runtime-actor-network-provenance.txt`. Arrows denote required provider paths, not logical implication or message sequence.

Request path. A write needs ingress, web, router, and its selected shard; a multi-shard operation needs every shard its semantics require. Adding required series stages cannot improve availability; independent stage availabilities multiply. Kubernetes schedules and restarts actors but does not supply the application's quorum or consistency protocol.

Replica pools. Proxy and web actors are parallel alternatives within series stages; each data shard has its own parallel replica gate. A usable alternative may improve pool availability, but shared

failure domains and higher quorum thresholds can erase that benefit. Shards are partitions, not interchangeable failover copies.

Stateless pool. Zero ready instances makes new requests unavailable. Otherwise compare offered work with sustainable capacity under the latency bound: overload may trigger rejection, queue growth, or deadline misses even while heartbeats continue. Round-robin request counts need not balance unequal-cost work.

Replicated shard. Majority-acknowledged writes require a current primary and a communicating quorum of eligible configured members; one external observer's heartbeats do not establish mutual reachability. A restarted member is not ready until it catches up.

Read and shared dependencies. Declare read consistency separately: a lost write quorum may still permit local stale reads, but a surviving copy does not prove linearizability. Ingress or routing failure can affect all shards; metadata failure depends on the operation and cached routing, so does not necessarily disable every read or write.

Sharding. Additional partitions can increase capacity when work distributes well; hot keys, shared dependencies, and cross-shard operations limit the gain. Replication protects each partition, not other shards.

Evaluate these dependency gates against timestamped observations and declared uptime/MTTR assumptions; availability is operation-specific, not a total replica count.

Use detector-derived $u_i(t) \in \{0, 1\}$ values and declared boundaries to combine instances through each operation's series, parallel, or threshold gate. For a series gate, every member must be up:

$$u_{\text{series}}(t) = \prod_i u_i(t).$$

For a parallel gate, at least one member must be up:

$$u_{\text{parallel}}(t) = 1 - \prod_i (1 - u_i(t)).$$

More generally, a replicated service declares a threshold k and is available when $\sum_i u_i(t) \geq k$. Its measured availability is the fraction of the observation window for which that predicate is true. These formulas operate on synchronized observed indicators, so simultaneous and correlated failures are preserved. Products such as $\prod_i A_i$ or $1 - \prod_i (1 - A_i)$ may estimate expected availability from marginal availabilities only under an explicit independence assumption; they must not replace the interval calculation when joint outage observations exist.

A dependency edge points from a consumer service or task to the provider gate it requires. At each outage boundary, failure propagation repeatedly marks a consumer unavailable when one of its series requirements is unavailable or when a parallel/threshold requirement has too few surviving providers. The fixed point for an initiating failure set F is $C(F)$, its unavailable- operation set and structural blast radius. Retain the propagation paths, initiating set F, affected shards, request classes, tenants, workload, affected tasks, client-visible outcomes, and recovery order. A degraded read path and an unavailable write path are distinct outcomes. This makes questions such as "what happens if the database and one authorization replica fail together?" answerable without pretending that all

outages are independent. Minimal cut sets identify the smallest initiating sets that disable each operation.

The standard reliability output includes per-instance and series/parallel/threshold service uptime, a time-aligned outage chart, and the dependency graph overlaid with observed or proposed cascades. Named service-down scenarios remain replayable when unobserved and must be labeled predictions, separately from measured uptime.

Calculate impact before estimating its frequency. Uptime estimates help assign frequencies or probabilities to these scenarios; MTTR estimates help assign recovery durations. Neither marginal uptime nor mean repair time determines a cascade by itself. The dependency graph determines which operations are exposed, while correlated failure and repair models determine how often and how long. If scenarios F_j form a mutually exclusive, exhaustive partition of a declared snapshot model, with probabilities π_j, then the expected number of unavailable operations is

$$\mathbb{E}[B] = \sum_j \pi_j \, |C(F_j)|.$$

A count weights every operation equally; a report may instead use an explicitly declared demand or business-impact weight. Overlapping outage scenarios cannot simply be added as if they were disjoint.

For a time-aligned trace, let $C(t)$ be the unavailable-operation set at time t, and let $\lambda_o(t)$ be offered requests per second for operation o. The offered-request exposure over window $[t_0, t_1]$ is

$$L = \int_{t_0}^{t_1} \sum_{o \in C(t)} \lambda_o(t) \, dt.$$

This exposure is measured in requests, not necessarily lost work: queues, retries, deadlines, and recovery determine which requests ultimately fail or complete late. For overload, measure unmet work and deadline violations separately rather than counting every request as unavailable.

Repair the operation, not just the process. Actor MTTR is not automatically service MTTR. Service recovery may include failure detection, replacement scheduling, volume attachment, election, replication catch-up, routing refresh, and backlog draining. These stages can overlap, so compose their event dependencies rather than blindly adding their means. The end boundary is restored operation readiness or the declared latency target, not merely receipt of the first heartbeat. Open incidents at the end of the window remain censored observations, not zero-duration repairs.

Calibrated forecasts require a declared failure/repair process, correlated domains, placement, workload and recovery policy. Keep those assumptions separate from observations; report affected operations, request exposure, deadline outcomes and recovery sequence, with a source and reducer for each forecast. Existing `DesiredOutput` entries establish provenance. This is an analysis contract, not a claim that a general Kubernetes deployment or forecast engine is implemented.

The generated artifact `paxos-quorum-shutdown-network.svg` gives a concrete majority-service example. A Kubernetes-like runtime manages a parallel pool of three stateless web actors; the one-of-three web pool is in series with a two-of-three database-replica gate. Each web actor has specified uptime $9999/10000$; each replica has specified uptime $999/1000$. If replica outages were independent and each replica had uptime p, the quorum's assumed uptime would be

$$A_{\text{2-of-3}} = 3p^2(1-p) + p^3.$$

This independent-marginal baseline excludes correlated shutdowns.

Stopping `kv-1` leaves two voters; stopping `kv-2` next leaves only `kv-3` in that shard, below quorum, and makes the dependent web service unavailable even if other shards remain healthy. Planned shutdowns, shared power, network partitions and common automation therefore belong in named joint scenarios, not independent uptime factors. Adding stateless web instances cannot restore the failed database quorum and may only accumulate waiting requests.

Work waiting on an instance inferred unavailable remains stalled, is rejected, or ends in an explicit timeout/cancellation outcome according to the requirement. An ordinary send or receive attributed to that instance after a detector's loss boundary is evidence that either recovery occurred or the detector/attribution model was wrong; it is not silently discarded as an invalid trace. Thus an outage commonly falsifies a nominal liveness promise (for example, "every accepted request eventually receives a response") even when safety remains intact.

The nominal protocol language and its failure extension must therefore be distinguished. The nominal language describes ordinary successful and rejected conversations. The failure extension adds heartbeat expectations, queue and message-progress observations, derived availability boundaries, stalled or censored work, explicit failure outcomes, and legal resume/retry behavior. Cascade analysis suppresses modeled progress for every instance inferred unavailable at the dependency fixed point, then reports which nominal safety and progress properties remain true, become false, or cannot be decided before the observation window closes.

The same dependency graph supports denial-of-service analysis. A PageRank-like centrality calculation, with dependency edges directed toward the consumers whose operation they enable, highlights nodes whose failure influence is reinforced by other important services. Edge weights may incorporate observed message traffic, affected task value, or measured cascade frequency, but the chosen weighting is part of the requirement and must be shown in the report. Centrality is a prioritization heuristic, not an availability proof: ordinary PageRank can miss a low-traffic series dependency, a cut vertex, or the exact number of replica failures that crosses a quorum threshold. The denial-of- service report therefore places the centrality ranking beside single-node and named multi-node cascade simulations, minimal service cut sets, lost task capacity, and client-visible outcomes. This answers both "which node looks important?" and the stronger question "what actually stops if an attacker makes this set unavailable?"

6.6 Infrastructure cost and demand-constrained supply

Hardware accrues per provisioned hour; licenses, managed services, and operations may cost per available hour. Neither is a task-event cost to multiply by request count:

```
cost Server
  |>.hardwarePool "general-purpose"
  |>.hardwareCostPerProvisionedHour 12 USD.cent
  |>.serviceCostPerAvailableHour 3 USD.cent
  |>.sustainableCapacityPerHour 3600000 bytes
```

For R replicas, provisioned hours H_r, available hours A_r, hardware rate c_h, and service rate c_s, the accounting-period infrastructure cost is

$$C(R) = \sum_{r=1}^{R} (H_r c_h + A_r c_s).$$

Supply planning is a constrained optimization, not "pick the cheapest server." Given per-replica capacity μ in work units per hour, required availability a, and work rate D within the declared latency boundary, feasibility requires

$$R\mu a \geq D.$$

This is only a necessary capacity condition: routing imbalance, queues, or latency can still make the observed system infeasible.

Route total offered rate λ_o across R servers; let $\lambda_{o,r}$ be server r's offered share:

$$\lambda_o = \sum_{r=1}^{R} \lambda_{o,r}, \qquad \lambda_{o,r} = \frac{\lambda_o}{R} \text{ only under ideal balance.}$$

Under ideal balance, nominal demand is D/R; verify observed instance loads and effective supply. Each replica incurs hardware cost while provisioned and service cost while healthy; compare the hourly total with the optimizer's result.

Executable code and finite-model tests: `optimalSupply?` in `LeanFM/Artifacts.lean` searches replica counts 1 through `maxReplicas`, rejects availability-adjusted capacity below demand, and selects the least expected hourly cost. `LeanFM/ArtifactsTests.lean` checks one feasible optimum and one infeasible demand. This bounded function does not compute observed routing imbalance, latency, or workload-window results; the full supply-planning XY output above remains a target, not generated evidence. Shard hashing can create a hot shard despite aggregate spare capacity; sticky routing and round-robin make different balance assumptions. Product replication, reserved pricing, autoscaling delay, and nonlinear tiers require explicit constraints.

Reuse the observation window and completion boundary above, adding hourly cost and per-server utilization or imbalance. Count only demand completed within the declared outcome and latency boundary; unbounded supply, backlog, cost, or latency does not guarantee finite-deadline service.

6.7 Specified dwell time

A `dwell` constrains modeled duration for a step or named production; observed event-pair durations are defined in Chapter 2:

```
emit "token requested" [field "code" code]
  |>.dwell 3.milliseconds

include exchangeAndFetch
  |>.dwell 12.milliseconds
```

Step dwell d advances the model clock from t to $t + d$ before timestamping its completion event. Production dwell constrains entry-to-exit time, not each contained event. Use exact values, bounded ranges, or weighted outcomes when supported by the timing information.

Dwell can drive modeled latency, throughput, timeout, and queue-pressure predictions, but does not create event order: `priorIds` supplies predecessors, while the grammar states which sequences and dependencies are legal.

6.8 Composing requirements means intersecting constraints

Scenario composition reuses the grammar chapter's operations for sequencing, alternatives, independent interleavings, repetition, and context-sensitive filtering (Chapter 5). Requirements add a different operation: `requireAll` intersects the generated trace language with independent constraints instead of adding message sequences.

```
def quoteSystem : SystemSpec :=
  generatedBy quoteOutcome
  |> requireAll
      [ correlate "offered quote matches request"
          (sameField "requestId")
      , forbid "confirmation without acceptance"
          (event "order confirmed" && notPreviously "quote accepted")
      , ctl "every accepted quote is answered"
          (AG (accepted --> AF (confirmed || rejected)))
      , timing "answer within service objective"
          (accepted --> within 2.seconds (confirmed || rejected))
      ]
```

Consistency requires a nonempty intersection. If it is empty, tooling should report a minimal conflicting requirement set and the shortest available counterexample prefix.

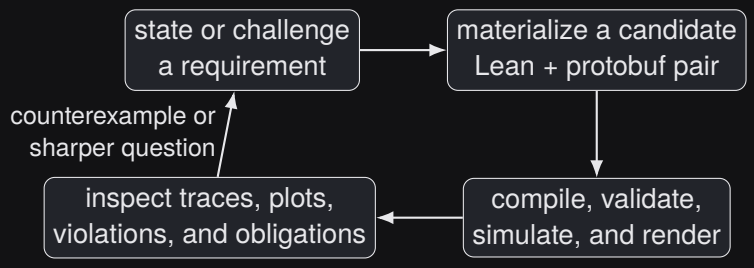
Chapter 7

Arguing Requirements through the Lean Web Server

This chapter documents the staging server that implements the workbench, its project isolation, and its interface for the two acceptance gates defined in the Preface. The evidence labels in Chapter 1 distinguish its executable paths from proposed and future interfaces; each named check retains its stated scope.

7.1 The argument loop

The interface resembles ChatGPT because natural language is convenient for finding omissions. The loop, however, has a formal boundary:



The human contribution is often adversarial. “The upload completes” should prompt: what observable event constitutes completion, who observes it, and does it mean receipt, durable storage, or policy acceptance? “The service handles load” should prompt: how many messages fit in each queue, how many `(session, task)` pairs may be active, and what happens at the bound? The assistant’s job is to turn each answer into structure, not merely agreeable prose.

A productive exchange might proceed as follows.

> **Person:** A submitted upload eventually completes.
> **Assistant:** I will treat an `uploadCompleted` response as completion.
> **Person:** No. Completion requires the storage actor’s acknowledgment, and the client must be able to distinguish rejection from timeout. Both queues are finite.
> **Assistant:** Then the grammar has accepted, rejected, and timed-out alternatives; acceptance contains a storage acknowledgment before the response. I will add queue capacities and a

bounded-liveness obligation.
Person: Also show the behavior with many sessions in flight, estimate the completion distribution from an event stream, and fail the requirement if queued bytes exceed the memory budget.

Every turn should change a named assumption, artifact, check, or open obligation. If it changes none of them, it has not improved the specification.

7.2 The server is a staging environment

`lake exe leanfm-server` serves the workbench on loopback. Every project owns one exclusive root (hosted layout: Section 7.8; portable contract: Chapter 15). Project-relative and canonical absolute paths are accepted only when their resolved targets remain beneath that root; traversal and escaping symlinks fail closed.

The three specifications have the separate roles defined in Section 15.4. HTTP, in-process channels, TCP and other adapters are implementation choices; validation, views, metrics, reports and OpenAPI project the same model. A language model may propose edits, but compilation and validation remain deterministic: prose establishes no proof.

7.2.1 The code that creates the TCP listener

An address value does not open a port. The running server performs four distinct operations: create a socket, bind it to an address, put it into listening mode, and repeatedly accept clients. This is the relevant executable pattern from `Server.lean`:

```
import Std.Internal.Async.TCP

open Std
open Std.Net

abbrev TcpServer := Std.Internal.IO.Async.TCP.Socket.Server

def bindAddr : SocketAddress :=
  SocketAddress.v4
    { addr := IPv4Addr.ofParts 127 0 0 1
    , port := 8080
    }

partial def serveLoop (server : TcpServer) : IO Unit := do
  let client ← (server.accept).block
  let _ ← IO.asTask (handleClient client) Task.Priority.dedicated
  serveLoop server

def main : IO Unit := do
  let server ← Std.Internal.IO.Async.TCP.Socket.Server.mk
  server.bind bindAddr
  server.listen 128
  IO.println "LeanFM server listening on http://127.0.0.1:8080"
  serveLoop server
```

Read the final definition as follows: `Server.mk` creates the TCP socket; `bind` attaches it to loopback port 8080; `listen 128` changes it into a listener with a pending-connection backlog of 128; and `serveLoop` calls `accept` for each connection. Only after `listen` succeeds is there a TCP listener. HTTP is the application protocol implemented by `handleClient`; there is no separate HTTP socket. The actual handler catches per-request errors before returning to the accept loop. This shortened excerpt leaves that response logic out so the listener lifecycle remains visible.

7.3 Using the agent in Codex

Codex can drive the complete bootstrap. Open the LeanFM repository as its workspace, or run `codex` from the repository root. The required book, live language references, source tree, and workflow from self-hosting through acceptance and generation are defined in Chapter 15.

7.4 MCP: the agent-facing entrance

The browser UI is for stakeholders; the Model Context Protocol endpoint is for an LLM agent. LeanFM serves stateless JSON-RPC requests at `POST /mcp`. On initialization, the server tells the agent that the book is durable project context—effectively a maintained section of the system prompt—rather than optional background reading. The agent can enumerate and read fixed resource URIs for `book/main.pdf`, its dark rendering, the canonical requirements, protobuf and implementation files, and both live language references.

This interaction is generated from the implemented MCP dispatch surface:

The first surface is intentionally read-only. It provides resource and prompt discovery, the requirements interrogation, the post-acceptance code-generation prompt, and canonical artifact validation. It exposes no arbitrary pathname and no tool can accept a requirements or implementation gate. Those are stakeholder decisions, not consequences of a successful tool call. MCP transports context and evidence; the existing LeanFM workflow controls revision and acceptance.

The bundled server listens only on loopback. A deployment that exposes MCP to a network must add authentication and authorization at the transport or reverse proxy before treating the resource contents as safe to publish. Statelessness does not make project data public, and a future writable MCP surface must bind every request to a user, project, revision and accepted snapshot.

A local Codex client can register the running endpoint and inspect the result:

```
codex mcp add leanfm --url http://127.0.0.1:8080/mcp
codex mcp list
```

The MCP interface does not replace the book; it makes the book, specifications and live checker discoverable to an agent client.

7.5 A tour of the LeanFM chat

Select a project in the left rail, then **Chat**. Its mounted workspace contains Project memory, Chat, Book, Grammar, Tasks and Source. Switching project or phase changes visibility while preserving drafts, selections, scroll positions, queues, timers, previews and fetched results. Active work continues



Figure 7.1: An agent obtains durable project context and deterministic evidence through MCP. Source: `LeanFM.MCP`; reducer and provenance: `book/generated/mcp-interaction-provenance.txt`.

while hidden. Each message has a timestamp; the header identifies the billed provider, model and non-secret key fingerprint, with response token usage shown per turn.

Chat accepts ordinary language, images, rough pseudocode and protocol questions before Book, Grammar, Tasks or Source are settled. A response may revise the book, ask a focused blocking question, derive a requirements view or create a scratch explanation; none accepts a gate. For each turn the model receives the selected project's complete editable book and canonical requirements and implementation specifications, then a bounded recent transcript. Neither LeanFM's own book nor another project supplies fallback context. This is one contextual conversation, without a hidden mode selector.

The ordinary outcome is a decision and a checked artifact, with no clarification question. The agent reads prior answers and durable choices, selects reasonable reversible defaults within the recorded requirements and authorization, and records those decisions. It briefly lists only choices the stakeholder is likely to challenge. A question must name a consequential unresolved effect on scope, observable behavior, compatibility, security/privacy, irreversible data or material spending, and explain why a recorded decision or authorized default cannot resolve it. The two explicit acceptance gates remain explicit.

A readiness review can report ready without inventing questions. Compiler syntax, formatting and routine check failures use the bounded diagnostic and recovery path below. They do not consume a stakeholder clarification turn. The agent retains the exit code, narrows the correction and verifies progress; an unchanged failure class cannot authorize another paid retry.

7.5.1 Progress, cancellation and the Operational trace

Every phase shows an **Agent running** banner with the active prompt's captured model, incrementing elapsed time, latest real stage and cooperative **Cancel** control. Status, Project memory, Source and the Operational trace are polled every two seconds; transient failures retain the last successful value and retry. The changing artifacts remain inspectable.

The Operational trace is appended before the next agent step. It records each bounded workspace action as started, completed, failed or rejected. A proposed write names its normalized project-relative destination, content kind and character count before validation, so a path failure says what LeanFM was trying to write. Failures are highlighted in the collapsed banner summary and the accumulated trace may be expanded before the turn completes. Cancellation is checked at provider and workspace-tool boundaries: no new mutation begins after the request is observed, while already completed durable edits remain truthful project state.

Prompts typed during a turn enter a visible removable project-local FIFO and execute exactly once in order. Each queued prompt captures the model visible at the time it is enqueued. In the composer, focus order is prompt editor, **Send**, then model selector. Thus **Tab**, **Enter** commits a typed prompt with the already visible model rather than opening the selector.

Structured clarification is also observable: a returned question logs its stable ID, and a composer submission while that question is pending logs that it bypassed the answer form. Both events appear in the turn trace and the project diagnostic journal. They retain the question ID and bounded text lengths, never the operator's prompt or answer text.

7.5.2 A private fault barrier and actionable errors

The visible trace is for stakeholder decisions; a separate bounded private engineering journal supports diagnosis across navigation, server restarts and discarded chat history. Each project stores it at `logs/runtime-diagnostics.jsonl`. It contains sanitized stage transitions, tool destinations, compiler diagnostics, failed required artifacts, warning results, terminal provider failures, authorized project HTTP errors and correlated unhandled server exceptions. A later agent request receives the twenty newest unresolved entries before choosing a recovery strategy. Verified matching writes and book builds remove superseded failure clusters and leave one compact resolution, so the journal remains memory rather than an ever-growing graveyard.

The React client is inside the same fault boundary. A React error boundary, global JavaScript exception listener, unhandled-promise listener, API wrapper, and explicit Mermaid and LaTeX renderer catches send an authenticated report back to the server. A report names the operation, error type and message, sanitized JavaScript stack, component stack when available, page, time and active project. The reporting request uses a recursion guard, bounded fields and at most twenty reports per browser minute. Project authorization is checked again on the server. Prompts, form and workspace-write bodies, credentials, cookies, arbitrary provider responses, URL queries and cross-project data are never diagnostic payloads.

Every error shown to a person still has a separate obligation: state what happened, recommend a concrete next action, provide a stable code and offer a bounded diagnostic safe to paste to Codex. An unknown proxy response such as “527 Something went wrong” is insufficient. The private journal normally means the person need not paste anything: LeanFM or Codex can inspect the project root, repair the cause and continue the original task.

Provider credit/spend/quota rejections, rate limits, incomplete output and local session safety limits receive distinct diagnostics. Response length establishes no credit balance. An incomplete Responses result retains reported usage but cannot execute returned tools or update the book. A malformed reply envelope never exposes its hidden book source as prose. One complete, unambiguous JSON object surrounded by prose or fences, and illegal JSON escapes before LaTeX math delimiters, can be normalized locally before ordinary validation and the book build. Arrays, multiple proposed objects and truncated objects are not salvaged. Other format failures retain the parser offset, character count, leading-token category and complete-object flag, not arbitrary response text. None triggers a paid retry.

Every provider round records its input, cached-input, output and cumulative usage, request and answer character counts, outcome and tool count before the next paid call. The session budget is checked between tool rounds and before bounded recovery or repair, not just when a chat turn begins. Reported usage is charged once, including failed validation. This is an observed-token brake, not a hard monetary cap: an in-flight request can exceed the remaining budget, and an unanswered request’s usage is unknown. Provider spending controls remain necessary. Saved API keys remain password-encrypted on disk across server restarts; only unlocked sessions disappear. Blank login key fields preserve those saved keys and never display them.

Repeated failure is bounded at the point of expense. Within an active turn, a second tool call with the same name and exact arguments after an error is not executed; it terminates that strategy before a third paid attempt. Provider timeouts are never automatically retried because a timed-out request may have consumed tokens or completed an earlier mutation. The agent instead reads the diagnostic, changes strategy, exhausts safe reversible alternatives, validates the result and resumes the requested task. Only missing authority, a destructive choice or an unavailable dependency should

return control to the user.

7.5.3 Requirements properties and implementation provenance

The Tasks page separates four views of one selected task: interaction productions, residual states, temporal properties and guarded constraints. A property is an observable requirement originating in `Requirements.lean`; it is not silently invented by `Implementation.lean`. The UI renders its actual CTL wrapper rather than labeling prose as CTL: `always` becomes $AG\,p$, `eventually` becomes $AF\,p$, `never` becomes $AG\,\neg p$, and `preparedFor` becomes $EF\,p$. A `where_` trace predicate is displayed with its stable identifier under the invariant $AG\,p$. Each card identifies its requirements origin.

The implementation mapping follows Section 15.4: its answers realize accepted behavior and cite their requirements. It cannot silently introduce a property or change the wire contract.

The assistant owns routine formal-language and implementation decisions. The interrogation checklist is an internal completeness check, not a questionnaire for the stakeholder. Choose, record and validate reversible defaults without interrupting the user. A question needs a major consequence that is hard to reverse and cannot be resolved within existing authorization. Formal approval describes behavior and material consequences in plain language. Chat shows the same revision-bound approval controls as Project memory once a candidate is checked; pressing one records the human decision without a provider request. Successful intermediate checks are progress toward the original goal. The assistant selects the next reasonable reversible step and continues through client-driven batches without asking whether to continue. The user can Cancel. Completion compares real artifacts and behavior with the accepted goal, including required negative behavior and conformance evidence; source existence or passing self-authored tests alone cannot establish that result. ChatUI obtains the language references and implementation guidance through LeanFM's read-only MCP dispatcher. The selected project's sandboxed compiler still checks its own canonical files; the host's MCP self-validation cannot substitute for a foreign project's evidence. The full-book MCP resource uses iterative base64 encoding so a large central PDF cannot exhaust the server's call stack.

Provider-supported web search can resolve public implementation questions without another stakeholder turn. Source links survive client batch boundaries and remain clickable. External pages supply evidence, never authority to change project boundaries, acceptance, secrets or spend controls. Adding a provider key through the signed-in account editor preserves the current login, usage, expiry and existing provider work. Replacing an existing secret retains the session-revocation policy. Blank key fields preserve their stored values; account passwords and keys are never returned to the browser.

Application delivery ends with a working `source/Makefile` launch target, not just a set of files. `make run` uses a bounded loopback port separate from LeanFM. The project retains `spec/runtime.json` and `spec/ProjectMemory.md`: the original goal, reversible defaults, known failure remedies, remaining gaps and PR base/review context survive erased chat. The `run` workspace check launches the actual command in a private network and checks readiness and its application page. It has no host credentials or host-network access, writes only project cache, and stops only its own process group. The host records protected source-bound evidence; changed source makes that evidence stale. A failed launch returns its concrete diagnostic to the same client-driven task. Passing the launch check establishes these bounded observations, not complete semantic refinement of the accepted specification.

An existing username and its correct password can update retained API keys. Login and account settings also offer an optional password change. The current password must verify and both new-

password fields must match. A single atomic credential envelope contains the new verifier and re-encrypted provider bundle, so an interrupted write cannot split the two. Old sessions are revoked; a credential revision checked at authentication and tool boundaries also rejects stale sessions in another service process sharing the vault. Additive provider saves preserve that revision. A legacy verifier file remains a compatibility mirror; it cannot override the envelope. The `?keys=1` link opens the secure account editor and submits no credential request by itself.

7.5.4 How a chat turn becomes a durable edit

Guided chat selects the artifact layer required by the request. Specification work uses project-confined workspace tools to update the canonical Lean and protobuf files; it is not forced to rewrite the whole book. The `leanfm_reference` tool returns the same live language and workflow references exposed through MCP. The `leanfm_check` tool compiles the project's named typed requirements and optional implementation mapping, validates their structure and protobuf declarations, and regenerates `spec/tasks.json` from that same requirements value. Compiler execution has no network, credentials or writable project source: only project-local scratch is writable. Failed validation preserves the last valid views. The generated catalog retains canonical input digests and reducer provenance; neither a check nor a PDF build accepts a stakeholder gate. A turn changing canonical specifications cannot report SUCCESS without a successful check of its final bytes.

The requirements reference includes a complete compilable request/response example with matching protobuf, task states, grammar, processes, properties, negative obligations and generated-view provenance. A reduced scope still needs these collections. In particular, `markdown` contains typed `RequirementMarkdown` records, not plain strings. The implementation reference likewise includes a compiled mapping for that example. `MessageCodegen.wireType` is a string naming the generated protobuf type. `ImplementationSpec.channels` is one `ChannelCodegen` record with explicit send, receive, drain and close semantics; capacities remain in the requirements' actor resource contracts. If the model tries to finish after an unchecked specification edit, the server returns the concrete compiler or structural diagnostic within the same native provider conversation. The model can correct the files and recheck without a second user message. This continuation retains the original cancellation, session, usage accounting and request budget. An unchanged candidate and diagnostic stop further paid work; an unavailable compiler stops immediately. A repaired current check can report success while retaining earlier failures in the operational trace. Neither recovery nor successful validation accepts a stakeholder gate.

When the turn revises the book, the provider's structured reply includes the complete proposed `book/main.tex`. The server enforces root and size boundaries, writes atomically and builds in the isolated book worker. Only a successful build commits the revision and sets `book_updated=true`. If compilation fails, the first bounded compiler diagnostic becomes visible before repair begins and the exact previous source is restored after every candidate. At most three repair requests may run, each with a short deadline. Each request receives only the compiler diagnostic and a numbered seventeen-line source excerpt, asks for one replacement line with at most 2000 output tokens, and is rejected if it targets another line or makes no change. A successful candidate is rebuilt before commit; exhaustion produces WARN and leaves the last valid book authoritative. Deterministic syntax mistakes such as literal ampersands and underscores should be corrected directly when their intended prose meaning is unambiguous rather than rediscovered through repeated paid model calls.

The same rule becomes stricter at the implementation boundary. After both acceptance gates and the security critique, an implementation request writes real buildable files, generated bindings,

and tests beneath the selected project's `source/` directory by using the project-confined workspace tools. The Source phase reads those files directly. A provider response that says "implemented," or pastes a Go listing into chat, cannot be SUCCESS while `source/` is empty. When a gate or material implementation decision is missing, the valid alternative is a focused selectable question naming that blocker—not an invented implementation and not another generic plan.

This instantiates Chapter 10: substantive turns can create validated artifacts, while malformed, oversized, cross-project, unbuildable or merely narrated artifacts cannot be reported as committed. The deterministic HTTP test checks a compiled book update, an actual `source/main.go` write and rejection of an implementation claim with empty Source.

7.5.5 Example: ask to see MCP move

The following is a useful prompt because it asks for both an explanation and a specific interaction technique. It is reproduced as entered; spelling does not need to be polished before the model can act.

```
i want to understand what MCP looks like.  make me an animation that i
can step forwrd or back in, showing messages passing back and forth in
dynamic interaction diagrams.  visualized right here in the chat ui
```

Here *MCP* means the Model Context Protocol, not MQTT and not a generic message broker. A faithful first tour uses a host or MCP client and an MCP server as lifelines. When the selected example invokes a real downstream tool, that tool may be a third lifeline; otherwise it is omitted. A representative trace is: the client sends JSON-RPC `initialize` with its version and capabilities; the server replies with its supported version, capabilities and identity; the client sends `notifications/initialized`, discovers tools with `tools/list`, then sends `tools/call` with a tool name and arguments and receives its result or error correlated by JSON-RPC ID.

This is the same kind of interaction shown durably in figure 7.1, but projected into a small review instrument. The inline view should show the participating lifelines, one selected message in motion, its method or result label, and a detail panel containing the request ID, direct cause, arguments or result summary. **Forward** completes the next event; **Back** restores the immediately preceding prefix; **Play/Pause** advances or suspends automatic stepping; **Restart** returns to the empty prefix; and a scrubber may select any event boundary directly. Keyboard left and right arrows are convenient equivalents when focus is inside the animation.

Stepping backward does not send an inverse protocol message. It changes which prefix of the same immutable example trace is visible. At step k, the canvas shows exactly events $e_1, \ldots, e_k$; advancing exposes e_{k+1} and stepping back hides e_k. Request and response remain correlated by their JSON-RPC identifier, while LeanFM's richer event representation can separately retain a direct causal pointer and predecessor links. Thus the animation may explain causality without falsely claiming that temporal adjacency alone pairs a response with a request.

7.5.6 What should appear in the response

The chat response should contain a short explanation followed by the working animation itself. Its HTML and JavaScript are transport material, not the answer: they must not fill the message as pages of source code. LeanFM expects an actual `leanfm-canvas` fenced block and removes that block from the prose before placing its canvas and self-contained script in an isolated frame. The

frame has no network access, cannot navigate the parent, and may load no external library or image. Stop and Restart controls outside the frame allow the stakeholder to halt an expensive or confusing animation. Unless the user specifically requests a text color, canvas labels, legends, and annotations use a contrasting foreground: light text on dark drawn backgrounds and dark text on light drawn backgrounds. Choose from the canvas drawing's actual background rather than the surrounding page theme.

There is one bounded repair for a common provider formatting error. If the model literally writes "fenced leanfm-canvas" and immediately follows it with exactly one canvas element and one script element, the client recovers that pair into the same sandbox instead of displaying it as noise. This exception does not turn chat into a general HTML renderer: unmarked or arbitrary HTML is escaped and remains inert text. A response that shows the raw program instead of an animation is a rendering failure, not SUCCESS. Resend preserves the original user request while allowing the corrected renderer and prompt contract to try the turn again.

The animation is initially a scratch explanation. It answers "what would this interaction look like?" quickly, but it is not yet a generated view of accepted requirements. If the discussion changes the project design, retain the originating prompt, bind the selected messages to a named scenario, update the typed grammar and regenerate the interaction diagram from that specification. Only that provenance-bearing derived view can serve as evidence at the requirements gate. If stakeholders instead want to share the exact motion, promote the chosen canvas revision to a project-owned GIF or MP4 with fixed frame timing and serve that media inline.

Chat's **Attach image** control accepts PNG, JPEG, GIF and static SVG files, shows a removable preview, and retains the selected images with the queued prompt and later Resend. The original and a digest manifest remain below the owning project's `artifacts/attachments/` directory. The provider receives normalized PNG pixels through its native visual-input protocol; a GIF is explicitly represented by its first frame. The current implementation bounds each original to 2 MiB and each source to 16 megapixels, and sends at most four current or recently referenced images per turn. External SVG resources and executable content are rejected. These controls realize the existing multimodal requirement; their negative-test mappings are recorded in `evidence/chat-images/security.md`.

Task interaction renderers must receive the owning specification along with the task. Looking up a project's task in LeanFM's host ChatUI grammar yields a misleading missing-grammar placeholder even when the project grammar is valid. The serializer and renderer now share that explicit specification, and regression evidence includes a foreign task whose ID collides with ChatUI.

Every still image displayed in Chat retains a reference that can leave the screen. An adjacent keyboard-accessible **Copy reference** control copies a project image as its stable project-relative artifact path. For a public-web image it copies Markdown containing the displayed HTTPS image URL followed by the visible source-page URL. It never copies image bytes, a data or ephemeral object URL, the host filesystem path, credentials or unrelated response text. This makes "use the image above" a durable artifact reference that can be pasted into a later Chat turn, the book or Codex even after conversation state has been discarded. Clipboard failures cross the same client fault barrier as other JavaScript errors.

7.5.7 Screenshot-first documentation

When the scenario runner is available, replace this procedural tour with a named capture of the production React DOM using a deterministic provider. Stop at a declared event prefix, such as prompt submission, MCP initialization, tool discovery, or Source artifacts. Each caption should name the

prompt, what to notice, and the claim demonstrated.

A screenshot is useful evidence only when it is reproducible. Its manifest retains the project and requirements revision, scenario and visible event IDs, viewport, font scale, theme, browser version, capture time, reducer and content digest. The image receives meaningful PDF alternative text. Headless capture runs with a disposable account and deterministic data, without a provider key, external network or arbitrary target URL. A missing or stale required capture fails the book build instead of silently leaving a blank figure.

Use captures selectively: keep formal definitions, security boundaries and qualifications in text when an image cannot express them. Until the runner produces these artifacts, this describes the intended replacement, not existing screenshots.

7.6 Local credentials and visible billing identity

The LeanFM login is not a provider login. One account password unlocks the user's OpenAI and Anthropic keys together. The `API keys` form adds or replaces either key; blank inputs preserve stored keys. Provider labels and keys are authenticated together inside a password-encrypted bundle, never returned to the browser or stored in plaintext. Existing single-key accounts migrate when keys change. Updates revoke older sessions; an already-running provider request may finish, but its returned tools cannot run afterward. The project list exposes the checked-in GoServer project under one canonical `GoServer` identity, not a separate "GoServer example" tab.

Account records are runtime state, not source. By default the salted password verifier, password-encrypted provider keys, private conversations, and session state remain beneath the operator's local `data/` directory. Git ignores that directory and continuous integration rejects any tracked file below it. Checked-in example projects therefore contain books, specifications, source, tests, and deliberately selected media, but never user logins or credentials.

The model selector combines models discovered for the unlocked providers. Its server-owned catalog binds each selection to one provider and the exact account-owned key; an unknown model or missing key fails without a paid call or a fallback to another provider. OpenAI Responses and Anthropic Messages use the same confined workspace tools, retry brakes, and book recovery. The localhost HTTP interface waits for the complete byte-counted request before dispatch. Its framing contract accepts UTF-8 with one Content-Length, bounds headers to 32 KiB and bodies to 8 MiB, and expires incomplete requests after 15 seconds. Ambiguous lengths, transfer encodings, and pipelined extra bytes are rejected without provider calls or mutations.

An authenticated header continuously identifies the provider, model, credential reference, and a one-way fingerprint of the exact key used for the next request. It also shows provider-reported session tokens used and tokens remaining in a finite local safety budget; exhaustion stops another request. That cross-provider token count is not a credit or dollar balance, because a model response does not expose such a balance. The UI says so explicitly. A provider project spend cap remains the authoritative protection against monetary overrun, while per-response token counts remain attached to the request that incurred them.

7.7 Cursor as an agent client

A Cursor subscription connects to LeanFM as an authenticated agent client, not as a model-provider API key. A project may commit `.cursor/mcp.json` with the local URL `http://127.0.0.1:8080/mcp`.

Cursor Agent opens the project root, reads `AGENTS.md`, and discovers LeanFM's book, canonical artifacts, language references, prompts and validation tools through MCP. Cursor continues to own its subscription authentication and billing; LeanFM receives no Cursor login secret. The current MCP surface remains read-only, so Cursor edits files using its ordinary project tools and cannot accept either gate through MCP.

For the inverse arrangement—the LeanFM browser driving a Cursor-backed conversation—Cursor CLI exposes the Agent Client Protocol over standard input and output. A `CursorACP` adapter can initialize and authenticate the local agent, create or load one session for each LeanFM project conversation, submit prompts, stream updates, answer explicit permission requests and cancel work. This is an implementation direction, not a claim about the running server. The adapter must isolate user, project, conversation, process and request identity; bound its queues; preserve tool and permission evidence; and terminate the process only after its queue is closed and drained. Cursor's authentication token is never imported into LeanFM's provider-key store.

7.8 One project, one directory

7.8.1 Accepted shared workspace and fresh start

The stakeholder accepted the requirements and separate implementation mapping for the fresh start on October 23, 2026. Ordinary projects live directly below `projects/<readable-project-name>` and have no user owner. The stakeholder has confirmed that the workspace opens before login; login is required for LLM calls and API-key management. Shared project contents include their books, specifications, source, artifacts and conversations. Password verifiers and encrypted API keys belong in the private application vault `.leanfm/accounts/`; they never become project files or model input.

The accepted typed component is `LeanFM/LLMGenerated/SharedProjects.lean`, exported by the canonical requirements and composed into the self-hosting product. The separate realization is in the canonical `LeanFM/LLMGenerated/Implementation.lean`. Running `lake env lean -run ReviewSharedProjects.lean` validates both candidates and derives the review, task views and provenance below `reports/shared-projects/`. Exact stakeholder decisions are recorded in `reports/shared-projects/acceptance.json`; a generator cannot accept a gate. The explicitly requested reset discarded five ordinary projects and the empty legacy scaffold, verified the preserved credential bytes and permissions, and removed `./data`. It preserved LeanFM's own source, book and specifications. The reset report is `reports/shared-projects/reset-result.json`.

`LEANFM_WORKSPACE_ROOT` selects the exclusive workspace directory and defaults to the server's current directory. Ordinary roots are discovered from their own validated manifests below `projects/`, independently of any credential account. A project contains its own book, canonical specifications, source, tests, media, conversation, evidence and caches. Deleting its root removes that project's state. The credential vault is excluded from every project capability. Foreign browser origins and unapproved Host values cannot mutate the loopback workspace; a conflicting project writer is rejected before mutation or provider work. Queued prompts retain their initiating credential identity and are held if that identity changes.

Directory labels come from project names, such as `projects/GoServer`, rather than opaque hashes. Spaces and punctuation become safe separators; collisions receive a short numeric suffix. Atomic directory reservation never adopts or overwrites an occupied root. The manifest's stable project identifier remains independent of the directory label, so relocating an existing root preserves its

ChatUI URLs, conversation and checked acceptance revisions. Discovery rejects duplicate identities instead of choosing an arbitrary root.

Project memory exposes the two independent acceptance controls for the current checked revisions. Source writes require both gates in server code; model tools cannot write gate records or checker evidence. Current gate state is supplied to subsequent model turns, so an accepted decision survives a cleared chat. Project-authored checks and media generators execute without network or host credentials and have only bounded project-local cache or scratch write access.

7.8.2 Credential accounts and project roots

The portable root and containment rules are in Chapter 15; this section records the hosted-runtime mapping. `LeanFM.ChatUI.Store.createProject` creates shared roots under `projects/<readable-project-name` writes a runtime `manifest.json`, and provisions the book, `spec/`, source, tests, artifacts, renders, events, reports, logs, cache, and conversations beneath that root. The manifest records all canonical project-relative paths and its acceptance-state path. Portable `project.json` roots are also discovered when their identity and configured paths validate; they are ordinary editable projects. The removed examples are not inputs to provisioning or validation.

7.9 What the Lean artifact must settle

The Lean artifacts are the semantic source for actors, framed messages, process projections, grammars, properties, proof obligations, and desired outputs. Canonical definitions are in Chapter 2 (event fields, correlation, predecessor order, causation, and elapsed time), Chapter 5 (legal conversations, actor projections, and bounded queue semantics), and Chapter 6 (choices, dwell, rates, scalability, reliability, and infrastructure cost).

One cross-cutting obligation deserves emphasis: message-count capacity alone does not bound memory when payload sizes vary. Derive queued bytes from encoded envelopes and declare a bounded footprint for active task state. A representative requirement is

$$\mathrm{AG}\big(\mathrm{queuedBytes}(a) + \mathrm{inFlightBytes}(a) \leq \mathrm{budget}(a)\big).$$

Here AG means every position on every permitted future, and the repeated a fixes the actor being measured. This is a modeling obligation, not a claim that the current validator proves the heap usage of an arbitrary implementation.

7.10 Interrogate, characterize, synthesize, replay

7.10.1 Characterization readiness

The bootstrap contract defines the opening question and durable project-root handoff (Chapter 15). Characterization tests whether desired claims are identifiable from evidence: scenario and task boundaries, terminal-to-start links, actor populations, clocks and units, observation windows, resource bounds, failure evidence, and probability provenance. Missing inputs remain indeterminate, not invented.

One event stream should produce the standard characterization. Its canonical inventory is `outOfBoxArtifacts` in `LeanFM/Characterization.lean`; `baselineDerivedProperties` records

each property's required fields, reducer, unit, visualization and boundary. A domain retains, replaces, extends or marks each applicable property indeterminate.

Specified distributions may drive predictions or generation, but are never presented as measurements; event-stream estimators are detailed in Chapter 2.

For generative similarity, reducers map the source stream to a named metric vector, with each histogram bin represented as an exact ratio. A candidate is serialized, decoded and reduced identically; it passes only when every required metric is within its declared relative tolerance. "Similar" therefore means equivalent under this finite characterization, not visually or textually similar. The grammar, invariants and byte language still constrain legal traces.

7.11 The protobuf artifact is part of the requirement

The parallel protobuf schema answers a question that a state diagram cannot: what bytes can cross a component boundary? Atomic payload messages and a `RequirementEnvelope.oneof` discriminator connect a grammar terminal to its wire representation. Validation checks, among other things, that field tags are nonzero and unique and that framing is explicit.

This pairing exposes useful disagreements. A Lean event may require a `priorIds` list while the wire schema provides only one predecessor. A latency property may depend on a clock that is never transmitted or recorded. A memory budget may be stated in bytes while an unbounded protobuf field remains. These are requirements defects, not serialization trivia.

The exact wire size is also an input to resource analysis:

```
queuedBytes actor observation :=
  (observation.inbound actor).sum encodedSize
  + (observation.outbound actor).sum encodedSize

memoryPressure actor observation :=
  queuedBytes actor observation
  + inFlightFootprint actor observation.activeWork
```

The first term can be computed from concrete envelopes. The second requires a declared abstraction of implementation state and must remain visibly labeled as such.

7.12 A ladder of checks

No single green check answers every kind of requirement. The workbench should present layered evidence: Lean/protobuf check well-formed syntax; structural validation checks actors, endpoints, terminals, framing, process coverage and proof obligations; grammar/state-machine checks cover ordering and temporal checks cover safety, reachability and bounded liveness; probability/dwell reducers predict outcomes and event-stream reducers estimate them; queue and byte-budget checks model capacity, backpressure, concurrency and memory; generated diagrams, state machines, plots and reports support review.

Use the evidence labels defined in Chapter 1. Predictions are not measurements, incomplete windows censor tasks, frequencies do not prove specified probabilities, and bounded traces do not prove unbounded liveness. Reports identify each result as proved, calculated, sampled or assumed.

7.13 How to argue well

The most useful prompts attack boundaries rather than ask for polish:

- "Which event and grammar branch make this claim true?"
- "With two concurrent sessions and tasks, what blocks on a full queue and which events correlate?"
- "Which random choice belongs to the user, and how does it change the downstream completion and latency distributions?"
- "Derive the plot from the event stream, including in-flight tasks."
- "Show the corresponding protobuf payload and its encoded-size bound."
- "Is this result proved, model-checked, simulated, measured, or assumed?"

Use the project-root, derived-view, event-boundary and completion contracts in Chapters 15, 3, 2 and 10. Conversation discovers the specification; durable artifacts retain it.

7.14 Client-driven bounded work

The accepted resumable-work component is recorded in `LeanFM/LLMGenerated/ResumableWork.lean`; its separate implementation mapping and acceptance hashes are beside it and in `reports/resumable-work/`. One user task crosses bounded provider batches through server-issued project checkpoints. The client continues the same task, while later prompts wait and Cancel stays available. A batch boundary is progress, never a total step limit. Checkpoint consumption and cached receipts prevent replay; caller, model, credential, token accounting, failure history and both human gates survive the boundary. Closing the client starts no further batch. Ambiguous provider failures pause for reconciliation rather than an automatic paid retry. Oversized provider context restarts from durable project files and verified tool receipts. The generated interaction and state views retain their prompt, reducer and boundary in the typed component. Tests use local provider fixtures, the real compiler and production Chrome; they do not prove generated applications correct.

Chapter 8

Worked Example: OAuth Authorization Code Flow

This chapter is a behavioural sketch inspired by the OAuth authorization-code flow. It is not a replacement for the OAuth specification and deliberately begins above HTTP and protobuf. Its purpose is to exercise scenario decomposition, parameter flow, visible evidence, alternatives, requirements, and rendering.

8.1 Actors, values, and observable events

The example has four participants: a person using a browser, an application, an authorization service, and a resource service. The browser is modelled separately from the person because redirects are observable messages between software participants.

```
inductive Actor where
  | user | browser | app | authorizationServer | resourceServer

opaque State       : Type
opaque AuthCode    : Type
opaque AccessToken : Type
opaque Resource    : Type
opaque Profile     : Type
opaque Password    : Type
opaque Bytes       : Type
```

Every terminal elaborates to the common event envelope. For example, the vague message “authorization redirected” becomes conceptually:

```
{ timeAt := t2, id := e2, priorIds := [e1]
, session := browserSession, task := oauthAuthorizationTask
, from := Actor.app, to := Actor.browser
, kind := "authorization redirected"
, fields := [field "state" state, field "redirectUri" callback]
}
```

Authentication detail used by Production 2

Before the redirect returns, the authorization service may authenticate the person. OAuth does not prescribe the authentication mechanism; we include a password branch to exercise knowledge flow without pretending that the password is an OAuth protocol parameter.

```
def authenticateAtAuthorizationServer (password : Password) : Scenario :=
  scenario "authenticate resource owner" do
    from_ .browser
    to_ .authorizationServer
    emitBytes "credentials submitted"
      (protectFor .authorizationServer
        (encode [field "password" password]))

    from_ .authorizationServer
    to_ .browser
    oneOf "authentication result"
      [ emitBytes "authentication accepted" (encode [tag 1])
      , emitBytes "authentication rejected" (encode [tag 0])
      ]
```

The wire bytes are specified even though the transport is not. The abstract `protectFor` constructor promises confidentiality to the named endpoint. A later refinement may realize that promise with TLS, message-level encryption, or a local trusted channel. Merely placing the password inside a field would not support the intended knowledge set: without protection, a network observer would also derive the password from the bytes.

8.2 Production 1: application request and redirect

```
def applicationRequestAndRedirect (resource : Resource) : Scenario :=
  scenario "application request and redirect" do
    let state <- declare State "state"
    let callback := Uri.https "app.example" "/oauth/callback"

    from_ .browser
    to_ .app
    emit "application request" [field "resource" resource]

    from_ .app
    to_ .browser
    emit "authorization redirected"
      [field "state" state, field "redirectUri" callback]

    export "state" state
    export "redirectUri" callback
    export "resource" resource
```

This production treats the application request and its redirect as one named unit. It establishes the correlation value `state` and exports it to the next production. Nothing here says how the application stores it.

8.3 Production 2: authorize, then grant or deny

```
def authorizationGranted : Scenario :=
  scenario "authorization granted" do
    use state <- imported State "state"
    let code <- declare AuthCode "code"

    from_ .authorizationServer
    to_ .browser
    emit "authorization code returned"
      [field "state" state, field "code" code]

    from_ .browser
    to_ .app
    emit "authorization callback"
      [field "state" state, field "code" code]
    export "code" code

def authorizationDenied : Scenario :=
  scenario "authorization denied" do
    use state <- imported State "state"
    from_ .authorizationServer
    to_ .browser
    emit "authorization error returned"
      [field "state" state, field "reason" "access_denied"]
    from_ .browser
    to_ .app
    emit "authorization error callback"
      [field "state" state, field "reason" "access_denied"]

def authorizeAndDecide (password : Password) : Scenario :=
  scenario "authorize and decide" do
    use state       <- imported State "state"
    use redirectUri <- imported Uri "redirectUri"

    from_ .browser
    to_ .authorizationServer
    emit "authorization requested"
      [field "state" state, field "redirectUri" redirectUri]

    include (authenticateAtAuthorizationServer password)
    oneOf "authorization decision"
      [ include authorizationGranted
      , include authorizationDenied
      ]
```

The authorization request and its two possible continuations form one production. At a more abstract rendering level, the entire production can be shown as one node; expanding it reveals authentication and the explicitly bracketed choice. The abstract contract makes both decisions possible. A product model may later attach an estimated or measured probability distribution without changing the set of legal traces.

8.4 Scenario 3: exchange the code and use the token

```
def exchangeAndFetch : Scenario :=
  scenario "exchange code and fetch resource" do
    use code     <- imported AuthCode "code"
    use resource <- imported Resource "resource"
    let token    <- declare AccessToken "accessToken"
    let profile  <- declare Profile "profile"

    from_ .app
    to_ .authorizationServer
    emit "token requested" [field "code" code]

    from_ .authorizationServer
    to_ .app
    emit "access token issued" [field "accessToken" token]

    from_ .app
    to_ .resourceServer
    emit "protected resource requested"
      [field "accessToken" token, field "resource" resource]

    from_ .resourceServer
    to_ .app
    emit "protected resource returned"
      [field "resource" resource, field "profile" profile]

    from_ .app
    to_ .browser
    emit "application response" [field "profile" profile]
```

The token itself is visible only at the endpoints where the abstract protocol says it is transmitted. The grammar makes no claim about the authorization server's database schema or the application's session representation.

8.5 Reading the scenarios together

```
def oauthAuthorization (resource : Resource) (password : Password) : Scenario :=
  scenario "OAuth authorization code flow" do
    include (applicationRequestAndRedirect resource)
    include (authorizeAndDecide password)
    after authorizationGranted include exchangeAndFetch

def oauthSystem (resource : Resource) (password : Password) : SystemSpec :=
  generatedBy (oauthAuthorization resource password)
  |> requireAll oauthRequirements
```

The two named productions provide the first abstraction level. A renderer may collapse either production to a single node or expand it to its messages and nested alternatives. The `after` line is a guarded nonterminal reference; the denial branch ends without exchanging a code.

8.6 The parallel protobuf alphabet

The scenario and its wire alphabet are maintained side by side. The scenario says which messages may occur, in what predecessor-order structure, and with which parameter relationships. The protobuf file assigns field numbers and concrete wire types to those same message atoms:

```
syntax = "proto3";

package leanfm.oauth;

enum Actor {
  ACTOR_UNSPECIFIED = 0;
  BROWSER = 1;
  APPLICATION = 2;
  AUTHORIZATION_SERVER = 3;
  RESOURCE_SERVER = 4;
}

message EventEnvelope {
  string event_id = 1;
  repeated string prior_ids = 2;
  string session_id = 3;
  string task_id = 4;
  uint64 clock_tick = 5;
  Actor src = 6;
  Actor dst = 7;

  oneof payload {
    ApplicationRequest application_request = 20;
    AuthorizationRedirect authorization_redirect = 21;
    AuthorizationRequest authorization_request = 22;
    CredentialsSubmitted credentials_submitted = 23;
    AuthenticationResult authentication_result = 24;
    AuthorizationCodeReturned authorization_code_returned = 25;
    AuthorizationCallback authorization_callback = 26;
    AuthorizationErrorReturned authorization_error_returned = 27;
    AuthorizationErrorCallback authorization_error_callback = 28;
    TokenRequest token_request = 29;
    AccessTokenIssued access_token_issued = 30;
    ProtectedResourceRequest protected_resource_request = 31;
    ProtectedResourceReturned protected_resource_returned = 32;
    ApplicationResponse application_response = 33;
  }
}

message ApplicationRequest {
  string resource = 1;
}

message AuthorizationRedirect {
  string state = 1;
  string redirect_uri = 2;
}

message AuthorizationRequest {
  string state = 1;
  string redirect_uri = 2;
}

message CredentialsSubmitted {
  bytes protected_credentials = 1;
}

message AuthenticationResult {
```

```
  bool accepted = 1;
}

message AuthorizationCodeReturned {
  string state = 1;
  bytes code = 2;
}

message AuthorizationCallback {
  string state = 1;
  bytes code = 2;
}

message AuthorizationErrorReturned {
  string state = 1;
  string reason = 2;
}

message AuthorizationErrorCallback {
  string state = 1;
  string reason = 2;
}

message TokenRequest {
  bytes code = 1;
}

message AccessTokenIssued {
  bytes access_token = 1;
}

message ProtectedResourceRequest {
  bytes access_token = 1;
  string resource = 2;
}

message ProtectedResourceReturned {
  string resource = 1;
  bytes profile = 2;
}

message ApplicationResponse {
  bytes profile = 1;
}
```

Each emitted scenario message maps to exactly one member of `EventEnvelope.payload`. The envelope carries the event identity, repeated `prior_ids`, `(session_id, task_id)` correlation key, monotonic clock tick, and endpoints outside the payload. The payload carries only the fields specific to that message.

Protobuf alone does not specify the protocol order, queue behavior, probability, or dwell time; those remain in the scenario. Conversely, the scenario does not invent byte tags. Validation checks that every scenario terminal has one protobuf variant, that its endpoints agree with the scenario, and that every referenced field has the expected protobuf type.

The schema determines valid protobuf encodings, but protobuf does not generally promise one canonical byte sequence. When byte-for-byte comparison is required, the refinement also fixes deterministic serialization and length-delimited message framing. An instantiated envelope can then be serialized and displayed as the actual bytes sent between the named actors.

8.7 Generate one legal trace and its artifacts

We can now run the specification in one direction. Grammar generation chooses productions and concrete field values; everything after that is a deterministic projection or reduction of the resulting events. For this run choose the authentication-accepted and authorization-granted alternatives and instantiate

```
session = "s1"             task = "oauth"
resource = "profile"       state = "x7"
redirectUri = "https://a/cb"
protectedCredentials = aa bb
code = 01 02               accessToken = 03 04
profile = 05 06
```

The production derivation is

```
oauth
=> applicationRequestAndRedirect ;
   authorizeAndDecide ;
   after(authorizationGranted, exchangeAndFetch)
=> applicationRequestAndRedirect ;
   authorizationRequested ;
   credentialsSubmitted ; authenticationAccepted ;
   authorizationGranted ; exchangeAndFetch
```

Because `authorizationGranted` was selected, the `after` guard admits `exchangeAndFetch`. Selecting the denial alternative would disable that production. Expanding every remaining nonterminal yields this terminal word:

The choice has now disappeared into ordinary terminals. Event `e5`'s payload bytes select the accepted authentication result. Event `e6`'s `authorizationCodeReturned` oneof tag, source, and bytes distinguish the grant branch from the error-returned branch. No independent "chooser" value is needed.

8.7.1 Envelope stream and concrete bytes

Assign monotonically increasing clock ticks and event identifiers. This example has a linear predecessor chain, so `priorIds(e1)` is empty and `priorIds(e`n`)` is `[e`$n-1$`]`. A fork or join would make these lists non-linear without changing the procedure. Deterministic protobuf serialization is the intended encoding; the companion fixture records one concrete stream:

```
e1   0  application_request          31  0a0265311a02733122056f6175746830013802a201090a0770726f66696c65
```

The companion fixture `examples/oauth-granted.bytes` retains all twelve envelopes (440 encoded bytes total; largest envelope 46 bytes). The excerpt above shows one complete envelope. The repository does not currently include a generator or round-trip check for this particular fixture, so it is a concrete example rather than evidence that serialization was reproduced or verified.

8.7.2 Event graph with stored backpointers

Parsing each envelope selects a payload alternative, checks its declared `src`/`dst`, and consumes one grammar terminal. For this selected trace, the useful concrete view is the event graph itself:

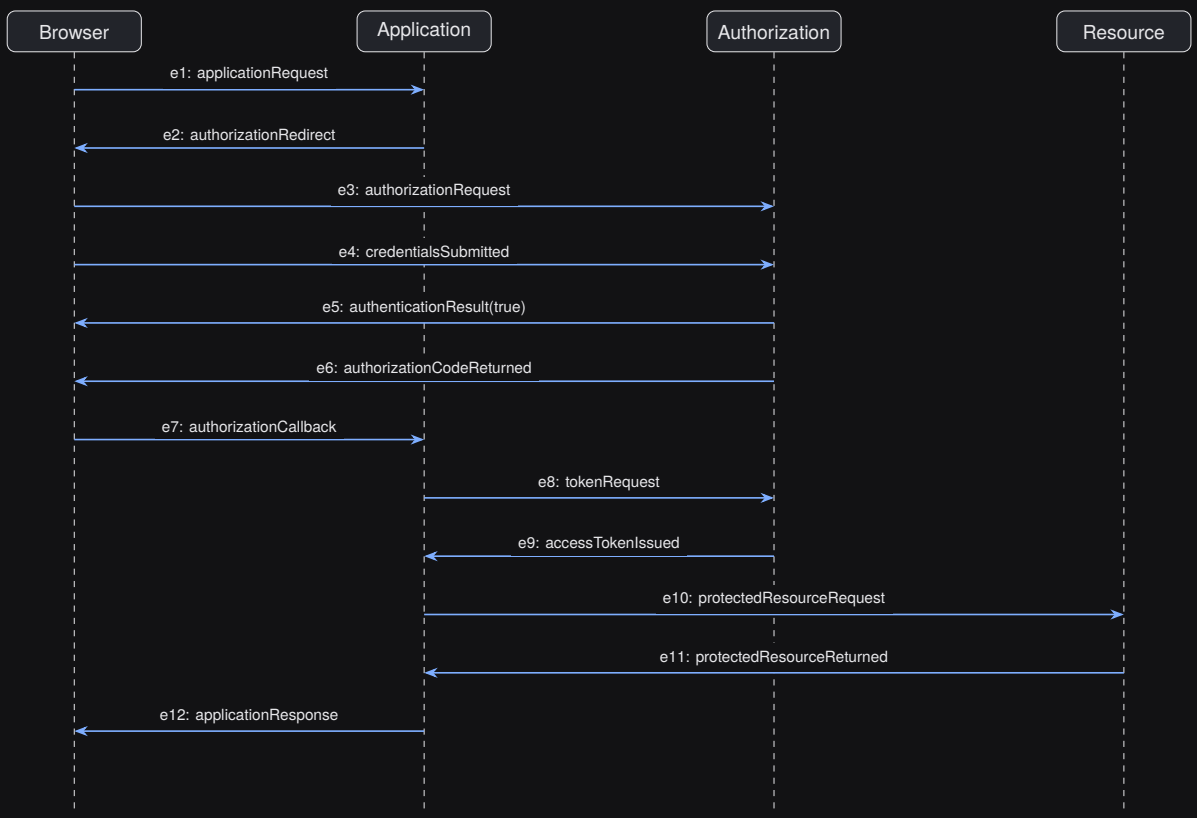


Figure 8.1: The selected terminal word as an interaction diagram. Event IDs label messages on arrows; no actor-to-actor arrow is written as implication-like text.



Figure 8.2: The accepted trace as an event graph. Arrows are stored backpointers: `e1` has none and each later event points to its immediate predecessor.

This graph is one selected path. The full residual state machine is derived from every legal grammar residual and also contains the rejected-authentication and denied-authorization branches; it should be rendered as a separate graph when those alternatives are under discussion.

8.7.3 Actor projections

Project each terminal onto its sender and receiver, preserving order and erasing the rest. For example, Resource receives `e10` and sends `e11`; the other local scripts follow the same endpoint filter. The example generator emits the shared interaction and predecessor diagrams, not replay scripts. The global word remains authoritative for cross-component order and parameter equality; a local projection does not reveal erased events.

8.7.4 Event order, queues, and memory

The `priorIds` fields generate a twelve-node order graph with edges $e_i \to e_{i+1}$. The interaction diagram is the same graph laid out by actor; the state-machine path lays it out by grammar residual. Neither is independently authored.

The message word does not uniquely determine a transport schedule. For a concrete queue artifact, choose the canonical schedule that performs, for each event, `produce; transport; receive` before generating the next event. With positive inbound and outbound capacities, each queue peaks at one message. The queued-byte peak for an actor is the largest encoded envelope waiting at that actor in this schedule, and the global queued-byte peak is 46. A different legal interleaving may have a larger peak and must be explored when checking the universal memory-budget property. The selected schedule is a witness, not the capacity proof.

8.7.5 Measurements, plots, and verdicts

From the clocks and correlation key, the generated task attempt is

```
{ key = ("s1", "oauth"),
  started = { kind = applicationRequest, timeAt = 0 },
  completed = { kind = applicationResponse, timeAt = 23 },
  outcome = succeeded, latency = 23 ticks }
```

It contributes one point `(maxConcurrency=1, latency=23)` to the latency XY plot, a cumulative-byte series ending at `(clock=23, bytes=440)`, and one successful slice to the outcome chart. It does *not* establish a 100 percent success probability. The grammar-level weight predicts 0.87 mass on the grant branch; estimating that probability requires many generated or observed sessions, including incomplete ones.

Trace-local monitors accept the matching `state=x7`, the code flow `01 02`, and the fact that no exchange follows denial. CTL obligations such as “every decision reaches the application” are not proved by this trace; the checker must examine every reachable grammar residual. The complete artifact pipeline is thus

```
grammar + field values
  -> legal terminal word
  -> typed EventEnvelope stream
  -> framed protobuf bytes
  -> partial-order graph + interaction diagram
```

```
  -> residual FSM path + per-actor projections
  -> queue schedule + byte-pressure samples
  -> task attempt + probability/latency/XY reducers
  -> replay fixture + trace-local property verdicts
```

Every arrow is reproducible. Only the initial branch and value generation, and the separately stated transport schedule, add choices.

8.8 Interaction diagrams by scenario

8.9 Requirements as observable constraints

```
def oauthRequirements : List Requirement :=
  [ require "callback state matches request"
      (on "authorization callback" (previouslySame "state"
        "authorization requested"))

  , forbid "code exchange after denial"
      (previously "authorization denied" && event "token requested")

  , require "issued token came from a code"
      (on "access token issued" (previouslySame "code" "token requested"))

  , ctl "every decision reaches the application"
      (AG (decisionAtAuthorizationServer --> AF callbackAtApplication))

  , ctl "a protected result is never returned without a token request"
      (AG (resourceReturned --> previously tokenRequested))
  ]
```

8.9.1 Who knows the password?

This OAuth instance applies the trace-derived knowledge closure from Section 13.5:

```
def passwordKnowers (trace : Trace WireEvent) : List Actor :=
  allActors.filter fun actor =>
    Derives (knowledgeFrom actor trace) (value password)

theorem oauth_password_knowers
    (trace : Trace WireEvent)
    (accepted : (oauthSystem resource password).Accepts trace)
    (initial : InitiallyKnown password [.browser, .authorizationServer])
    (honest : Uncompromised [.browser, .authorizationServer])
    (protection : PerfectProtection protectFor) :
    passwordKnowers trace = [.browser, .authorizationServer] := by
  exact knowledgeClosureOfProtectedField accepted initial honest protection
```

The browser itself knows a typed password. A narrower boundary requires a trusted credential actor; compromise, logging, a visible key, or failed protection changes the result.

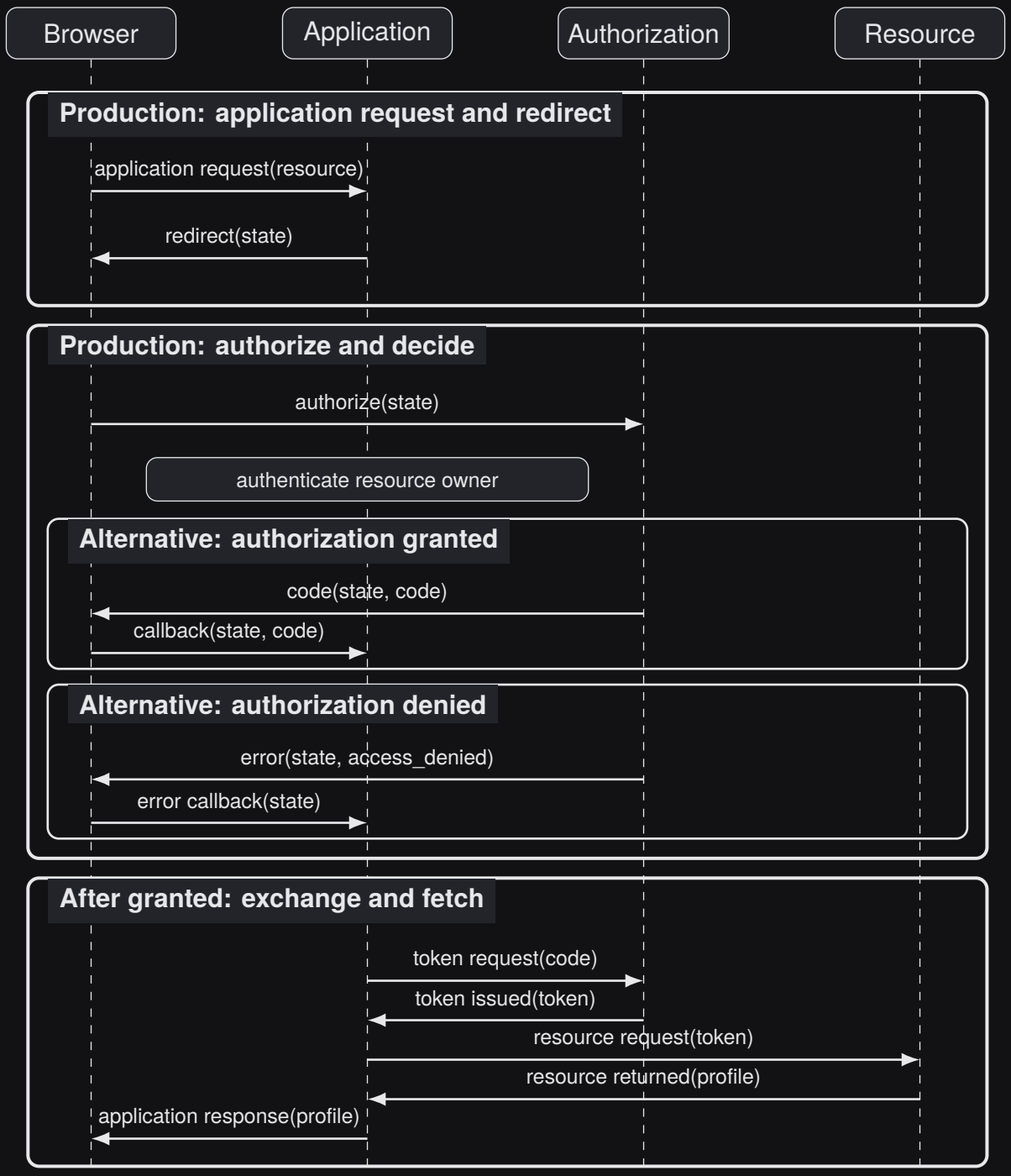


Figure 8.3: OAuth interaction view. Production frames provide collapsible abstraction boundaries. The authorization production contains its request and both alternatives; the guarded exchange production is enabled only by the granted alternative.

8.9.2 Byte-level refinement without choosing a transport

Every vague OAuth message can be assigned a canonical byte term before HTTP or protobuf is chosen:

```
refineMessage "authorization requested" where
  term := tuple
    [ tagBytes 0x01
    , namedBytes "state" (encode state)
    , namedBytes "redirectUri" (encode callback)
    ]
  bytes := encodeCanonical term

refineMessage "token requested" where
  term := tuple [tagBytes 0x02, namedBytes "code" (encode code)]
  bytes := encodeCanonical term
```

This fixes framing, tags, and value relationships at an abstract byte-algebra level. A subsequent HTTP mapping might put those bytes into form fields; a protobuf mapping might use numbered fields. Either concrete alphabet must prove that decoding and re-encoding preserve the abstract terminal and its parameter bindings.

The use of `previously` is deliberately a trace predicate in this first version. It can later receive dedicated past-time syntax without changing the event model.

8.10 Probabilities and charts

The authorization service need not be random. The user supplies random input by granting or denying consent; the browser, queues, transport, and authorization service then react deterministically to the input they receive:

```
def consentDecision : Scenario :=
  weightedOneOf "consent decision"
    [ weighted (87 / 100) authorizationGranted
    , weighted (13 / 100) authorizationDenied ]

def oauthCharts : List ChartSpec :=
  [ pie "authorization outcomes" do
      source events; groupBy terminalOutcome; value count
  , line "end-to-end authorization latency" do
      source (correlate events by "state")
      x clock; y (elapsed "authorization requested" "application response")
  ]
```

No separate field says that the user “owns” this choice. The grant and denial branches resolve to different consent-event bytes, and that terminal’s source identifies the user-facing actor that emitted them. After those bytes select a branch, every transition may have probability one and the global model is still probabilistic. The grant branch carries probability mass 0.87 through the outbound queue, transport, the server’s inbound queue, and the deterministic grant response. The denial branch similarly carries mass 0.13. Deterministic processing transports probability mass; it does not remove it.

For the simplified path with no loss or timeout,

$$P(\text{grant callback}) = 0.87 \cdot 1 \cdot 1 \cdot 1 = 0.87, \qquad P(\text{denial callback}) = 0.13.$$

Here P denotes probability; each factor of one is a deterministic enqueue, transport, or server step. Dwell affects latency, not probability mass.

Unassigned scheduling choices (such as which enabled actor runs next) are nondeterministic, while consent and measured outcomes supply probabilities. The composed system is an MDP: analysis fixes or quantifies over a scheduling policy before propagating consent mass (see Chapter 6).

For several sessions in flight, the MDP state contains every actor's bounded queues and its per-`(session, task)` residuals. One user's random event can change later queue pressure and scheduling possibilities for other sessions, so the probabilistic state is the composed system state rather than a special marker attached only to the user actor.

The percentages remain model inputs or measurements, not truths inferred from the grammar. The grammar defines which behaviours are possible; the probabilistic refinement says how mass is distributed over them.

8.11 Residual state machine

This automaton is intentionally behavioural. It does not contain browser cookies, database rows, server threads, or cryptographic library state. Those details may be essential to an implementation proof, but they are refinements of this contract, not the contract itself.

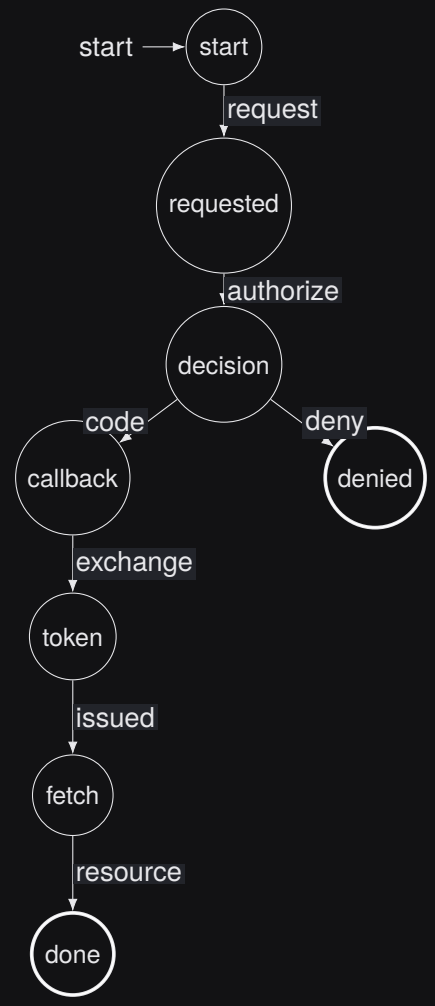


Figure 8.4: OAuth state machine derived from legal grammar continuations.

Chapter 9

CTL over Generated Futures

A grammar generates a branching transition system. CTL speaks directly about that branching, but its vocabulary becomes clearer when placed between basic modal logic and linear temporal logic. Path quantifiers distinguish *there exists a future* from *every future*, while temporal operators distinguish the next state, eventuality, invariance, and until.

The author's modal-logic background used here was learned mostly from Kröger and Merz [5]. This citation identifies the source of that education; it is not a claim that the authors originated modal or temporal logic.

9.1 Basic modal logic

Propositional logic evaluates a formula at one world or state. Basic modal logic adds an accessibility relation R between worlds and two operators:

$$\begin{aligned} w \models \Box p \quad &\text{iff} \quad p \text{ holds at every } v \text{ such that } wRv, \\ w \models \Diamond p \quad &\text{iff} \quad p \text{ holds at some } v \text{ such that } wRv. \end{aligned}$$

Thus $\Box p$ and $\Diamond p$ read as "necessarily p" and "possibly p." Under classical negation they are dual: $\Diamond p \equiv \neg\Box\neg p$. The symbols do not fix the meaning of accessibility. Depending on the model, an accessible world may be a next computation state, a world compatible with an actor's knowledge, a normatively permitted world, or something else.

The basic normal modal logic K allows arbitrary Kripke frames. Additional frame conditions validate additional principles: reflexivity, transitivity, symmetry, and the Euclidean condition lead to familiar stronger systems. Those principles must not be imported merely because the English word "necessarily" sounds strong. LeanFM uses a directed computation-successor relation and does not silently treat it as reflexive, symmetric, or epistemic.

9.2 LTL: time along one path

Linear Temporal Logic (LTL) describes one linear path with operators including X (next), F (eventually), G (always), and U (until). Thus Fp says that p occurs at some suffix of the selected path, while Gp says that p holds at every suffix of that path.

LTL's temporal words answer *where along this path*; they do not themselves choose between one possible branch and every branch of a transition system. A model-checking convention often asks whether every initial path satisfies an LTL formula. That outer convention should not be confused with a path quantifier in the LTL syntax. For example, asking whether some legal execution eventually reaches recovery and asking whether all legal executions eventually reach recovery require different branching claims even though both contain the words "eventually recovery."

LeanFM does not currently implement LTL. It uses CTL because the requirements deliberately combine two independent axes:

- *always* or *eventually* says where along forward time a predicate must hold; and
- *necessarily* or *possibly* says whether it must hold on every permitted forward continuation or may hold on at least one.

This choice arose from trying to combine temporal logic with alethic logic while investigating whether all the familiar modal families could be combined. The construction retained by LeanFM is specifically *temporally alethic*: possible means possible now or at some point along a legal future. Ordinary language uses "always" both for universal modality and for temporal persistence. CTL keeps those coordinates visible long enough to prevent the ambiguity from disappearing inside an English gloss.

9.3 CTL: path modality combined with time

CTL pairs a path quantifier with each temporal operator. E is existential: there is at least one permitted continuation. A is universal: every permitted continuation satisfies the temporal condition. In the vocabulary of modal logic, E supplies temporal possibility and A supplies temporal necessity.

"Necessary" and "possible" are future-relative readings: they quantify over legal computations from the current observation, not timeless logical worlds. The temporal operators and path quantifiers do different work: F versus G means eventuality versus invariance along a continuation; E versus A means some versus all permitted continuations.

The four core combinations and their checking dualities are:

$EF\,p$: possibly eventually p, $\quad AF\,p \equiv \neg EG\,\neg p$: on every future, eventually p,
$EG\,p$: p always on some future, $\quad AG\,p \equiv \neg EF\,\neg p$: p holds on every future at every position.

These are exact CTL dualities. A ranges over all legal futures, while G ranges over all positions along one future; their English glosses must keep those scopes distinct. The identities support counterexample checking: test $AG\,p$ by searching for $EF\,\neg p$, and test $AF\,p$ by searching for $EG\,\neg p$. A witness refutes the universal property; absence of one establishes it only within the finite checked model.

9.3.1 Least upper bounds and greatest lower bounds

Eventuality and strong until choose the least solution of their recursive equations. Over the lattice of state sets, begin with the empty set and repeatedly add states justified by the previous approximation:

$$\mathrm{Sat}(EF\,p) = \mu Z.(P \cup \mathrm{Pre}_\exists(Z)) = \bigcup_{n\geq 0} Z_n, \qquad Z_0 = \varnothing.$$

The same construction gives existential until:

$$\mathrm{Sat}(p\ EU\ q) = \mu Z.(Q \cup (P \cap \mathrm{Pre}_\exists(Z))).$$

AF and strong AU are also least fixed points, using universal predecessor. A cycle that never reaches the target therefore adds nothing to the least solution.

Invariance and weak until make the dual choice. They begin with all candidate states and remove unsupported states, producing the greatest fixed point—the greatest lower bound of descending approximations:

$$\mathrm{Sat}(EG\,p) = \nu Z.(P \cap \mathrm{Pre}_\exists(Z)) = \bigcap_{n\geq 0} Y_n.$$

LeanFM implements these choices with cycle-aware searches rather than by materializing every approximation set. An unresolved revisit is false for `EF`, `EU`, `AF`, and strong `AU`, selecting the least fixed point. A predicate-preserving revisit is true for `EG` and weak until, selecting the greatest fixed point. `AG` is checked as $\neg EF\,\neg p$, and `AF` as $\neg EG\,\neg p$. Thus least upper bound and greatest lower bound are paired implementation choices; neither describes every CTL operator by itself.

9.3.2 Current asymptotic cost

The smaller model boundary is not itself a faster model-checking algorithm. Let V be the number of reachable observations, E the successor edges, L the number of nodes in the checked CTL formula, and P the number of simple paths inspected by a search. The current Lean implementation carries a path-local `List` of visited observations. Membership in that list is linear in the current path length, and completed subgraphs are not memoized between branches. A single temporal search therefore costs $O(E_P + VP)$ time, where E_P counts edge visits across the inspected paths, and $O(V + L)$ stack and path memory. P can be exponential—and in dense graphs worse than 2^V—even though the input graph contains only V states. On an acyclic tree, where each state has one path from the root, the list membership overhead alone is bounded by $O(V^2)$ and is often smaller for a shallow tree.

Read this as: “the present checker may revisit the same state once for every different path that reaches it; its cost is governed by paths, not merely by states and edges.” Formula nesting may repeat those searches because subformula results are also not cached.

The intended indexed evaluator first assigns stable state identifiers, stores predecessor and successor adjacency, and memoizes the set satisfying each subformula. Fixed-point worklists then give a target bound of $O(L(V + E))$ time and $O(LV + E)$ space for explicit finite CTL checking. That is an implementation target, not the complexity of the current code. The target must be accompanied by equivalence tests against the small recursive evaluator and by measurements on specified graph families.

No comparison with TLC follows from these bounds. LeanFM often constructs a smaller observable-event graph because it deliberately omits irrelevant private variables; a carefully abstracted TLA+ model may omit them too. A claim that LeanFM is faster than TLA+ requires the same

protocol, properties, finite bounds, hardware, checker settings, and reported state/edge counts. Until that experiment exists, the book claims only a different modeling boundary, not a performance victory.

An observation is immutable once recorded. A value changes only by moving forward to a successor observation; it is never updated retroactively inside the current state. This does not mean that values are monotone—a queue may grow and later shrink—but every change occurs on a forward transition.

Formula	Reading	Typical use
$EF\,p$	possibly, eventually p	temporal possibility / witness
$AF\,p$	necessarily, eventually p	termination / response
$EG\,p$	possibly, always p	viable invariant path
$AG\,p$	necessarily, always p	safety invariant
$EX\,p$	possibly, next p	enabled successor / one-step witness
$AX\,p$	necessarily, next p	required one-step consequence
$(p\ AU\ q)$	every path keeps p until q	constrained progress
$(p\ EU\ q)$	some path keeps p until q	possible progress
$(p\ AW\ q)$	every path: p until q, or p forever	necessary weak until
$(p\ EW\ q)$	some path: p until q, or p forever	possible weak until

For example, $AF\,p$ is not merely "eventually p" and $EF\,p$ is not a weaker spelling of the same linear claim. The first requires p on every legal future generated from the current trace prefix; the second needs one legal future as a witness. Branching is part of the requirement.

Strict CTL requires every temporal operator to be governed by A or E. CTL* is a more general branching-time language in which state formulas and path formulas can be nested more freely; LeanFM does not implement CTL*. This matters when translating prose: not every arbitrary alternation of "on all paths" and "eventually" is a formula of the small CTL datatype used here.

9.4 Projecting to LTL-style F and G

Requirements often want the simpler LTL surface forms Fp and Gp. LeanFM may display or accept those forms only with an explicit path-policy boundary. On a selected linear execution, the branching coordinate has already been fixed, so AFp and EFp both project to Fp, while AGp and EGp both project to Gp. Under the common model-checking convention that an LTL formula must hold on every initial path, Fp corresponds to AFp and Gp corresponds to AGp.

Consequently, presenting frequently used AF and EG obligations as F and G is a deliberate interface reduction, not a CTL equivalence on a branching model. The durable artifact retains the exact originating CTL formula so the checker can still use existential witnesses, negation, and the correct counterexample search.

9.5 Persistence and recurrence

Nested CTL operators express two common requirements:

$$AF\,AG\,p, \qquad AG\,AF\,p.$$

Read the first, symbol by symbol: “On every path, eventually reach a state from which, on every path, p always holds.” This is eventual permanent safety: for example, eventually finish key rotation and never again accept the old key. Read the second: “On every path, at every state, on every path from that state, eventually p holds.” This is universal recurrence, often pronounced “always eventually p”: for example, service keeps returning to a ready state.

For witness searches their duals are:

$$\neg(AF\,AG\,p) \equiv EG\,EF\,\neg p, \qquad \neg(AG\,AF\,p) \equiv EF\,EG\,\neg p.$$

Read: “Failure of eventual permanent safety means some path always retains some future reaching not-p. Failure of universal recurrence means some future reaches a state with some path on which not-p holds forever.”

The corresponding existential forms are $EF\,EG\,p$ and $EG\,EF\,p$. Read: “Some path eventually reaches a state with some path keeping p forever,” and “some path always visits states from which some path eventually reaches p.” In the second formula those eventual witnesses may leave the outer path; it does not require infinitely many occurrences of p on that same path.

These forms cover useful LTL-style persistence and recurrence, not all LTL. Under universal LTL path semantics, GFp corresponds to $AGAFp$ on the usual total transition model. But $AFAGp$ is stronger than pathwise FGp: it requires eventually reaching a state where *all* continuations stay safe, not merely a separate stabilization point on each path. Terminal states require the same explicit continuation convention used by the core checker.

9.6 Knowledge, obligation, and permission

General modal logic need not have only one accessibility relation. A multimodal model can use $\Box_t/\Diamond_t$ for temporal reachability, $K_a/\widehat{K}_a$ for what actor a knows or considers possible, and a different pair for permission or obligation. Dynamic logic can index modalities by programs. Tense logics can add past-directed relations. These systems share the Kripke-style idea of evaluating modal formulas through accessibility, but their modalities are not interchangeable.

LeanFM’s CTL A and E range only over legal forward computations generated by the current model. They do not mean:

- probability one versus positive probability;
- known versus merely believed by an actor;
- obligatory versus permitted by a policy; or
- physically necessary versus logically possible.

Knowledge and policy relations are documented modelling techniques, not built-in operators. Projects may define them explicitly using the relations below. The Kerberos video [7] illustrates their use; it does not verify the protocol.

For knowledge, let W be the declared set of legal histories and $\mathrm{obs}_a(h)$ actor a’s local observation of history h:

$$K_a\varphi(h) \iff \forall h' \in W.\ (\mathrm{obs}_a(h) = \mathrm{obs}_a(h') \Rightarrow \varphi(h')).$$

Read: “Actor a knows φ at history h if and only if, for every history h' in W, if a observes the same thing in h and h', then φ holds at h'.” Observations must declare visible messages, private keys,

clock information, and retained memory. With the current history included in W, knowledge is factive: what is known is true in that model. Belief needs a different relation; it must not inherit this guarantee accidentally.

$$\widehat{K_a}\varphi \equiv \neg K_a \neg \varphi.$$

Read: "a considers φ possible exactly when a does not know that φ is false." This is epistemic possibility, not temporal reachability.

For policy, let $\mathrm{Ideal}_a(h)$ be the declared policy-compliant futures from h, and let ψ be a condition on a whole future path:

$$O_a\psi(h) \iff \forall \pi \in \mathrm{Ideal}_a(h).\ \psi(\pi), \qquad P_a\psi(h) \iff \exists \pi \in \mathrm{Ideal}_a(h).\ \psi(\pi).$$

Read: "a is required to satisfy ψ at h exactly when every ideal future π satisfies ψ; a is permitted to satisfy ψ exactly when some ideal future satisfies ψ." Require a nonempty set of ideal futures: contradictory policies must be reported, not certified by vacuous quantification. Actual legal executions may violate policy; detecting that violation is part of the analysis. These are obligations under a named policy, not judgments of morality.

$$P_a\psi \equiv \neg O_a \neg \psi.$$

Read: "ψ is permitted exactly when its negation is not required." This duality applies only when both operators use the same declared policy relation. For example:

$$O_a F\ \mathrm{audit}(r)(h) \iff \forall \pi \in \mathrm{Ideal}_a(h).\ \exists i \geq 0.\ \mathrm{audit}(r, \pi_i).$$

Read: "a must eventually audit request r at h exactly when, for every ideal future π, there is a position i, zero or later, where r is audited." Here F is a path operator; a project implementation must distinguish state formulas from path conditions.

Temporal, epistemic, and policy quantification can then be combined without confusing their domains. A legal future may be possible, unknown to Alice, and forbidden by policy at the same time. Probability, belief, strategic ability, and program-indexed modalities remain separate design work; adding named relations does not implement every modal logic.

Past predicates remain useful for security: a ticket must have been issued before it is accepted. Compile bounded past conditions into forward monitor state where an equivalence can be established. Evaluate "previously issued" before updating the monitor with the current event. On concurrent histories, declare whether "before" means causal ancestry or a chosen linearization; screen order is not evidence. Unbounded past logic is not thereby implemented.

Symbolic uncompromised-key assumptions do not establish computational cryptographic security. General knowledge and policy checking is deferred.

9.7 Possibility means preparedness

There is a practical consequence to treating possibility as a first-class temporal mode. When a requirement declares $EF\,p$—"maybe possible p"—it does more than acknowledge that a model checker can find an accidental path to p. It says that p is a supported future for which the system must prepare.

A `possibly` obligation therefore contains three linked artifacts:

1. a legal grammar branch and witness trace reaching the condition;
2. a handling plan in `Requirements.lean`; and
3. generated implementation code whose justification cites that possibility requirement.

For example, "a bakery delivery may fail" requires a decodable `DeliveryFailed` message, a legal failure branch, and code that performs the specified refund or notification behavior. Removing every failure branch does not implement reliability; it makes the real failure trace out-of-spec.

Possibility is distinct from probability. A possible condition may have a small estimated probability, an expected distribution, or no known probability at all. Conversely, assigning positive probability to a condition makes its branch possible. The handling obligation comes from supported possibility, not from whether the estimated percentage crosses a threshold.

The separate operational obligation applies to $EG\,p$—"p always in some cases." It identifies at least one future on which p persists. The system must pay attention to the conditions that keep that future viable: monitor them, expose their loss, and, where required, control choices that would destroy the invariant path. $EG\,p$ alone does not promise that every execution preserves p; that stronger claim is $AG\,p$.

9.8 Strong implication and the Chewbacca argument

The name "Chewbacca defense" is a cultural reference to the *South Park* episode "Chef Aid," whose courtroom parody uses an irrelevant claim about Chewbacca to distract from the conclusion at issue. The reference supplies a memorable example of non sequitur; it is not a technical term originating in modal or temporal logic.

Material implication is weak in the precise sense that $p \to q$ is true whenever p is false. That is useful inside a state, but it permits a vacuous global claim when no reachable state ever satisfies the antecedent. LeanFM names the non-vacuous temporal combination

$$p \Rightarrow_s q \quad \stackrel{\text{def}}{=} \quad EF\,p \;\wedge\; AG(p \to q).$$

The first conjunct requires a legal future witnessing p. The second requires that, necessarily always, every reachable occurrence of p also satisfies q. Together they imply $EF\,q$: the p witness cannot avoid q.

The deliberately absurd test uses

$$p = \text{"Chewbacca lives on Endor"},$$
$$q = \text{"you must acquit"}.$$

A Chewbacca argument cannot obtain q merely by mentioning p. To enter the requirements as strong implication, the model must admit a reachable p witness and establish $AG(p \to q)$ over every permitted continuation. If the observable grammar supplies no relation between residence and acquittal, that second obligation has no justification and is rejected. Logic cannot determine the meaning or relevance of arbitrary English propositions by itself; the useful point is that neither

a missing antecedent nor a single rhetorical occurrence of it can make the formal implication pass vacuously.

In Lean the definition is direct:

```
def stronglyImplies (p q : CTL S) : CTL S :=
  CTL.and (CTL.ef p) (CTL.ag (CTL.implies p q))
```

The path quantifier is part of the until operator. We write `AU` and `EU` as binary algebraic operators, not an outer `A` or `E` wrapped around a bracketed until formula. Consequently a subexpression never changes meaning merely because another expression surrounds it:

```
def boundedUntilTerminal : CTL Observation :=
  atom withinCapacity AU atom isTerminal

def someAuthorizedContinuation : CTL Observation :=
  atom hasValidAuth EU atom isSucceeded
```

9.9 Weak until: AW and EW

Strong until requires the right operand eventually to occur. Weak until permits it never to occur, provided the left operand continues to hold. Thus

$$p\ AW\ q \text{ : on every future, } p \text{ holds until } q\text{, or } p \text{ remains forever,}$$
$$p\ EW\ q \text{ : on some future, } p \text{ holds until } q\text{, or } p \text{ remains forever.}$$

Read the first line as: “on every future, p holds until q, or if q never occurs, p remains forever.” Read the second with “on some future.” AW and EW are the universal-path and existential-path weak-until operators.

The boundary state satisfying q need not also satisfy p; p is required only before q. On a future where q never occurs, every state must satisfy p. This makes weak until useful for requirements such as “necessarily remain available until shutdown, while shutdown is permitted but not required.”

For existential paths, $p\,EW\,q$ is equivalent to $(p\,EU\,q) \vee EG\,p$. The analogous-looking $(p\,AU\,q) \vee AG\,p$ is *not* a correct definition of AW on a branching system. Some futures may reach q while other futures preserve p forever; neither global disjunct need hold even though every individual future satisfies weak until. LeanFM therefore implements `AW` directly.

```
def serviceMayRunUntilShutdown : CTL Observation :=
  atom available AW atom shutdown

def someWorkerCanRemainReady : CTL Observation :=
  atom ready EW atom retired
```

This convention also makes tree structure explicit. In `r && (p AU q)`, the right operand is the complete `AU` expression; there is no context-sensitive reinterpretation of `[p U q]`.

For a request event carrying identifier r, an expected response property might say that on every continuation a matching success or failure eventually occurs. Authentication might say that every successful use of a session has an earlier issuance event with the same session identifier.

9.10 Past claims

CTL is future-facing at the current state, but security and protocol claims often refer to evidence in the past: “this capability was previously issued” or “this response matches an earlier request.” Since states are trace prefixes, past facts can initially be expressed as predicates over the accumulated trace. This gives a clear baseline without prematurely committing the surface language to a particular past-time temporal logic.

There is still an open design question. Explicit past operators may make properties more readable and allow better checking algorithms. The project should compare trace predicates, CTL with past operators, and compilation of finite-history monitors into state. The criterion is maintainability of specifications as the event schema evolves.

9.11 Fairness and time bounds

AF does not by itself distinguish a participant that is perpetually denied a turn from one that violates the protocol. Fairness assumptions must therefore be named, scoped, and visible in theorem statements. Similarly, “eventually” is not “within 200 milliseconds.” Quantitative bounds belong in an explicit timed layer rather than being smuggled into timestamps and informal prose.

Chapter 10

Feature Completion as Positive and Negative Proof

A request to show a message, list source, accept a transaction, render a diagram or produce an MP4 states a desirable observation. The change can also introduce stale state, disclosure, resource exhaustion, weaker authentication, corruption or success without an artifact. Completion therefore requires a paired argument:

1. a *positive invariant* states what the system must do; and
2. applicable *negative invariants* state what the system must never do.

Review before declaring completion, in proportion to the boundary crossed: button spacing rarely needs a cryptographic audit; filesystem access needs authorization and path analysis. A material risk without an accepted control blocks completion.

10.1 The logical shape of a feature

Let $good_f$ mean the observable contract of feature f, and let $bad_{f,1}, \ldots, bad_{f,n}$ name reachable failures introduced or exposed by that feature. A common safety-shaped completion condition is

$$AG\, good_f \;\wedge\; \bigwedge_{i=1}^{n} AG\, \neg bad_{f,i}.$$

The first conjunct is positive: the requested behavior remains true throughout every permitted execution. Each remaining conjunct is negative: no permitted execution reaches a forbidden observation. The equivalent counterexample form

$$AG\, \neg bad \;\equiv\; \neg EF\, bad$$

is operationally useful. A negative test tries to construct the $EF\, bad$ witness: another user's project path, a stale response after a project switch, an oversized upload, a duplicate response, or a secret in a log. Finding such a witness refutes the claimed completion.

Not every feature is purely invariant. “A submitted request eventually gets a terminal response” is a liveness obligation such as $AG(submitted \rightarrow AF(complete \vee failed))$. A feature review must not replace liveness with safety or hide fairness and availability assumptions. Positive and negative requirements classify intent; the temporal operator must still match the claim.

10.2 Stability review before security review

Before editing code, identify the existing invariant and review the applicable ways an environment change could break it:

Regression. Which accepted behavior, file format, route, proof, or UI state can this change invalidate? Preserve conforming behavior and add a regression test at the narrowest durable boundary.

Failure. What happens when a dependency, renderer, compiler, provider, filesystem operation, or network request fails? Failure must become an explicit terminal observation rather than a false success or endless wait.

Concurrency. Can a late response, simultaneous session, queued prompt, or overlapping write be attributed to the wrong project or revision? Correlation identity and revision checks must survive reordering.

Resources. What bounds memory, bytes, queue occupancy, recursion, process time, media dimensions, and response size? A new feature must not silently turn a finite resource contract into an unbounded one.

Durability and recovery. Which state is authoritative after restart or partial failure? Atomic writes, rebuildable indexes, and project-local ownership make recovery inspectable.

Compatibility. Which clients, stored revisions, generated artifacts, and build environments must continue to work? Version or migrate a changed representation rather than reinterpreting old bytes silently.

Retain concrete invariants at the changed boundary, briefly recording when a dimension is non-material.

10.3 Security as forbidden observable behavior

“A user never reads a project they do not own” states protected behavior; “use authentication” only names a mechanism. Before selecting a control, identify:

1. the trust boundary crossed by the feature;
2. the asset being protected;
3. the attacker’s relevant capability; and
4. the forbidden observable outcome.

Applicable forbidden behaviors include unauthorized access, cross-user or cross-project disclosure, secret persistence or logging, injection, path or symlink escape, replay, request/response confusion, privilege expansion, executable media, resource exhaustion and fail-open behavior. State testable invariants, such as “a source path absent from the authorized project inventory never returns bytes.”

Map each material negative invariant to a control—canonical-path confinement, authenticated project identity, an allowlisted inventory, a byte limit or an atomic revision comparison—and a negative test or proof. A mechanism alone does not establish the forbidden transition’s absence.

10.4 The feature-completion ledger

For each feature, retain the following compact traceability record in the typed requirements and implementation mapping:

Field	Required content
Prompt	The stakeholder request whose completion is being claimed.
Positive claim	Observable always or liveness property, with actor, scenario, boundary, and revision.
Negative claims	Applicable never or always-not properties; omitted families are explicitly judged non-material.
Controls	Concrete implementation mechanisms mapped one-to-one or many-to-one to the negative claims.
Evidence	Positive conformance tests plus negative tests, proofs, generated traces, and named checks.
Residual risk	Accepted limitation, assumption, deferred mitigation, or an explicit completion blocker.

Both sides need evidence: forbidding failures does not demonstrate the requested feature, and a successful demonstration does not rule out disclosure or corruption.

10.5 Worked example: the project Source view

Consider the request “show implementation files in the Source tab.” The positive claim is that selecting an authorized project lists its implementation files and selecting one readable UTF-8 file displays its content. Project switching clears the old selection, so a late response cannot repopulate the new project with stale content.

The feature crosses a filesystem boundary. Project source, ignored account records, repository metadata, and provider credentials are protected assets. A user can choose a URL path and can guess names that the visible tree never listed. Applicable negative invariants therefore include:

- a user never reads another user’s project source;
- a guessed ignored or metadata path never returns bytes;
- a path or symlink never escapes the authorized source root;

- a binary or oversized file never becomes an unbounded text response;
- a project switch never retains the previous project's source state; and
- an empty source directory never accompanies a claim that implementation succeeded.

Controls are an authenticated explicit project ID, canonical root resolution, exact source-inventory membership, UTF-8 reads, a finite response limit and project-keyed React state. Negative tests request another account's file and a guessed `.git/config`; both return not-found without revealing existence. The positive test creates, lists and opens a project-local file and checks its exact content.

10.6 Worked example: chat auto-scroll

Even a small convenience has stability boundaries. The positive invariant is that initial conversation load and every newly rendered user, assistant, or failure message reveal the bottom of the chat message pane. The implementation must scroll that pane rather than the entire page, must not move the independent project sidebar, and must reset with the project-keyed chat component. No new authorization boundary is crossed, so a full threat model would be noise; the project-isolation and stale-response invariants remain material.

10.7 When the result is not complete

An unobservable or untested positive claim, missing durable negative requirement, uncontrolled risk or material control without a negative test or proof leaves the ledger incomplete. So does a liveness claim missing its fairness or availability assumptions, failed or skipped validation, or a check of a substitute artifact. Report the blocker and return to the relevant gate; stakeholder-visible residual risk requires acceptance there. A green build or persuasive diagram does not fill these gaps.

Chapter 11

Lean as the Rendered Language

The user-facing notation should render to a conservative subset of Lean. There is no second opaque specification file: domain declarations are Lean definitions, expected behaviours are Lean values, and requirements are Lean propositions or checked queries. Rendering the proposed scenario DSL into Lean is a **future interface**; its listings are **proposed notation**. Current **executable code** is the typed LeanFM API, and a **Lean proof** claim applies only when its theorem and proof are checked in the declared build.

Throughout this book, every displayed mathematical formula is accompanied by a literal English reading that accounts for each operator, index, bound, and new glyph. The reading is part of the notation discipline: it must map back to the symbols one-to-one rather than merely paraphrase the formula's intent.

```
def checkout : Grammar CheckoutEvent :=
  request >>> (accepted <|> rejected)

def eventuallyAnswered : CTL Observation :=
  CTL.ag (CTL.implies requestPending
    (CTL.af (CTL.or requestAccepted requestRejected)))
```

Most semantic machinery should be imported from a fixed library. A system author normally supplies only:

- typed actors, identifiers, event kinds, and payloads;
- parameterised grammar definitions;
- projections from trace prefixes to observable propositions;
- named temporal requirements and proofs or decidable checks.

This boundary matters for automation. Generated Lean remains small enough to review, while library proofs and algorithms are reused rather than regenerated.

11.1 An exploratory Lean proof fixture

`book/logic.lean`, reproduced completely below, explores a regularized assignment for $1+2+3+\cdots$. For each finite cutoff n, its theorem equates the triangular prefix $n(n+1)/2$ plus the defined correction

`Tail_S(n)` with $-1/12$. This identity under that definition does not assert convergence of ordinary partial sums. The optional example is separate from protocol/trace semantics and excluded from `make book`; `lake env lean book/logic.lean` stops here because Mathlib is absent. Its inclusion therefore claims no successful proof check. A validity objection should identify a circular dependency, unchecked assumption or interpretation mismatch; kernel acceptance establishes only the stated theorem under its imports and axioms.

```
import Mathlib

set_option linter.style.whitespace false
set_option linter.unusedVariables false

/- Justify the origin of F using real differentiation. -/
noncomputable def Ar (x : ℝ) : ℝ := 1 / (1 - x)
noncomputable def Fr (x : ℝ) : ℝ := 1 / (1 - x)^2
def F (x : ℤ) : ℚ := 1 / (1 - x)^2

theorem deriv_Ar_eq_Fr (x : ℝ) (hx : x ≠ 1) :
    deriv Ar x = Fr x := by
  have h : 1 - x ≠ 0 := sub_ne_zero.mpr hx.symm
  have hd := ((hasDerivAt_id x).const_sub 1).inv h
  simp only [id_eq, neg_neg] at hd
  change HasDerivAt
    (fun y : ℝ => (1 - y)⁻¹)
    (1 / (1 - x)^2) x at hd
  unfold Ar Fr
  simpa only [one_div] using hd.deriv

theorem F_eq_Fr (x : ℤ) :
    (↑(F x) : ℝ) = Fr ↑x := by
  unfold F Fr
  norm_cast

/- B just needs to work, but this was how we found it.  -/
def Tail_B (n : ℕ) : ℚ :=
  ((2 * (n : ℚ) + 1) * (-1)^n) / 4

theorem whyTailB (n : ℕ) :
    Tail_B 0     = F (-1) ∧
    Tail_B (n+1) = Tail_B n - ↑(n+1) * ↑(-1)^n := by
  rw[F]
  unfold Tail_B
  norm_num
  ring

def Sum_B (n : ℕ) : ℚ := F (-1) - Tail_B n
def Bf (n : ℕ) : ℚ := (Sum_B + Tail_B) n
def B : ℚ := Bf 0

def Sum_S (n : ℕ) : ℚ :=
  n * (n + 1) / 2

theorem selfSimilarSumS (n : ℕ) :
    (Sum_S  (2*n) - Sum_B  (2*n) = 4*Sum_S  n) :=  by
  rw[Sum_B]
  rw[Sum_S]
  unfold F Sum_S Tail_B
  norm_num
  linarith

def Tail_S (n : ℕ) : ℚ :=
  (4 * Sum_S n - (Sum_S (2*n) + Tail_B (2*n))) / 3 - Sum_S n

theorem whyTailS (n : ℕ) :
```

```
    Tail_S n =
      (4*Sum_S n - (Sum_S (2*n) + Tail_B (2*n)))/3 - Sum_S n := by
  rw[← selfSimilarSumS n]
  rw[Tail_B]
  unfold Tail_S Sum_S Sum_B Tail_B F
  norm_num
  ring_nf

def Sf (n : ℕ) : ℚ := Sum_S n + Tail_S n
def S : ℚ := Sf 0

theorem selfSimilarS (n : ℕ) :
    (Tail_S (2*n) - Tail_B (2*n) = 4*Tail_S n) ∧
    (S            - B            = 4*S) := by
  rw[whyTailS]
  rw[B]
  unfold S Sf Tail_S Bf Tail_B Sum_S Sum_B F Tail_B
  norm_num
  ring_nf

/-- The final Goal, now explicitly powered by the calculus/geometry proofs -/
theorem Goal (n : ℕ) :
    -1/12 = n * (n + 1) / 2 + Tail_S n := by
  rw [whyTailS n]
  unfold Tail_B Sum_S
  norm_num
  ring_nf
```

For a LeanFM runtime judgment, define the double turnstile $\models$ relative to the declared transition model and initial state. It means "at every reachable state, the premises require at least one of these conclusions." Write it in a rule-like form, familiar from Prolog:

$$a \wedge b \wedge \cdots \wedge w \;\models\; x \vee \cdots \vee z.$$

Its executable definition is that the following CTL formula holds at the declared initial state:

$$AG\big((a \wedge b \wedge \cdots \wedge w) \rightarrow (x \vee \cdots \vee z)\big).$$

Read: "at every reachable state, if a and b and so on through w all hold, then x or so on through z holds." A means every continuation; G means every position along it, including the current state. This is entailment within the declared runtime model, not a redefinition of general entailment across every possible model. The model and initial state must accompany the judgment. The $\wedge$ symbols collect premises with "and"; the $\vee$ symbols collect alternative conclusions with "or." The conclusion need not be the same in every state.

For code, represent the left side as a list of premises and the right side as a list of conclusions. A counterexample is a reachable state where every premise is true and every conclusion is false:

```
counterexample(state, premises, conclusions) :=
  premises.all(p => evaluate(p, state)) &&
  !conclusions.any(q => evaluate(q, state))

entails(successors, initialState, premises, conclusions) :=
  !anyReachable(successors, initialState,
    state => counterexample(state, premises, conclusions))
```

The evaluator and successor relation come from the declared model. A sampled trace checks only that sample, not all reachable states.

The same counterexample condition can be constructed symbolically:

$$\neg EF(a \wedge b \wedge \cdots \wedge w \wedge \neg x \wedge \cdots \wedge \neg z).$$

Read: “there is no reachable state with a and b and so on through w, and not x and so on through not z.” This is exactly the AG implication above, using $AG\,p = \neg EF\,\neg p$, read “always p is no reachable not-p.”

With a single conclusion, the analogy is the Prolog rule `x :- a, b, ..., w`: x follows when all its premises hold. The displayed disjunctive conclusion is more general than an ordinary single-head Prolog rule; $\models$ does not specify Prolog’s execution or search order. The compact notation $\Gamma \models p$, read “Gamma semantically entails p,” collects the premises into a context Γ and names the conclusion p. The distinction remains: $\Gamma \vdash p$, read “Gamma derives p,” asserts a derivation. Our runtime judgment checks that no reachable state makes all the premises true and p false. General semantic entailment uses all models of the assumptions instead; it must not be confused with this explicitly scoped CTL test.

For the proof-theoretic background behind derivations, natural deduction, sequent calculus, normalization, and cut elimination, see Mancosu, Galvan, and Zach’s *An Introduction to Proof Theory* [8].

11.2 From specification to service

Lean can host executable code, including a small web server. That makes the formal artifact part of the running project rather than a document beside it. The design target is for typed definitions to drive validation endpoints, test-trace generation, protocol schemas, visualisations, and runtime monitors; current coverage is partial, as Chapter 14 records. Extraction and integration choices must preserve the distinction between a checked behavioural model and an unverified infrastructure layer.

Chapter 12

Scope and Open Questions

The implementation backlog and audit findings live in `CRITIQUE.md`. This chapter distinguishes current evidence from open technical questions using the labels defined in Chapter 1; proposed backlog items are **future interfaces**, not implemented behavior.

Self-hosting consolidation

The canonical requirements entry point now composes the existing LeanFM ChatUI specification with a proposed revision-bound requirements-to-PR workflow. The Worker protocol remains an example, not a specification of the whole product. `make selfhost-check` validates the typed composition and regenerates `SELF_HOSTING_REVIEW.md`, including its interaction diagram and state machine. A component registry and fresh-session handoff preserve decisions without requiring conversation history.

The consolidation proposes checker outcomes, acceptance invalidation, probability interpretations and PR evidence. Earlier proposals also cover modal indexes; built-in knowledge and policy checking is now deferred. These records are not implemented features or proofs. Error protocols, formula semantics and complete traceability remain prerequisites for a complete one-shot specification.

12.1 Proof obligations

The initial metatheory should make the following questions precise:

- Does a projected local behaviour faithfully implement its part of the global grammar?
- Under what finiteness restrictions is CTL checking decidable and executable?
- Which guards can be compiled into finite monitor state?
- When does modular composition preserve an established temporal property?
- Which fairness and timing assumptions are required by a liveness theorem?
- Which observation and policy relations justify each knowledge, permission, and obligation assertion, and which bounds make their combination with CTL executable?

Part II

Security Protocol Exercises from the CSP Approach

Chapter 13

From CSP Roles to Multiparty Grammars

This part revisits examples from Ryan et al. [12] and its official Casper companion scripts. The goal is not to transliterate CSP operators. It is to use well-known security protocols as demanding exercises for the grammar-based method: multiple roles, fresh values, encrypted components, an active intruder, calculated knowledge sets, agreement, and messages whose legality depends on the preceding trace.

Status: this chapter is a design exercise, not an account of implemented LeanFM features. Its ‘RoleSpec‘, ‘WireTerm‘, ‘composeRoles‘, and attacker-deduction listings are proposed notation; the current executable model has ‘GrammarExpr‘ and finite trace/CTL checks, but no role-projection or composition engine, symbolic byte algebra, or attacker-knowledge evaluator. The checklist below is a set of proposed library requirements, not implemented capabilities.

13.1 Can actors really be specified separately?

Yes, provided composition is defined as synchronization over the same observable events rather than as concatenation of independent scripts. Each actor accepts a language over its own projection of the global event alphabet.

```
structure RoleSpec where
  actor    : Actor
  accepts  : Grammar (ProjectedEvent actor)
  initialKnowledge : Knowledge actor

def composeRoles (roles : List RoleSpec) : GlobalGrammar :=
  synchronize roles where
    sendReceiveAgree := sameEventId && sameBytes && oppositeEndpoints
```

For a global trace t, let $t\restriction_a$ retain events sent or received by actor a. The composition accepts exactly when every local role accepts its view:

$$L(\mathrm{composeRoles}(R_1, \ldots, R_n)) = \{t \mid \forall i,\ t\restriction_{a_i} \in L(R_i)\}.$$

Read this as: “the language L accepted by the composition of roles R_1 through R_n is the set of global traces t such that, for every role index i, trace t restricted to actor a_i belongs to the language

accepted by role R_i." Here n is the number of roles, i is one role's index, the vertical bar means "such that," and $\upharpoonright$ means "restricted to."

One global message is therefore not duplicated into unrelated send and receive events. It has a single identity, byte string, sender, and recipient; both endpoint grammars constrain it. An unmatched send, an unexpected receive, or disagreement about bytes makes the intersection empty at that trace prefix.

13.1.1 Local state without global implementation state

A role may bind parameters from the events it observes. Those bindings are a parser environment for the trace, not a claim about implementation memory. A role can say "the nonce in this reply equals the nonce I previously sent" without specifying a table, variable layout, or retry mechanism.

13.2 A transport-neutral byte algebra

Security claims need more than message names. They need an algebra that says which bytes expose which values. We still need not choose HTTP, protobuf, ASN.1, or a particular cipher suite.

```
abbrev Bytes := ByteArray

inductive WireTerm where
  | atom    : Name -> Bytes -> WireTerm
  | tuple   : List WireTerm -> WireTerm
  | encrypt : KeyRef -> WireTerm -> WireTerm
  | sign    : KeyRef -> WireTerm -> WireTerm

structure WireEvent where
  timeAt : ClockTimestamp
  id     : EventId
  priorIds : List EventId
  session : SessionId
  task   : TaskId
  from   : Actor
  to     : Actor
  term   : WireTerm
  bytes  : Bytes
  encodedBy : bytes = encodeCanonical term
```

At this level, encryption is a symbolic constructor with an encoding into bytes. The intruder may concatenate, split visible tuples, encrypt with known keys, and decrypt only with a derivable inverse key. A concrete refinement must replace this idealized algebra with a transport and cryptographic implementation whose security assumptions justify the abstraction.

13.3 Yahalom as three separate actors

Yahalom is an excellent first exercise because an initiator, responder, and trusted server exchange fresh nonces and receive a session key. The companion model states secrecy for the session key and agreement claims involving the nonces and key.

Let K_A and K_B be long-term keys shared by the server with Alice and Bob, N_A and N_B fresh nonces, and K_{AB} the proposed session key. The following actor grammars use symbolic wire terms whose canonical encodings are byte strings.

13.3.1 Alice, the initiator

```
def aliceRole (alice bob server : Actor) (ka : SharedKey) : RoleSpec :=
  role alice "Yahalom initiator" do
    let na <- fresh Nonce "na"

    from_ alice; to_ bob
    send (atom "initiator nonce" na.bytes)

    from_ server; to_ alice
    receive (encrypt ka.ref
      (tuple [actor bob, sessionKey "kab", nonce na, nonce "nb"]))
      where_ field "na" == na
    let kab <- take SessionKey "kab"
    let nb  <- take Nonce "nb"
    let ticketForBob <- take WireTerm "ticketForBob"

    from_ alice; to_ bob
    send ticketForBob
    send (encrypt kab.ref (nonce nb))
```

Alice's `where_` clause is the context-sensitive step: the received nonce must equal the value generated earlier. The surface structure is context-free, but legality depends on the binding environment accumulated from the prefix.

13.3.2 Bob, the responder

```
def bobRole (alice bob server : Actor) (kb : SharedKey) : RoleSpec :=
  role bob "Yahalom responder" do
    from_ alice; to_ bob
    receive (atom "initiator nonce" ?naBytes)
    let na <- take Nonce "na"
    let nb <- fresh Nonce "nb"

    from_ bob; to_ server
    send (encrypt kb.ref
      (tuple [actor alice, nonce na, nonce nb]))

    from_ alice; to_ bob
    receive (encrypt kb.ref (tuple [actor alice, sessionKey "kab"]))
    let kab <- take SessionKey "kab"
    receive (encrypt kab.ref (nonce nb)) where_ field "nb" == nb
```

Bob need not know Alice's local control state. He constrains only the events visible at his boundary and the values learned from them.

13.3.3 The trusted server

```
def serverRole (alice bob server : Actor)
    (ka kb : SharedKey) : RoleSpec :=
  role server "Yahalom key server" do
    from_ bob; to_ server
    receive (encrypt kb.ref
      (tuple [actor alice, nonce "na", nonce "nb"]))
    let na  <- take Nonce "na"
    let nb  <- take Nonce "nb"
    let kab <- fresh SessionKey "kab"

    let bobTicket := encrypt kb.ref
      (tuple [actor alice, sessionKey kab])
    from_ server; to_ alice
    sendMany
      [ encrypt ka.ref
          (tuple [actor bob, sessionKey kab, nonce na, nonce nb])
      , bobTicket
      ]
```

13.3.4 Global assembly and context conditions

```
def yahalom : GlobalGrammar :=
  composeRoles
    [ aliceRole .alice .bob .server keyAlice
    , bobRole .alice .bob .server keyBob
    , serverRole .alice .bob .server keyAlice keyBob
    , intruderRole .ivo
    ]
  |> where_ (distinct [.alice, .bob, .server, .ivo])
  |> where_ (freshValuesNeverRepeat ["na", "nb", "kab"])
  |> where_ (longTermKeysKnownOnlyToOwnersAndServer)
```

The three honest roles can be authored and checked separately. Composition exposes incompatibilities: for example, if Alice expects the server's first encrypted tuple to name Bob but the server grammar could name another responder, the shared global event cannot satisfy both projections.

13.4 The attacker is another grammar plus deduction

An active network attacker observes public wire events, blocks them, replays known bytes, and emits any term derivable from current knowledge. It does not learn the plaintext of an encryption merely because its encoded bytes are observable.

```
inductive Derives : Knowledge -> WireTerm -> Prop where
  | initial : term in knowledge -> Derives knowledge term
  | tuple   : (forall x in xs, Derives knowledge x) ->
              Derives knowledge (.tuple xs)
  | project : Derives knowledge (.tuple xs) -> x in xs ->
```

```
              Derives knowledge x
  | encrypt : Derives knowledge key -> Derives knowledge body ->
              Derives knowledge (.encrypt key body)
  | decrypt : Derives knowledge (.encrypt key body) ->
              Derives knowledge (inverse key) -> Derives knowledge body

def intruderRole (ivo : Actor) : RoleSpec :=
  adversary ivo do
    observe publicWireEvents
    mayBlock
    mayReplay
    maySend fun knowledge term => Derives knowledge term
```

This is the grammar analogue of the CSP/Dolev–Yao attacker: possible hostile messages are generated from observable history and constrained by a context-sensitive derivability relation.

13.5 Knowledge and agreement become theorems

Knowledge is calculated for every actor from every accepted trace, not merely the happy-path scenario. A traditional secrecy claim is recovered by comparing that set with the actors permitted to know the value. For Yahalom:

```
theorem yahalom_session_key_knowers
    (trace : Trace WireEvent)
    (accepted : yahalom.Accepts trace)
    (honest : Uncompromised [.alice, .bob, .server]) :
    knowers (sessionKey accepted.kab) trace = [.alice, .bob, .server] := by
  -- proof uses the fixed symbolic-crypto library
  exact knowledgeClosureFromKeyOwnership accepted honest

theorem yahalom_bob_agrees_with_alice
    (trace : Trace WireEvent)
    (accepted : yahalom.Accepts trace)
    (bobDone : completed trace .bob) :
    exists run, peer run == .alice &&
      run.na == bobDone.na && run.nb == bobDone.nb := by
  exact agreementFromNonceChallenge accepted bobDone
```

The exact proof scripts depend on the library still to be built. The important design constraint is already visible: theorem statements refer only to accepted event traces, knowledge derived from those traces, and explicit compromise assumptions.

13.5.1 Put calculated knowledge and policy predicates inside CTL

For routine protocol documentation, knowledge need not begin as a new modal operator. Define ordinary state predicates by reducing the accepted event history, then place those predicates inside the existing CTL operators:

```
def knows (actor : Actor) (value : Secret) (s : ProtocolState) : Bool :=
  value in knowledgeClosure actor s.events s.compromise
```

```
def permitted (policy : Policy) (actor : Actor)
    (action : Action) (s : ProtocolState) : Bool :=
  policy.allows actor action s

def required (policy : Policy) (actor : Actor)
    (action : Action) (s : ProtocolState) : Bool :=
  policy.requires actor action s
```

These are functions that label a state. CTL then says when those labels may or must hold. The following formulas are useful documentation because each has a direct English pronunciation:

$$AG(\neg \,\mathrm{knows}(\mathit{Attacker}, K_{AB})).$$

$$EF(\mathrm{knows}(\mathit{Bob}, K_{AB})).$$

$$AG(\mathrm{accepts}(\mathit{Bob}) \rightarrow (\mathrm{knows}(\mathit{Bob}, K_{AB}) \wedge \mathrm{knows}(\mathit{Bob}, \mathrm{peer}(\mathit{Alice})))).$$

$$AG(\mathrm{knows}(a, s) \rightarrow \mathrm{permitted}(\mathit{DisclosurePolicy}, a, \mathrm{possess}(s))).$$

$$AG(\mathrm{validTicketReceived}(r) \rightarrow (\mathrm{required}(\mathit{AuditPolicy}, \mathit{Server}, \mathrm{audit}(r)) \wedge AF\ \mathrm{audited}(r))).$$

$$EF(\mathrm{knows}(\mathit{Attacker}, K_{AB})).$$

The last formula is the counterexample search for the first. This is often the most practical security check: ask the existential engine for a trace that derives the forbidden knowledge. Likewise, a universal policy claim should be checked by searching for a reachable violation:

$$AG(\mathrm{discloses}(a, s) \rightarrow \mathrm{permitted}(P, a, \mathrm{disclose}(s))) \;\equiv\; \neg EF(\mathrm{discloses}(a, s) \wedge \neg\, \mathrm{permitted}(P, a, \mathrm{disclose}(s))).$$

This state-local `knows` predicate is not the full epistemic modality $K_a\varphi$ over indistinguishable histories defined in Chapter 9. It tests whether one declared deduction closure contains a value, an executable check for many secrecy and authorization claims; use K_a when the requirement compares histories, such as anonymity or “Alice knows that Bob knows.”

13.6 Otway–Rees: guards are not optional decoration

The Otway–Rees companion model introduces a run marker M and guards that the two participants differ. Each participant encrypts its nonce, marker, and actor names under its long-term server key. The server returns separate encrypted components.

This example adds three useful `where_` obligations:

```
where_ alice != bob
where_ response.field "marker" == request.field "marker"
where_ encryptedPart.names == [alice, bob]
```

The grammar shape remains a small sequence of messages. The accepted language is not context-free once equality of an unbounded marker and actor parameters matters. Keeping these tests as named guards preserves the readable protocol skeleton while making the extra context explicit.

The example also asks whether an observer can distinguish executions after one candidate session key is substituted for another. That cannot be answered by a single knowledge set: it is a relational property over pairs of trace distributions, beyond a single CTL state formula, and should be represented as a separate theorem family.

13.7 Needham–Schroeder: public-key lookup as behaviour

The seven-message public-key example makes key discovery observable. Alice requests Bob's public key from a server, receives a signed binding, sends an encrypted nonce and identity to Bob, and later answers Bob's nonce. Bob performs the symmetric key lookup for Alice.

Actor separation is particularly valuable here:

- the key server grammar promises signed actor–public-key bindings;
- Alice's grammar accepts Bob's key only where the signature verifies and the named actor is Bob;
- Bob imposes the corresponding constraint for Alice;
- the nonce-response messages reuse previously bound values;
- the intruder grammar may substitute any derivable public key or replay any signed binding, but cannot manufacture a server signature.

```
receive (sign serverSigningKey (tuple [actor bob, publicKey "pkb"]))
  where_ verifiesWith serverPublicKey
  where_ field "actor" == bob

send (encrypt pkb (tuple [nonce na, actor alice]))

receive (encrypt aliceSecretView (tuple [nonce na, nonce "nb"]))
  where_ field "na" == na
```

This demonstrates why "all behaviour is observable" must not mean "all plaintext is public." The globally recorded event exposes endpoints, timing, kind, and wire bytes. Which symbolic values an actor can recover from those bytes is governed by the deduction relation.

13.8 What these exercises demand from the library

The three case studies produce a concrete implementation checklist:

1. actor-local grammar definitions and trace projection;
2. synchronization by common event identity and bytes;
3. typed symbolic variables, exports, imports, and prefix-dependent guards;
4. a transport-neutral term-to-byte relation;
5. explicit initial knowledge and compromise assumptions;
6. attacker closure under symbolic construction and decryption;
7. per-value knowledge sets, agreement, and relational indistinguishability theorem schemas;
8. counterexamples rendered simultaneously as a global interaction diagram and as each actor's local projection.

These are not security-specific embellishments. They are a rigorous test of whether multiparty context-sensitive grammars can remain modular while expressing the information flow on which real behavioural claims depend.

Part III

Machine Reference

Chapter 14

The LeanFM Web-Server Protocol

This chapter is the protocol reference for the LeanFM web server. It is primarily machine-oriented context for an implementation agent and may be read as a maintained portion of the system prompt. The normative behavioral source is `LeanFM/ChatUI/Requirements.lean`; the byte vocabulary is `LeanFM/ChatUI/Requirements.proto`; and transport selection is `LeanFM/ChatUI/Implementation.lean`. The tables and grammars below are generated from those files. They are not a second, hand-maintained API design.

Two versions of the protocol must not be confused. The *accepted protocol* is the complete typed grammar and intended HTTP mapping. The *running adapter* is the subset currently dispatched by `LeanFM/ChatUI/Server.lean`. A route in the accepted table is not a claim that its handler already exists.

14.1 One exchange across four representations

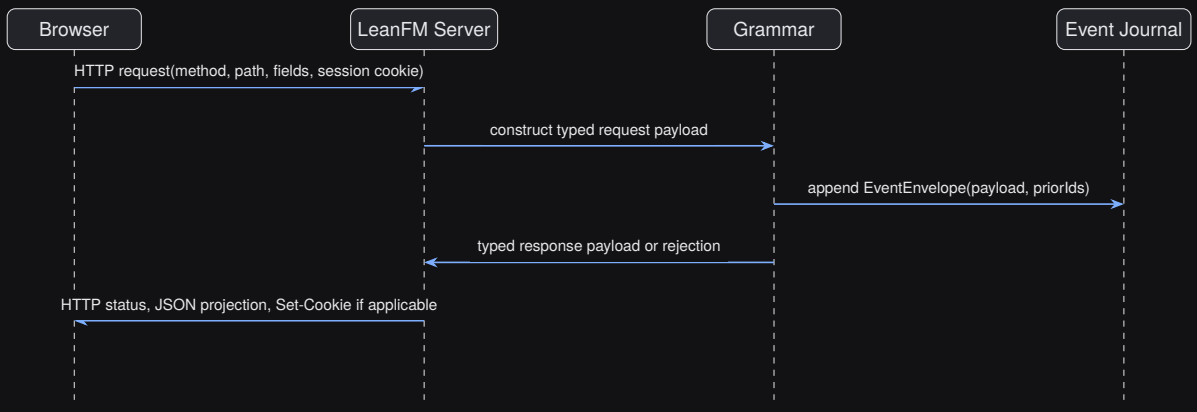


Figure 14.1: HTTP is an adapter around the typed LeanFM conversation. Source, reducer, question, and unit are recorded in `generated/web-protocol-provenance.txt`.

The figure summarizes the round trip: authenticate and decode HTTP into a grammar-permitted protobuf value, record its `EventEnvelope`, then project the selected response or rejection back to HTTP. Provider messages remain in-process, not invented HTTP calls. Protobuf fixes the typed value and stable tags; HTTP chooses the browser representation, so their bytes differ. The current

adapter uses `application/x-www-form-urlencoded` for POST requests and JSON for responses; session establishment also issues a secure cookie. The journal stores length-delimited binary `EventEnvelope`s. A future binary HTTP codec may change representation without changing message values or correlation fields. Field-level conversion follows.

14.1.1 Conservative default HTTP projection

LeanFM can generate an HTTP projection and its OpenAPI description when `Implementation.lean` does not provide an explicit binding. The task grammar supplies request/response pairing: an initiating browser-to-server message is the request, its correlated finite server-to-browser alternatives are responses, and their protobuf schemas supply the operation schemas. An explicit implementation binding always overrides the default.

The default is deliberately conservative. A field may appear in a path or query only when the implementation explicitly classifies it as a bounded, URL-safe scalar selector. Bytes, repeated fields, nested messages, secrets, large or unbounded strings, and any request whose arguments would be unnatural as a URL use `POST`; the protobuf value is carried in the request body. This prevents a convenient generator from leaking credentials, exceeding practical URL limits, or inventing ambiguous encodings. Generated encoders and decoders must preserve the protobuf value exactly.

Unless an explicit transport binding selects another codec, the default body encoder and decoder use the protobuf JSON mapping with `Content-Type: application/json`. This default is part of the generated adapter contract, not a change to the protobuf value. The server publishes the resolved OpenAPI document at `GET /openapi.yaml`; this is the default discovery endpoint for generated HTTP operations and schemas.

The generated OpenAPI operation records the resolved method and path, body content type, request schema, every grammar-permitted response schema and status, authentication requirement, correlation fields, and LeanFM requirement identifier. OpenAPI documents the transport projection; the protobuf and task grammar remain the definitions of values and legal ordering.

14.2 The conversion, field by field

For a request binding `httpRequest(method, pathTemplate)`, conversion is defined in this order:

1. Match the HTTP method exactly and split the path into decoded segments.

2. For each `{field}` in the template, take the protobuf field of that name, encode it as one path segment, and reject slashes, traversal, or a value that escapes the authenticated project root.

3. For the currently implemented POST subset, encode remaining browser input fields as UTF-8 form values. A protobuf `bytes` field is opaque bytes at the protocol boundary; a form adapter must use an explicitly declared textual encoding and must never place credentials in a URL, trace, diagnostic, or returned object.

4. Recover the authenticated user from the `leanfm_session` HttpOnly, SameSite cookie. The cookie is transport authorization, not a field silently added to every protobuf message. Its resolved user and session identifiers constrain which project-scoped message may be constructed.

5. Validate required fields and task `where_` constraints. On success, assign the envelope metadata and the oneof payload. On failure, emit the named rejection alternative when the grammar has one; otherwise return an HTTP error without pretending a successful protocol event occurred.

For a response binding `httpResponse(RequestName)`, conversion reverses the projection. The request name selects the finite legal response set. Scalar fields become JSON strings, booleans, or numbers; repeated fields become JSON arrays; nested messages become objects; and `bytes` become an explicitly identified download or Base64 field rather than accidental UTF-8. Secrets are redacted before JSON rendering. Success commonly uses 200, creation uses 201, authentication failure uses 401 or 403, invalid input uses 400, missing owned objects use 404, revision or credential conflicts use 409, and provider failure uses 502. The named response message, not the status alone, determines the grammar transition.

Before projection, request and response are independent framed events. In that limited structural sense the stream resembles a multiplexed protocol such as HTTP/2: several exchanges may be in flight and correlation is explicit. This is not a claim that the event bytes are HTTP/2 frames. Every response envelope must set `cause_id` to the exact request event identifier within the same session and task. The HTTP adapter follows that causal backpointer to recover the exchange and emit the response on the connection that carried the request. It never pairs by adjacency, arrival order, or a process-global "current request." The separate `prior_ids` list constrains legal interleavings and is not used as a substitute for causal correlation.

HTTP projection is valid only for a causally closed exchange. Every projected request must be named by exactly one terminal response's `cause_id`, and every projected response must name one compatible request in the same session and task. Failure is still a response event: it carries an HTTP error status and points to the request. An orphan request, a response with no cause, an incompatible or cross-session cause, or two terminal responses for one request is an unchecked HTTP call and fails validation rather than becoming an underspecified adapter behavior. Protocols that deliberately permit interim or streaming responses must model those nonterminal messages explicitly and still identify one terminal response.

14.3 The running adapter

The following table is the implemented ChatUI subset. Static assets, diagnostic reports, generated source downloads, MCP, and health checks are document resources or tool protocols outside the ChatUI task grammar; they must not be mistaken for user-task protobuf events.

POST `/api/accounts`
Request: AccountCreateSubmitted. Response: AccountCreated or AccountCreateRejected. Encoding: application/x-www-form-urlencoded.

POST `/api/login`
Request: LoginSubmitted. Response: LoginAccepted or LoginRejected. Encoding: application/x-www-form-urlencoded.

GET `/api/me`
Request: authenticated session. Response: user identity JSON. Encoding: cookie only.

GET `/api/projects`
Request: ProjectListRequested. Response: ProjectListReturned. Encoding: cookie plus path.

POST `/api/projects`
Request: ProjectCreateRequested. Response: ProjectCreated. Encoding: application/x-www-form-urlencoded.

GET `/api/projects/{project_id}/conversation`
Request: project identifier. Response: conversation JSON. Encoding: path field plus cookie.

GET `/api/projects/{project_id}/tasks`
Request: project identifier. Response: task model JSON. Encoding: path field plus cookie.

GET `/api/projects/{project_id}/artifacts`
Request: ArtifactListRequested. Response: ArtifactListReturned. Encoding: path field plus cookie.

POST `/api/projects/{project_id}/chat`
Request: ChatPromptSubmitted. Response: ChatResponseRendered or ChatTurnFailed. Encoding: application/x-www-form-urlencoded.

Every `/api/` route except public account creation and login requires a valid in-memory session. Project paths additionally resolve the authenticated owner's project root. A missing session yields 401; an unknown or unauthorized project is deliberately represented as 404. The server binds to loopback by default. Network exposure requires an authenticating reverse proxy or an equivalent transport boundary.

14.4 The accepted HTTP projection

This is the complete intended browser mapping from the accepted implementation specification. Rows absent from the running-adapter table remain implementation obligations.

POST `/api/login`
Request: `LoginSubmitted`. Response: `LoginAccepted`, `LoginRejected`.

POST `/api/accounts`
Request: `AccountCreateSubmitted`. Response: `AccountCreated`, `AccountCreateRejected`.

GET `/api/projects/{project_id}/tasks/{task_id}`
Request: `TaskModelRequested`. Response: `TaskModelReturned`.

GET `/api/projects/{project_id}/tasks/{task_id}/sessions/{session_id}/events`
Request: `MessageSequenceRequested`. Response: `MessageSequenceReturned`.

GET `/api/projects/{project_id}/events/{event_id}`
Request: `ProtobufObjectRequested`. Response: `ProtobufObjectReturned`.

POST `/api/projects/{project_id}/research`
Request: `WebResearchRequested`. Response: `WebResearchReturned`, `WebResearchFailed`.

POST `/api/projects/{project_id}/markerboard/assist`
Request: `MarkerboardAssistRequested`. Response: `MarkerboardSyntaxHelpReturned`, `MarkerboardPatchPro`

POST `/api/projects/{project_id}/markerboard/assist/{artifact_id}/accept`
Request: `MarkerboardPatchAccepted`. Response: `MarkerboardDraftUpdated`.

POST `/api/projects/{project_id}/markerboard/assist/{artifact_id}/reject`
Request: `MarkerboardPatchRejected`. Response: `HTTP204(nopayload)`.

GET `/api/projects`
Request: `ProjectListRequested`. Response: `ProjectListReturned`.

POST `/api/projects`
Request: `ProjectCreateRequested`. Response: `ProjectCreated`.

GET `/api/projects/{project_id}`
Request: `ProjectOpenRequested`. Response: `ProjectOpened`, `ProblemInquiryPresented`.

DELETE `/api/projects/{project_id}`
Request: `ProjectDeleteRequested`. Response: `ProjectDeleted`.

POST `/api/projects/{project_id}/specification/problem`
Request: `ProblemStatementSubmitted`. Response: `AmbiguityQuestionPresented`, `RequirementsReviewPrese`

POST `/api/projects/{project_id}/specification/ambiguities/{ambiguity_id}`
Request: `ClarificationSubmitted`. Response: `ProjectBookRevised`.

POST `/api/projects/{project_id}/specification/requirements/{requirements_revision}/accept`
Request: `SpecificationRequirementsAccepted`. Response: `ImplementationReviewPresented`.

POST `/api/projects/{project_id}/specification/implementation/{implementation_revision}/accep`
Request: `SpecificationImplementationAccepted`. Response: `ImplementationAuthorizationConfirmed`.

POST `/api/credentials`
Request: `ProviderCredentialSubmitted`. Response: `ProviderCredentialAccepted`, `ProviderCredentialRej`

POST `/api/projects/{project_id}/conversations/{conversation_id}/messages`
Request: `ChatPromptSubmitted`. Response: `ChatResponseRendered`, `ChatTurnFailed`.

POST `/api/projects/{project_id}/markerboard/mermaid`
Request: `MermaidDraftSaved`. Response: `MermaidPreviewRendered`.

POST `/api/projects/{project_id}/markerboard/media`
Request: `MediaUploaded`. Response: `MediaRendered`.

POST `/api/projects/{project_id}/artifacts/{artifact_id}/promotions`
Request: `ArtifactPromoted`. Response: `CandidateRequirementUpdated`.

GET `/api/projects/{project_id}/artifacts`
Request: `ArtifactListRequested`. Response: `ArtifactListReturned`.

GET `/api/projects/{project_id}/artifacts/{artifact_id}/revisions/{revision}`
Request: `ArtifactOpenRequested`. Response: `ArtifactOpened`.

Messages between `LeanFMServer` and `ModelProvider` map to the bounded in-process channels `provider.requests` and `provider.results`. They retain their protobuf message identities and request identifiers even though they do not cross HTTP. This is important for concurrency: two users can use different provider credentials because the key, session, user, project, conversation, and request identities remain request-local instead of being stored in a process-wide environment variable.

14.5 Accepted message endpoints

Every accepted event is a length-delimited `EventEnvelope`; payload tags 20 and above identify its concrete message. This generated index records each message's sender and receiver. The canonical schema that follows defines field names, tags, types, and cardinality.

`LoginSubmitted`
UserBrowser sends to LeanFMServer.

`LoginAccepted`
LeanFMServer sends to UserBrowser.

`LoginRejected`
LeanFMServer sends to UserBrowser.

`AccountCreateSubmitted`
UserBrowser sends to LeanFMServer.

`ProviderKeyCheckRequested`
LeanFMServer sends to ModelProvider.

`ProviderKeyVerified`
ModelProvider sends to LeanFMServer.

`ProviderKeyRejected`
ModelProvider sends to LeanFMServer.

`AccountCreated`
LeanFMServer sends to UserBrowser.

`AccountCreateRejected`
LeanFMServer sends to UserBrowser.

`TaskModelRequested`
UserBrowser sends to LeanFMServer.

`TaskModelReturned`
LeanFMServer sends to UserBrowser.

`MessageSequenceRequested`
UserBrowser sends to LeanFMServer.

`MessageSequenceReturned`
LeanFMServer sends to UserBrowser.

`ProtobufObjectRequested`
UserBrowser sends to LeanFMServer.

`ProtobufObjectReturned`
LeanFMServer sends to UserBrowser.

`WebResearchRequested`
UserBrowser sends to LeanFMServer.

`WebResearchDispatched`
LeanFMServer sends to ModelProvider.

`WebSearchPerformed`
ModelProvider sends to LeanFMServer.

`WebPageOpened`
ModelProvider sends to LeanFMServer.

`WebResearchReturned`
LeanFMServer sends to UserBrowser.

`WebResearchFailed`
LeanFMServer sends to UserBrowser.

`MarkerboardAssistRequested`
UserBrowser sends to LeanFMServer.

`MarkerboardSyntaxHelpReturned`
LeanFMServer sends to UserBrowser.

`MarkerboardAssistDispatched`
LeanFMServer sends to ModelProvider.

`MarkerboardAssistReturned`
ModelProvider sends to LeanFMServer.

`MarkerboardPatchProposed`
LeanFMServer sends to UserBrowser.

`MarkerboardPatchAccepted`
UserBrowser sends to LeanFMServer.

`MarkerboardPatchRejected`
UserBrowser sends to LeanFMServer.

`MarkerboardDraftUpdated`
LeanFMServer sends to UserBrowser.

`ProjectListRequested`
UserBrowser sends to LeanFMServer.

`ProjectListReturned`
LeanFMServer sends to UserBrowser.

`ProjectCreateRequested`
UserBrowser sends to LeanFMServer.

`ProjectCreated`
LeanFMServer sends to UserBrowser.

`ProjectOpenRequested`
UserBrowser sends to LeanFMServer.

`ProjectOpened`
LeanFMServer sends to UserBrowser.

`ProjectDeleteRequested`
UserBrowser sends to LeanFMServer.

`ProjectDeleted`
LeanFMServer sends to UserBrowser.

`ProblemInquiryPresented`
LeanFMServer sends to UserBrowser.

`ProblemStatementSubmitted`
UserBrowser sends to LeanFMServer.

`AmbiguityQuestionPresented`
LeanFMServer sends to UserBrowser.

`ClarificationSubmitted`
UserBrowser sends to LeanFMServer.

`ProjectBookRevised`
LeanFMServer sends to UserBrowser.

`RequirementsReviewPresented`
LeanFMServer sends to UserBrowser.

`SpecificationRequirementsAccepted`
UserBrowser sends to LeanFMServer.

`ImplementationReviewPresented`
LeanFMServer sends to UserBrowser.

`SpecificationImplementationAccepted`
UserBrowser sends to LeanFMServer.

`ImplementationAuthorizationConfirmed`
LeanFMServer sends to UserBrowser.

`ProviderCredentialSubmitted`
UserBrowser sends to LeanFMServer.

`ProviderCredentialAccepted`
LeanFMServer sends to UserBrowser.

`ProviderCredentialRejected`
LeanFMServer sends to UserBrowser.

`ChatPromptSubmitted`
UserBrowser sends to LeanFMServer.

`ModelRequestIssued`
LeanFMServer sends to ModelProvider.

`ModelResponseReturned`
ModelProvider sends to LeanFMServer.

`ChatResponseRendered`
LeanFMServer sends to UserBrowser.

`ChatTurnFailed`
LeanFMServer sends to UserBrowser.

`ConversationCheckpointRequested`
UserBrowser sends to LeanFMServer.

`ConversationCheckpointBlocked`
LeanFMServer sends to UserBrowser.

`ConversationCheckpointCommitted`
LeanFMServer sends to UserBrowser.

`ConversationHistoryCleared`
LeanFMServer sends to UserBrowser.

`MermaidDraftSaved`
UserBrowser sends to LeanFMServer.

`MermaidPreviewRendered`
LeanFMServer sends to UserBrowser.

`MediaUploaded`
UserBrowser sends to LeanFMServer.

`MediaRendered`
LeanFMServer sends to UserBrowser.

`ArtifactPromoted`
UserBrowser sends to LeanFMServer.

`CandidateRequirementUpdated`
LeanFMServer sends to UserBrowser.

`ArtifactListRequested`
UserBrowser sends to LeanFMServer.

`ArtifactListReturned`
LeanFMServer sends to UserBrowser.

`ArtifactOpenRequested`
UserBrowser sends to LeanFMServer.

`ArtifactOpened`
LeanFMServer sends to UserBrowser.

14.5.1 Canonical byte schema

The following complete schema is normative. It fixes envelope and payload tag numbers, repeated and nested cardinality, enum values, and field names. A reconstruction must reproduce these values rather than assigning tags in book order or replacing nested messages with opaque bytes.

```
syntax = "proto3";

package leanfm.chatui;

message EventEnvelope {
  string event_id = 1;
  repeated string prior_ids = 2;
  string session_id = 3;
  string task_id = 4;
  uint64 time_at_ms = 5;
  string src = 6;
  string dst = 7;
  // Direct causal parent for any event, including nonterminal and intermediate
  // events. Empty for a causal root. Unlike prior_ids, this does not constrain
  // otherwise legal concurrent interleavings. Renderers may use the causal
  // subtree as nested interaction boundaries.
  optional string cause_id = 8;

  oneof payload {
    LoginSubmitted login_submitted = 20;
    LoginAccepted login_accepted = 21;
    LoginRejected login_rejected = 22;
    ProjectListRequested project_list_requested = 23;
    ProjectListReturned project_list_returned = 24;
    ProjectCreateRequested project_create_requested = 25;
    ProjectCreated project_created = 26;
    ProjectOpenRequested project_open_requested = 27;
    ProjectOpened project_opened = 28;
    ProjectDeleteRequested project_delete_requested = 29;
    ProjectDeleted project_deleted = 30;
    ProviderCredentialSubmitted provider_credential_submitted = 31;
    ProviderCredentialAccepted provider_credential_accepted = 32;
    ProviderCredentialRejected provider_credential_rejected = 33;
    ChatPromptSubmitted chat_prompt_submitted = 34;
    ModelRequestIssued model_request_issued = 35;
    ModelResponseReturned model_response_returned = 36;
    ChatResponseRendered chat_response_rendered = 37;
    ChatTurnFailed chat_turn_failed = 38;
    MermaidDraftSaved mermaid_draft_saved = 39;
    MermaidPreviewRendered mermaid_preview_rendered = 40;
    MediaUploaded media_uploaded = 41;
    MediaRendered media_rendered = 42;
    ArtifactPromoted artifact_promoted = 43;
    CandidateRequirementUpdated candidate_requirement_updated = 44;
    ArtifactListRequested artifact_list_requested = 45;
    ArtifactListReturned artifact_list_returned = 46;
    ArtifactOpenRequested artifact_open_requested = 47;
    ArtifactOpened artifact_opened = 48;
    AccountCreateSubmitted account_create_submitted = 49;
    ProviderKeyCheckRequested provider_key_check_requested = 50;
    ProviderKeyVerified provider_key_verified = 51;
    ProviderKeyRejected provider_key_rejected = 52;
    AccountCreated account_created = 53;
    AccountCreateRejected account_create_rejected = 54;
    TaskModelRequested task_model_requested = 55;
    TaskModelReturned task_model_returned = 56;
    MessageSequenceRequested message_sequence_requested = 57;
    MessageSequenceReturned message_sequence_returned = 58;
    ProtobufObjectRequested protobuf_object_requested = 59;
    ProtobufObjectReturned protobuf_object_returned = 60;
    WebResearchRequested web_research_requested = 61;
    WebSearchPerformed web_search_performed = 62;
    WebPageOpened web_page_opened = 63;
```

```
    WebResearchReturned web_research_returned = 64;
    WebResearchFailed web_research_failed = 65;
    WebResearchDispatched web_research_dispatched = 66;
    MarkerboardAssistRequested markerboard_assist_requested = 67;
    MarkerboardSyntaxHelpReturned markerboard_syntax_help_returned = 68;
    MarkerboardAssistDispatched markerboard_assist_dispatched = 69;
    MarkerboardAssistReturned markerboard_assist_returned = 70;
    MarkerboardPatchProposed markerboard_patch_proposed = 71;
    MarkerboardPatchAccepted markerboard_patch_accepted = 72;
    MarkerboardPatchRejected markerboard_patch_rejected = 73;
    MarkerboardDraftUpdated markerboard_draft_updated = 74;
    ProblemInquiryPresented problem_inquiry_presented = 75;
    ProblemStatementSubmitted problem_statement_submitted = 76;
    AmbiguityQuestionPresented ambiguity_question_presented = 77;
    ClarificationSubmitted clarification_submitted = 78;
    ProjectBookRevised project_book_revised = 79;
    RequirementsReviewPresented requirements_review_presented = 80;
    SpecificationRequirementsAccepted specification_requirements_accepted = 81;
    ImplementationReviewPresented implementation_review_presented = 82;
    SpecificationImplementationAccepted specification_implementation_accepted = 83;
    ImplementationAuthorizationConfirmed implementation_authorization_confirmed = 84;
    ConversationCheckpointRequested conversation_checkpoint_requested = 85;
    ConversationCheckpointBlocked conversation_checkpoint_blocked = 86;
    ConversationCheckpointCommitted conversation_checkpoint_committed = 87;
    ConversationHistoryCleared conversation_history_cleared = 88;
  }
}

enum ContentKind {
  CONTENT_KIND_UNSPECIFIED = 0;
  MARKDOWN = 1;
  LATEX = 2;
  MERMAID = 3;
  GRAPHVIZ_DOT = 4;
  IMAGE = 5;
  VIDEO = 6;
  CODE = 7;
  FILE = 8;
  LEANFM_VIEW = 9;
  URL_CITATION = 10;
}

message ChatContentPart {
  string part_id = 1;
  ContentKind kind = 2;
  string text = 3;
  string artifact_id = 4;
  uint64 artifact_revision = 5;
  string media_type = 6;
  bytes inline_content = 7;
  string alt_text = 8;
  string url = 9;
  string title = 10;
}

message LoginSubmitted {
  string username = 1;
  bytes credential = 2;
  bytes openai_api_key = 3;
  bytes anthropic_api_key = 4;
  bytes new_password = 5;
  bytes new_password_confirm = 6;
}
message LoginAccepted { string user_id = 1; string session_id = 2; }
message LoginRejected { string reason = 1; }

message ProblemInquiryPresented { string project_id = 1; string question = 2; }
message ProblemStatementSubmitted { string project_id = 1; string statement = 2; }
message AmbiguityQuestionPresented { string project_id = 1; string ambiguity_id = 2; string question = 3; repeated string affected_artifacts
       = 4; }
message ClarificationSubmitted { string project_id = 1; string ambiguity_id = 2; string answer = 3; }
message ProjectBookRevised { string project_id = 1; string book_revision = 2; repeated string decision_ids = 3; string light_pdf = 4; string
       dark_pdf = 5; }
message RequirementsReviewPresented { string project_id = 1; string requirements_revision = 2; repeated string derived_artifact_ids = 3;
      repeated string open_ambiguity_ids = 4; }
message SpecificationRequirementsAccepted { string project_id = 1; string requirements_revision = 2; }
message ImplementationReviewPresented { string project_id = 1; string requirements_revision = 2; string implementation_revision = 3; string
      traceability_report = 4; }
message SpecificationImplementationAccepted { string project_id = 1; string requirements_revision = 2; string implementation_revision = 3; }
message ImplementationAuthorizationConfirmed { string project_id = 1; string requirements_revision = 2; string implementation_revision = 3;
      }

message AccountCreateSubmitted {
  string username = 1;
  bytes login_credential = 2;
  string provider = 3;
  bytes api_key = 4;
  bytes openai_api_key = 5;
  bytes anthropic_api_key = 6;
```

```
}
message ProviderKeyCheckRequested { string provider = 1; string verification_id = 2; }
message ProviderKeyVerified { string verification_id = 1; string provider_account = 2; }
message ProviderKeyRejected { string verification_id = 1; string reason = 2; }
message AccountCreated { string user_id = 1; string credential_ref = 2; }
message AccountCreateRejected { string reason = 1; }

message TaskModelRequested { string project_id = 1; string task_id = 2; string requirements_revision = 3; }
message TaskModelReturned {
  string task_id = 1;
  string interaction_artifact_id = 2;
  string state_machine_artifact_id = 3;
  string message_sequence_artifact_id = 4;
  repeated string constraint_ids = 5;
}
message MessageSequenceRequested { string project_id = 1; string task_id = 2; string session_id = 3; }
message MessageSequenceReturned { string task_id = 1; string session_id = 2; repeated string event_ids = 3; }
message ProtobufObjectRequested { string project_id = 1; string event_id = 2; }
message ProtobufObjectReturned { string event_id = 1; string protobuf_type = 2; bytes encoded_object = 3; string redacted_json = 4; }
message WebResearchRequested { string project_id = 1; string conversation_id = 2; string request_id = 3; string question = 4; repeated
      string supplied_urls = 5; }
message WebResearchDispatched { string request_id = 1; string model = 2; string research_mode = 3; }
message WebSearchPerformed { string request_id = 1; repeated string queries = 2; }
message WebPageOpened { string request_id = 1; string url = 2; string title = 3; }
message WebResearchReturned { string request_id = 1; repeated ChatContentPart parts = 2; repeated string source_urls = 3; }
message WebResearchFailed { string request_id = 1; string reason = 2; }
message MarkerboardAssistRequested { string project_id = 1; string artifact_id = 2; uint64 base_revision = 3; string diagram_kind = 4;
      string source = 5; string instruction = 6; string mode = 7; string credential_ref = 8; }
message MarkerboardSyntaxHelpReturned { string artifact_id = 1; uint64 base_revision = 2; string help_markdown = 3; }
message MarkerboardAssistDispatched { string request_id = 1; string artifact_id = 2; uint64 base_revision = 3; string diagram_kind = 4;
      string source = 5; string instruction = 6; }
message MarkerboardAssistReturned { string request_id = 1; string proposed_source = 2; string explanation = 3; uint64 input_tokens = 4;
      uint64 output_tokens = 5; }
message MarkerboardPatchProposed { string artifact_id = 1; uint64 base_revision = 2; string proposed_source = 3; string explanation = 4; }
message MarkerboardPatchAccepted { string artifact_id = 1; uint64 base_revision = 2; string proposed_source = 3; }
message MarkerboardPatchRejected { string artifact_id = 1; uint64 base_revision = 2; }
message MarkerboardDraftUpdated { string artifact_id = 1; uint64 revision = 2; string source = 3; }

message ProjectListRequested { string user_id = 1; }
message ProjectListReturned { repeated string project_ids = 1; }
message ProjectCreateRequested { string display_name = 1; }
message ProjectCreated { string project_id = 1; string display_name = 2; }
message ProjectOpenRequested { string project_id = 1; }
message ProjectOpened { string project_id = 1; string current_requirements_revision = 2; }
message ProjectDeleteRequested { string project_id = 1; string confirmation = 2; }
message ProjectDeleted { string project_id = 1; }

message ProviderCredentialSubmitted {
  string provider = 1;
  bytes api_key = 2;
  string retention = 3;
}
message ProviderCredentialAccepted { string credential_ref = 1; string billing_source = 2; }
message ProviderCredentialRejected { string reason = 1; }

message ChatPromptSubmitted {
  string project_id = 1;
  string conversation_id = 2;
  string message_id = 3;
  repeated ChatContentPart parts = 4;
  repeated string attachment_ids = 5;
  string credential_ref = 6;
  string model = 7;
}
message ModelRequestIssued {
  string project_id = 1;
  string conversation_id = 2;
  string request_id = 3;
  string model = 4;
  string research_mode = 5;
}
message ModelResponseReturned {
  string request_id = 1;
  repeated ChatContentPart parts = 2;
  uint64 input_tokens = 3;
  uint64 cached_input_tokens = 4;
  uint64 output_tokens = 5;
  uint64 context_window_tokens = 6;
}
message ChatResponseRendered {
  string conversation_id = 1;
  string message_id = 2;
  repeated ChatContentPart parts = 3;
  repeated string artifact_ids = 4;
  uint64 input_tokens = 5;
  uint64 cached_input_tokens = 6;
  uint64 output_tokens = 7;
  uint64 context_window_tokens = 8;
  string usage_warning = 9;
```

```
  uint64 response_time_ms = 10;
}
message ChatTurnFailed { string request_id = 1; string reason = 2; }
message ConversationCheckpointRequested { string project_id = 1; string conversation_id = 2; string through_message_id = 3; }
message ConversationCheckpointBlocked { string project_id = 1; string conversation_id = 2; string checkpoint_id = 3; string reason_code = 4;
       repeated string blocking_artifact_ids = 5; }
message ConversationCheckpointCommitted { string project_id = 1; string conversation_id = 2; string through_message_id = 3; repeated string
      artifact_revisions = 4; string manifest_digest = 5; }
message ConversationHistoryCleared { string project_id = 1; string conversation_id = 2; string cleared_through_message_id = 3; uint64
      retained_message_count = 4; }

message MermaidDraftSaved {
  string project_id = 1;
  string artifact_id = 2;
  uint64 revision = 3;
  string diagram_kind = 4;
  string source = 5;
}
message MermaidPreviewRendered { string artifact_id = 1; uint64 revision = 2; bytes svg = 3; }
message MediaUploaded {
  string project_id = 1;
  string artifact_id = 2;
  string media_type = 3;
  bytes content = 4;
}
message MediaRendered { string artifact_id = 1; string media_type = 2; string content_url = 3; }
message ArtifactPromoted { string artifact_id = 1; uint64 revision = 2; string target_status = 3; }
message CandidateRequirementUpdated { string requirements_revision = 1; repeated string source_artifact_ids = 2; }

message ArtifactListRequested {
  string project_id = 1;
  string kind_filter = 2;
  string scenario_filter = 3;
  string status_filter = 4;
}
message ArtifactSummary {
  string artifact_id = 1;
  uint64 current_revision = 2;
  uint64 accepted_revision = 3;
  string status = 4;
  string validation = 5;
  string media_type = 6;
  string title = 7;
  string scenario_id = 8;
  string diagram_kind = 9;
  string updated_by = 10;
  uint64 updated_at_ms = 11;
}
message ArtifactListReturned { repeated ArtifactSummary artifacts = 1; }
message ArtifactOpenRequested { string artifact_id = 1; uint64 revision = 2; }
message ArtifactOpened {
  string artifact_id = 1;
  uint64 revision = 2;
  string status = 3;
  string media_type = 4;
  bytes content = 5;
  string provenance = 6;
}
```

Envelope metadata remain distinct. `prior_ids` stores immutable predecessor links; any `map[Event][]Event` successor index is derived, not serialized as mutable truth. `session_id` separates concurrent conversations, `task_id` selects the grammar, and `src`/`dst` must match the message schema. As defined above, `cause_id` is a separate acyclic direct-cause relation usable by roots, intermediate messages, nested task boundaries, and terminals. Its tree groups renderer boxes without ordering concurrent children.

14.6 Complete LeanFM task grammar

The following is the whole accepted ChatUI conversation grammar, rendered from the typed `TaskGrammar` values. Pronounce `A ! B : Message` as “A sends Message to B.” `then` is ordered composition; `choose/or` is an explicit alternative; `optional` permits zero or one occurrence; `fork/join all` waits for every branch; and `race/join first` accepts the first completion and cancels the remaining branches. A message not admitted by the selected task’s current residual grammar is illegal even if its protobuf bytes decode.

14.7 Authenticate an individual user

Task: `authenticate_user`. Entry: `signed_out`; terminals: `authenticated, rejected`.

```
UserBrowser ! LeanFMServer : LoginSubmitted
then
choose
  LeanFMServer ! UserBrowser : LoginAccepted
or
  LeanFMServer ! UserBrowser : LoginRejected
```

14.8 Create an account using a user-supplied provider API key

Task: `create_account`. Entry: `registration`; terminals: `created, rejected`.

```
UserBrowser ! LeanFMServer : AccountCreateSubmitted
then
LeanFMServer ! ModelProvider : ProviderKeyCheckRequested
then
choose
  ModelProvider ! LeanFMServer : ProviderKeyVerified
  then
  LeanFMServer ! UserBrowser : AccountCreated
or
  ModelProvider ! LeanFMServer : ProviderKeyRejected
  then
  LeanFMServer ! UserBrowser : AccountCreateRejected
```

14.9 List, create, open, and delete owned projects

Task: `manage_projects`. Entry: `ready`; terminals: `listed, created, opened, deleted`.

```
choose
  UserBrowser ! LeanFMServer : ProjectListRequested
  then
  LeanFMServer ! UserBrowser : ProjectListReturned
or
  UserBrowser ! LeanFMServer : ProjectCreateRequested
  then
  LeanFMServer ! UserBrowser : ProjectCreated
or
  UserBrowser ! LeanFMServer : ProjectOpenRequested
  then
  LeanFMServer ! UserBrowser : ProjectOpened
or
  UserBrowser ! LeanFMServer : ProjectDeleteRequested
```

```
  then
  LeanFMServer ! UserBrowser : ProjectDeleted
```

14.10 Turn a fresh project conversation into accepted requirements and implementation

Task: `develop_project_specification`. Entry: `project_loaded`; terminals: `ready_to_implement`.

```
LeanFMServer ! UserBrowser : ProblemInquiryPresented
then
UserBrowser ! LeanFMServer : ProblemStatementSubmitted
then
repeat
  choose
    empty
  or
    LeanFMServer ! UserBrowser : ProjectBookRevised
  then
  LeanFMServer ! UserBrowser : AmbiguityQuestionPresented
  then
  UserBrowser ! LeanFMServer : ClarificationSubmitted
then
choose
  empty
or
  LeanFMServer ! UserBrowser : ProjectBookRevised
then
LeanFMServer ! UserBrowser : RequirementsReviewPresented
then
UserBrowser ! LeanFMServer : SpecificationRequirementsAccepted
then
LeanFMServer ! UserBrowser : ImplementationReviewPresented
then
UserBrowser ! LeanFMServer : SpecificationImplementationAccepted
then
LeanFMServer ! UserBrowser : ImplementationAuthorizationConfirmed
```

14.11 Select user-funded or platform-funded model access

Task: `configure_provider`. Entry: `unconfigured`; terminals: `configured, rejected`.

```
UserBrowser ! LeanFMServer : ProviderCredentialSubmitted
then
choose
  LeanFMServer ! UserBrowser : ProviderCredentialAccepted
```

```
or
  LeanFMServer ! UserBrowser : ProviderCredentialRejected
```

14.12 Conduct one durable project-scoped requirements turn

Task: `argue_requirements`. Entry: `ready`; terminals: `rendered, failed`.

```
UserBrowser ! LeanFMServer : ChatPromptSubmitted
then
LeanFMServer ! ModelProvider : ModelRequestIssued
then
choose
  ModelProvider ! LeanFMServer : ModelResponseReturned
  then
  LeanFMServer ! UserBrowser : ChatResponseRendered
or
  LeanFMServer ! UserBrowser : ChatTurnFailed
```

14.13 Compact a project conversation into durable specifications before clearing it

Task: `checkpoint_conversation`. Entry: `retained`; terminals: `cleared, blocked`.

```
UserBrowser ! LeanFMServer : ConversationCheckpointRequested
then
choose
  LeanFMServer ! UserBrowser : ConversationCheckpointBlocked
or
  LeanFMServer ! UserBrowser : ConversationCheckpointCommitted
  then
  LeanFMServer ! UserBrowser : ConversationHistoryCleared
```

14.14 Sketch and promote multimodal requirement artifacts

Task: `markerboard`. Entry: `editing`; terminals: `candidate`.

```
choose
  UserBrowser ! LeanFMServer : MermaidDraftSaved
  then
  LeanFMServer ! UserBrowser : MermaidPreviewRendered
  then
  UserBrowser ! LeanFMServer : ArtifactPromoted
  then
  LeanFMServer ! UserBrowser : CandidateRequirementUpdated
or
```

```
  UserBrowser ! LeanFMServer : MediaUploaded
  then
  LeanFMServer ! UserBrowser : MediaRendered
  then
  UserBrowser ! LeanFMServer : ArtifactPromoted
  then
  LeanFMServer ! UserBrowser : CandidateRequirementUpdated
```

14.15 Inspect current, accepted, superseded, and stale artifacts

Task: `inspect_artifacts`. Entry: `closed`; terminals: `listed, opened`.

```
UserBrowser ! LeanFMServer : ArtifactListRequested
then
LeanFMServer ! UserBrowser : ArtifactListReturned
then
choose
  empty
or
  UserBrowser ! LeanFMServer : ArtifactOpenRequested
  then
  LeanFMServer ! UserBrowser : ArtifactOpened
```

14.16 Inspect one task's linked interaction, state machine, constraints, and protobuf events

Task: `inspect_task_model`. Entry: `closed`; terminals: `linked, sequence_visible, object_visible`.

```
UserBrowser ! LeanFMServer : TaskModelRequested
then
LeanFMServer ! UserBrowser : TaskModelReturned
then
choose
  empty
or
  UserBrowser ! LeanFMServer : MessageSequenceRequested
  then
  LeanFMServer ! UserBrowser : MessageSequenceReturned
  then
  choose
    empty
  or
    UserBrowser ! LeanFMServer : ProtobufObjectRequested
    then
    LeanFMServer ! UserBrowser : ProtobufObjectReturned
```

14.17 Search the public web, follow supplied HTTP URLs, and return clickable citations

Task: `research_web`. Entry: `ready`; terminals: `returned, failed`.

```
UserBrowser ! LeanFMServer : WebResearchRequested
then
LeanFMServer ! ModelProvider : WebResearchDispatched
then
choose
  ModelProvider ! LeanFMServer : WebSearchPerformed
  then
  ModelProvider ! LeanFMServer : WebPageOpened
  then
  LeanFMServer ! UserBrowser : WebResearchReturned
or
  LeanFMServer ! UserBrowser : WebResearchFailed
```

14.18 Recall diagram syntax or propose AI-generated detail for one markerboard revision

Task: `assist_markerboard`. Entry: `editing`; terminals: `helped, updated, rejected`.

```
UserBrowser ! LeanFMServer : MarkerboardAssistRequested
then
choose
  LeanFMServer ! UserBrowser : MarkerboardSyntaxHelpReturned
or
  LeanFMServer ! ModelProvider : MarkerboardAssistDispatched
  then
  ModelProvider ! LeanFMServer : MarkerboardAssistReturned
  then
  LeanFMServer ! UserBrowser : MarkerboardPatchProposed
  then
  choose
    UserBrowser ! LeanFMServer : MarkerboardPatchAccepted
    then
    LeanFMServer ! UserBrowser : MarkerboardDraftUpdated
  or
    UserBrowser ! LeanFMServer : MarkerboardPatchRejected
```

14.19 The grammar as an executable canvas

The complete grammar is one navigable canvas rather than one permanently huge drawing. A task or production node may be closed to its typed boundary or opened to show its interaction messages,

residual states, parameters, constraints, paired measurements and nested productions. Opening and closing is presentation state only. It never changes the grammar or conceals a failing obligation.

Each editable parameter has one typed definition and explicit dependency edges. Changing a queue capacity, actor population, timeout, probability or other parameter marks every transitive dependent stale and regenerates the affected productions, reducers, proofs, codecs, charts and views. The canvas shows that cascade before the revised artifact can be accepted.

A production known to cross HTTP may attach its transport renderer at that production. The binding names the method, path template, request and response protobuf types, status alternatives, content type, byte encoder and byte decoder. Generated fixtures emit the exact request and response bytes and read them back. Conformance requires decoded emitted values to equal their originals and their message atoms to be accepted by the selected residual grammar. Thus the high-level canvas can be progressively opened all the way down to bytes without confusing transport with the message value.

14.20 What is checked, and what remains

Lean elaboration checks the actors, message schemas, task states, transitions, grammars, constraints, desired-output provenance, and implementation mappings. The chapter generator rejects an accepted HTTP request with no named response mapping and rejects incomplete running-route records. The ordinary repository checks also validate the protobuf source against the typed requirements.

This does not yet prove that every accepted route has a handler: the running table makes that gap explicit. Nor does the current server yet append every browser exchange to the specified binary event journal. Completing those handlers and the schema-specific codec remains implementation work. Until then, the accepted grammar is the protocol contract, the running table is the honest conformance statement, and neither may be substituted for the other.

Chapter 15

The LeanFM Project Bootstrap Contract

This chapter defines the minimum machine-readable context that makes a LeanFM project self-hosting. It belongs in the machine reference because an agent must apply it before it can turn the rest of the book into checked artifacts. The bootstrap defines the language and its tools; it must not contain a hidden copy of the product requirements or implementation.

The intended handoff is deliberately simple: send another person the URL of this PDF, ask Claude, Gemini, Codex, or another capable construction agent to read it, and ask that agent to make the software. New ChatUI projects are provisioned from built-in book and specification templates; they do not depend on an example project directory. The live language-reference tools and read-only MCP surface supply the generic authoring types and commands named below. An agent must not copy the existing LeanFM application source or prior ChatUI specifications as an answer key.

15.1 Required project inputs

A copyable project root contains the following stable inputs:

`book/main.pdf`
: Central product-design argument and maintained system-prompt context.

`book/main-dark.pdf`
: Accessible rendering of the same source.

`LeanFM/`
: Versioned generic authoring types and deterministic validators. Product actors, messages, tasks and routes do not belong in this directory.

`bootstrap/`
: Programs that print the language references and interrogation checklist, validate the conventional specification, and derive review views from its typed value.

`lean-toolchain` **and** `lakefile.toml`
: The exact compiler selection and module roots needed to run the bootstrap locally.

`project.json`
: Project identity, stable artifact paths, acceptance state and the classification of design versus language inputs.

All project-owned state and acceptance decisions belong below the exclusive root: inputs, conversations, artifact revisions, source, tests, generated views, traces, reports, logs, and project-specific caches. Choose the root before session work; temporary files and atomic promotion stay inside it. Reject configured outputs and symlinks that escape. Export declared artifacts explicitly; import by copying into the root or recording an immutable, content-addressed dependency. Shared tools and caches may live outside only without project-owned state. A global index must be reconstructible from manifests, and deleting one root must remove all its state without an external search.

The project root is also the complete input to a fresh agent conversation. The agent discovers the stable paths from the manifest and repository instructions; the stakeholder need not paste a transcript or restate a bootstrap procedure. If the root has no durable problem statement, LeanFM creates the minimal editable book shell and begins with the visible question “What problem are you trying to solve?” The answer starts requirements discovery. Named ambiguity questions, book revisions, validation evidence and the two acceptance decisions then accumulate beneath that same root.

15.2 Conventional generated interface

The requirements candidate lives at `Specification/Requirements.lean`, imports `LeanFM.Artifacts`, and defines:

```
namespace LeanFMProject.Requirements
def spec : LeanFM.RequirementSpec := ...
def all : List LeanFM.GeneratedRequirement := [.requirement spec]
end LeanFMProject.Requirements
```

The byte vocabulary lives beside it at `Specification/Requirements.proto`. After requirements acceptance, the separate implementation mapping lives at `Specification/Implementation.lean`. Capitalization is significant because these are Lean module paths, not informal filenames.

15.3 Required commands and order

The project supplies deterministic equivalents of these commands:

```
make reference       # print the complete requirements-language reference
make interrogation   # print questions that expose missing observations
make validate        # elaborate and run LeanFM plus protobuf validation
make render          # derive interaction and residual-state review views
make check           # validate, then render
```

An agent runs `reference` and `interrogation` before authoring. It must not infer constructors from prose or invent a smaller Lean lookalike. The validator consumes the exported `RequirementSpec`; successful Lean elaboration by itself is not LeanFM validation. The renderer also consumes that same value. A separately written diagram is not evidence that it matches the requirements.

15.4 Acceptance gates

The first gate accepts observable requirements; the second separately accepts their realization mapping. Both require explicit stakeholder acceptance, not a successful build, model response or

attractive diagram. Application source and tests follow only after both gates and the security review in Chapter 10.

`Requirements.lean` defines the contract. Its accepted multi-actor wire grammar is the conformance oracle: complete decoded traffic is legal or has a counterexample; an incomplete capture cannot establish conformance. `Requirements.proto` fixes byte values without choosing transport. `Implementation.lean` records choices needed to realize accepted behavior—for example queues, transports, journals or browser mechanisms—each mapped to its requirement. Answers to implementation questions belong there when they leave observable behavior unchanged. An answer changing messages, fields, ordering or another observable must instead revise Requirements and return to the first gate. Remove implementation answers made unnecessary by later requirements changes rather than preserving them as requirements.

The bootstrap therefore solves a circularity without weakening the gates. The book explains what is being built; the generic distribution makes the LeanFM language executable; accepted project artifacts preserve the decisions. None is a substitute for the other two.

15.5 Construction is not a runtime dependency

The construction agent is not part of the delivered server. After construction, the user starts LeanFM, visits its root URL in an ordinary browser, creates or logs into an account, supplies a model-provider key, and conducts the requirements argument through LeanFM's own ChatGPT-like interface. The server talks to the selected model provider through its documented message protocol; it does not require the construction agent, its conversation, or local tooling.

Consequently, a reconstruction is not successful merely because it can be driven from a coding-agent terminal. It must build a standalone executable, serve the complete browser workflow, start without a platform API key, isolate concurrent users and their credentials, and pass its conformance tests without Codex being present.

15.6 Build, runtime, and publication dependency inventory

The repository does not treat one developer's workstation as an implicit build input. The machine-readable `toolDependencyInventory` in `LeanFM/ChatUI/Requirements.lean` is normative; this section is its human review view. "Required" means that the default check, server, book, or stakeholder-media path invokes the dependency. "Optional" means the named workflow is outside the default product build. JavaScript versions are fixed by `ui/package-lock.json`, Python ranges by `scripts/chatui_requirements.txt`, and Lean and Go by `lean-toolchain` and the top-level `Makefile`.

Lean and Lake — required.
: Lean 4.28.0 from `lean-toolchain` compiles the DSL, validators, renderers, tests, and server. Lake resolves modules and drives native linking.

C toolchain and libc headers — required.
: A compiler and linker supported by Lean 4.28 link the native executables emitted by Lean.

GNU Make, Bash, and coreutils — required.
The workflows use `timeout`, `cmp`, `cp`, `tar`, `gzip`, and `sha256sum` for bounded commands and verified tool downloads.

Node and npm — required.
Node 22.12 or newer and `npm ci` type-check and bundle the React UI. The lockfile fixes React, ReactDOM, TypeScript, Vite, KaTeX, and their transitive packages.

Python packages — required.
Python 3.12 or newer with Cryptography 41–50, Pillow 10–12, and Protobuf 5–6 protects retained keys, performs protobuf round trips and reductions, and generates animated GIFs. CairoSVG 2.7–2.x and defusedxml 0.7–0.x normalize static SVG attachments without external resources; Pillow supplies bounded raster decoding.

Protocol Buffers compiler — required.
A `protoc` supporting proto3 and Python output generates the temporary byte-round-trip binding. Target-language generators are declared separately when needed.

Graphviz — required.
`dot` with PNG output rasterizes generated DOT diagrams during `make diagrams`.

TeX Live and pdfLaTeX — required.
Install the base, recommended, extra, pictures, and recommended-font collections. The source uses `lmodern`, TeX Gyre Heros, geometry, microtype, xcolor, graphicx, tagpdf, AMS math, mathtools, booktabs, longtable, enumitem, listings, natbib, imakeidx, TikZ, pgfplots, hyperref, and cleveref.

Go — required.
Go 1.23.12, whose archive hash is fixed by the Makefile, compiles and tests the repository Go/Paxos target.

Bubblewrap — required on the Linux server.
`bwrap` with namespace and bind-mount support runs project-local Python/Pillow GIF generators without network access.

FFmpeg — required for shareable video.
FFmpeg with `libx264` H.264 and AAC encoders converts GIF/WebM to validated MP4, adds fast-start metadata, and muxes optional narration.

curl, ripgrep, and iproute2 — required for development checks.
`curl`, `rg`, and `ss` exercise HTTP contracts, select validation evidence, and identify the local server.

Java and TLA+ tools — optional.
A Java runtime and the checksummed `tla2tools.jar` 1.7.4 run the separate `tla-check` example.

Manim and Piper — optional.
A project-local video Python environment and selected voice model render explanatory videos and synthetic narration.

Target compilers and SDKs — conditional.
The accepted `ImplementationSpec` inventories every compiler, code generator, SDK, runtime, native library, and version bound before source generation.

An agent can reconcile this declaration with the current host and report each item as satisfied, missing, incompatible, or optional. Installing or upgrading host software remains an explicit bounded action; inspection alone never silently mutates the machine. DuckDB is not currently invoked by a checked LeanFM target and is therefore not a build dependency. If a DuckDB-backed reducer is accepted, its CLI or library, binding, storage format, and version contract must first enter this inventory and the implementation mapping. The same rule prevents a locally installed compiler or media program from silently becoming part of the product contract.

15.7 Local and server-hosted references

The local commands are authoritative when no server is running. A running LeanFM server additionally exposes the requirements and implementation language references, the interrogation checklist, and read-only MCP context without requiring a user login. These public endpoints contain language documentation, not project or account data. Project artifacts, conversations, credentials and acceptance mutations remain authenticated and owner-scoped.

Selected Books and Further Reading

These annotations identify the role each work plays in this book; they are not reviews of the books.

Richards and Hamburger, *Logic and Language Models for Computer Science*.
A source of the author's background in formal languages and logic, not the originators of those methods [11].

Gensler, *Introduction to Logic*.
Background for symbolization, validity, proof, quantification, and modal reasoning [3].

Ryan et al., *The Modelling and Analysis of Security Protocols*.
The CSP and protocol-analysis source revisited in the protocol exercises [12].

Kröger and Merz, *Temporal Logic and State Systems*.
The author's principal source for temporal-logic background [5].

Lamport, *Specifying Systems*.
A reference point for the book's comparison with TLA+ state and action specifications [6].

Gunther, *Guerrilla Capacity Planning*.
The cited background for the Universal Scalability Law and capacity planning [4].

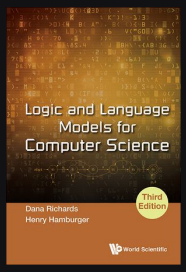
Menascé, Almeida, and Dowdy, *Performance by Design*.
Background for the queueing and capacity-planning discussion [9].

Mancosu, Galvan, and Zach, *An Introduction to Proof Theory*.
Further reading on natural deduction, sequent calculus, normalization, and cut elimination [8].

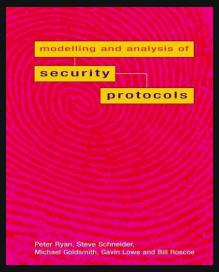
Pavlovic, *Programs as Diagrams*.
Related further reading; the current book does not rely on a specific claim from it [10].

Book-cover snapshots

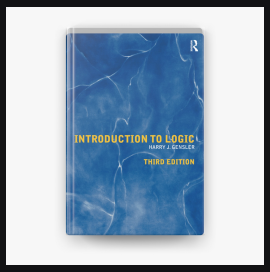
These cover snapshots identify the referenced editions.



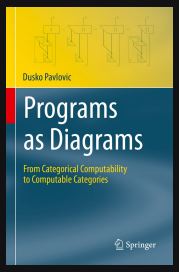
Logic and Language Models for Computer Science
Dana Richards and Henry Hamburger [11]



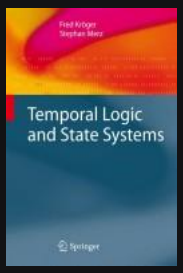
The Modelling and Analysis of Security Protocols
Peter Ryan, Steve Schneider, Michael Goldsmith, Gavin Lowe, and Bill Roscoe [12]



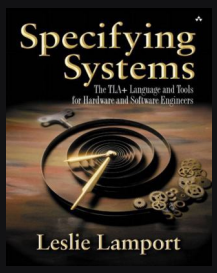
Introduction to Logic
Harry J. Gensler [3]



Programs as Diagrams
Dusko Pavlovic [10]



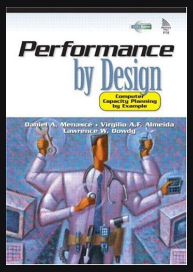
Temporal Logic and State Systems
Fred Kröger and Stephan Merz [5]



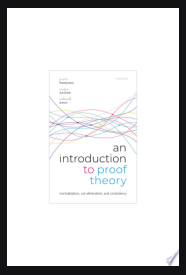
Specifying Systems
Leslie Lamport [6]



Guerrilla Capacity Planning
Neil J. Gunther [4]



Performance by Design
Daniel A. Menascé, Virgilio A. F. Almeida, and Lawrence W. Dowdy [9]



An Introduction to Proof Theory
Paolo Mancosu, Sergio Galvan, and Richard Zach [8]

Cover-image sources, accessed September 25 and 30, 2026: World Scientific edition via Kobo for Richards and Hamburger; Open Library for Ryan et al.; Apple Books for Gensler; and Kobo's Springer edition for Pavlovic. The Kröger and Merz cover snapshot is from Open Library. The Lamport and Menascé–Almeida–Dowdy snapshots are from Open Library; the Gunther snapshot is from Springer Nature; and the Mancosu–Galvan–Zach snapshot is from Google Books. Cover artwork remains the property of its respective publisher.

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